# Decoupled Precoder and Receiver Design for Massive MIMO Multiple LEO Satellite Communication

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***Abstract*—In this paper, we investigate the decoupled designs of precoders and receivers for both downlink (DL) and uplink (UL) in massive multiple-input multiple-output (MIMO) multiple low earth orbit (LEO) satellite communication systems. We first establish the beam based satellite channel model, where the beam matrix is constructed with sampled steering vectors. Then, we propose a decoupled precoder and receiver design for both DL and UL, which allows precoders and receivers to be designed independently at each satellite and user terminal (UT), respectively, with only local statistical channel state information (sCSI). Moreover, with the established beam based channel model, the design of space domain DL precoder and UL receiver can be converted into that of lower-dimensional beam domain vectors with only local sCSI, and the resulting space domain precoder and receiver are beam structured. Furthermore, we propose a low-complexity design and implementation for the beam structured DL precoder and UL receiver by exploiting properties of the beam matrix, significantly reducing the computational complexity. Simulation results validate the proposed approaches.**

***Index Terms*—Multi-satellite communication, massive MIMO, decoupled precoder and receiver design, statistical CSI.**

## I. Introduction

SATELLITE communication (SATCOM) is increasingly recognized as a transformative solution for bridging connectivity gaps [1], [2], [3], [4], [5]. Low Earth orbit (LEO) satellites positioned at altitudes between 500 and 2,000 km offer significant benefits over their geostationary (GEO) counterparts, including reduced latency, lower pathloss, and decreased power demands [6], [7], [8], [9]. Recent progress in satellite manufacturing, alongside declining launch costs and strong private investment, has made large-scale LEO constellations more practical than ever [10]. Satellite constellations are expected to be vital in future networks, complementing and extending terrestrial 5G systems through non-terrestrial networks (NTN).

### A. Related Work

Multi-beam satellites have played an important role in SATCOM, serving a number of ground user terminals (UTs) with multi-beam phased array antennas, which generate a limited number of spot beams with slow adaptability [11], [12], [13]. This makes it difficult to efficiently adapt to real-time link variations in satellite networks. Massive multiple-input multiple-output (MIMO) technology has been integral to the current 5G networks and is poised to underpin the forthcoming 6G systems [14], [15], [16], [17], [18]. For SATCOM, massive MIMO is utilized to improve connectivity, reliability, and data throughput [19], [20], [21], [22]. In [19], a massive MIMO transmission approach for single SATCOM systems is proposed, where the designs of downlink (DL) precoding and uplink (UL) receiving are investigated by using the statistical channel state information (sCSI). Inspired by [19], there are several works investigating massive MIMO SATCOM systems, including UL [20], and DL [21] transmissions as well as channel estimation [22]. In [20], an UL transmit design is developed to approach the ergodic sum rate capacity. In [21], a DL transmit design is proposed to maximize the ergodic sum rate, which can be converted into a precoder design. In [22], a channel estimation approach is proposed for massive MIMO SATCOM by utilizing the property of the satellite channel. Note that multi-satellite case is not considered in the above works.

The recent trend towards deploying large-scale satellite constellations, comprising hundreds or even thousands of satellites, has attracted considerable attention for its potential to significantly enhance system throughput and coverage [23], [24], [25], [26]. In [27], a low-complexity joint user scheduling and hybrid beamforming design method is proposed to

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improve the system spectral efficiency for multi-satellite cooperative systems. A deep machine learning based joint channel prediction and precoding design is proposed in [28] for multibeam LEO satellite Internet of Things. The beamspace MIMO for multi-satellite systems is investigated in [29], where a single ground station is served by multiple satellites, and the DL precoder and receiver are designed by using the location information. It should be noted that the aforementioned works assume that the links between satellites and receivers are perfectly synchronized. In [30], each satellite applies a time-advance mechanism so that the desired signals arrive synchronously at the target UT, whereas signals intended for other UTs remain asynchronous at that UT, and a cooperative beamforming method is proposed with sCSI to maximize the achievable rate.

For massive MIMO orthogonal frequency division multiplexing (OFDM) multi-LEO satellite systems, in contrast to [30] which performs synchronization at the satellites, [31] and [32] perform the time and frequency synchronizations at each UT. Specifically, in [31], a spatial receive processing approach with frequency and time compensations at each UT for each satellite is proposed, and a joint design of DL precoders and receivers based on sCSI is proposed to maximize the DL ergodic sum rate. In [32], a linear precoding technique combined with frequency and time compensations at each UT for each satellite is proposed to mitigate the effects of asynchronizations. Additionally, an Riemannian conjugate gradient based joint design of UL precoders and receivers with sCSI is proposed to maximize the UL ergodic sum rate.

### B. Motivations and Contributions

Even though the above works [31], [32] can achieve comparable performance to the cooperative transmission with genie aided synchronization and perfect instantaneous CSI (iCSI), their coupled DL/UL precoder and receiver designs necessitate numerous iterative steps to achieve local optimum and demand extensive inter-satellite information exchange. This results in considerable feedback and signaling overhead, especially in large-scale satellite constellations. To deal with these critical challenges, in this paper, we propose a decoupled design of precoder and receiver for both DL and UL in massive MIMO-OFDM multi-LEO satellite systems, where the designs are decoupled across satellites as well as between the transmitter and receiver sides, enabling each satellite and UT to perform independent design based on its local sCSI. We summarize our major contributions as follows.

- We establish the beam based channel model (BBCM) for massive MIMO SATCOM systems by exploiting the spatial single path property of satellite channels, where the steering vector is modeled as a beam matrix with a phase-shifted partial discrete Fourier transform (DFT) structure multiplied by a diffusion vector.
- We propose closed-form decoupled precoder and receiver (DPR) designs for both DL and UL in massive MIMO multi-LEO satellite systems. The proposed DL DPR and UL DPR can be designed independently at each satellite and each UT with only their local sCSI.
- With the proposed BBCM, we propose a DL beam structured precoder (BSP) design and an UL beam structured receiver (BSR) design for massive MIMO multi-LEO satellite systems. Specifically, we demonstrate that under certain conditions, the designs of DL precoder and UL receiver in the space domain can be converted into those of lower-dimensional beam domain vectors with only local sCSI, and the resulting space domain precoder and receiver are beam structured.
- We demonstrate that the computations of lower-dimensional beam domain vectors only involve real-valued matrices, and propose low-complexity designs and implementations for the proposed DL BSP and UL BSR by exploiting the properties of the beam matrices.

### C. Organization

The remaining sections are organized as follows. In Section II, we present the massive MIMO multi-LEO satellite system model including the system configuration, the proposed DL/UL BBCM, signal models, and the statistical properties of the satellite channel. Section III proposes the DL/UL DPR design with only local sCSI. Section IV proposes the DL BSP and the UL BSR, where the designs of space domain DL precoder and UL receiver are converted into that of lower-dimensional beam domain vectors with only local sCSI. A low-complexity design and implementation approach for the proposed DL BSP and UL BSR is presented in Section V. Section VI presents the simulation results. Section VII concludes this paper. The main symbols used in this paper are listed in Table I for ease of reference.

### D. Notations

Throughout this paper, scalars, vectors, and matrices are denoted by lowercase letters, bold lowercase letters, and bold uppercase letters, respectively. The set of real-valued matrices with dimensions $M \times N$ is denoted by $\mathbb{R}^{M\times N}$, and the set of complex-valued matrices with the same size is denoted by $\mathbb{C}^{M\times N}$. Conjugation, transpose, and conjugate transpose operations are, respectively, expressed as $(\cdot)^*$, $(\cdot)^T$, and $(\cdot)^H$. The element located at position $(m,n)$ within matrix $\mathbf{M}$ is expressed as $[\mathbf{M}]_{m,n}$, and the $l$-th element of vector $\mathbf{n}$ is identified as $[\mathbf{n}]_l$. We use $[\mathbf{m}]_{\mathcal{N}}$ to denote the sub-vector of the elements with the index set $\mathcal{N}$ of $\mathbf{m}$. The expectation operator is expressed as $\mathbb{E}\{\cdot\}$. Element-wise multiplication is given by $\odot$. The tensor product operation is given by $\otimes$. $(\cdot)^+$ is the Moore–Penrose inverse. For any set $\mathcal{N}$, the cardinality is expressed as $|\mathcal{N}|$, and the set difference between $\mathcal{M}$ and $\mathcal{N}$ is denoted as $\mathcal{M}\backslash\mathcal{N}$. A zero-mean circularly symmetric complex Gaussian distribution with covariance matrix $\mathbf{M}$ is given by $\mathcal{CN}(\mathbf{0},\mathbf{M})$. The vectorization operator $\mathrm{vec}(\mathbf{M})$ converts matrix $\mathbf{M}$ into a column vector by column-wise concatenation, while the inverse operation $\mathrm{uvec}(\mathbf{m})$ reconstructs the original matrix from its vectorized form $\mathbf{m}$. We adopt $k$, $i$ to denote the UT indices and adopt $s$, $j$ for the satellite indices.

## II. System Model

In this section, we first present the system configuration of the multi-satellite system with massive MIMO. Then, we

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TABLE I
SYMBOL LIST

| Notation | Definition |
|---|---|
| $M_{\mathrm{x}}$, $M_{\mathrm{y}}$ | Numbers of satellite antennas at x- and y- axes |
| $N_{\mathrm{x}'}$, $N_{\mathrm{y}'}$ | Numbers of UT antennas at x- and y- axes |
| $g^{\mathrm{x}}_{k,s}$, $g^{\mathrm{y}}_{k,s}$, $d^{\mathrm{x}'}_{k,s,\ell}$, $d^{\mathrm{y}'}_{k,s,\ell}$ | Directional cosines |
| $\mathbf{g}^{\mathrm{p}}_{k,s}$, $\mathbf{a}^{\mathrm{p}}_{\mathrm{x}}(g^{\mathrm{x}}_{k,s})$, $\mathbf{a}^{\mathrm{p}}_{\mathrm{y}}(g^{\mathrm{y}}_{k,s})$ | Array response vectors at the satellite side for $\mathrm{p} \in \{\mathrm{dl}, \mathrm{ul}\}$ |
| $\mathbf{d}^{\mathrm{p}}_{k,s,\ell}$, $\mathbf{a}^{\mathrm{p}}_{\mathrm{x}'}(d^{\mathrm{x}'}_{k,s,\ell})$, $\mathbf{a}^{\mathrm{p}}_{\mathrm{y}'}(d^{\mathrm{y}'}_{k,s,\ell})$ | Array response vectors at the UT side |
| $\mathbf{H}^{\mathrm{p}}_{k,s}$ | Channel matrix in OFDM subcarriers |
| $\beta^{\mathrm{p}}_{k,s}$ | Channel power |
| $\mathbf{R}^{\mathrm{ut,p}}_{k,s}$ | Channel correlation matrix at the UT side |
| $\mathbf{R}^{\mathrm{sat,p}}_{k,s}$ | Channel correlation matrix at the satellite side |
| $F^{\mathrm{sat}}_{\mathrm{x}}$, $F^{\mathrm{sat}}_{\mathrm{y}}$ | Fine factors at the satellite side |
| $F^{\mathrm{ut}}_{\mathrm{x}'}$, $F^{\mathrm{ut}}_{\mathrm{y}'}$ | Fine factors at the UT side |
| $\mathbf{V}^{\mathrm{p}}_{\mathrm{x}}$, $\mathbf{V}^{\mathrm{p}}_{\mathrm{y}}$, $\mathbf{V}^{\mathrm{p}}$ | Beam matrices at the satellite side |
| $\mathbf{U}^{\mathrm{p}}_{\mathrm{x}'}$, $\mathbf{U}^{\mathrm{p}}_{\mathrm{y}'}$, $\mathbf{U}^{\mathrm{p}}$ | Beam matrices at the UT side |
| $\mathcal{B}^{\mathrm{dia,p}}_{k,s}$, $\bar{\mathcal{B}}^{\mathrm{dia,p}}_{k,s}$ | Beam index sets at the satellite side |
| $\mathbf{\Omega}^{\mathrm{p}}_{k,s}$, $\bar{\mathbf{\Omega}}^{\mathrm{p}}_{k,s}$ | Beam selection matrices at the satellite side |
| $\tilde{\mathbf{g}}^{\mathrm{p,x}}_{k,s}$, $\tilde{\mathbf{g}}^{\mathrm{p,y}}_{k,s}$, $\tilde{\mathbf{g}}^{\mathrm{p}}_{k,s}$, $\check{\mathbf{g}}^{\mathrm{p}}_{k,s}$ | Diffusion vectors at the satellite side |
| $\tilde{\mathbf{d}}^{\mathrm{p,x}'}_{k,s,\ell}$, $\tilde{\mathbf{d}}^{\mathrm{p,y}'}_{k,s,\ell}$, $\tilde{\mathbf{d}}^{\mathrm{p}}_{k,s,\ell}$ | Diffusion vectors at the UT side |
| $\tilde{\mathbf{H}}^{\mathrm{p}}_{k,s}$, $\check{\mathbf{H}}^{\mathrm{p}}_{k,s}$ | Beam domain channel matrices |
| $p^{\mathrm{p}}_{k,s}$ | Transmit power |
| $\mathbf{w}^{\mathrm{p}}_{k,s}$ | Normalized space domain precoder |
| $\mathbf{c}^{\mathrm{p}}_{k,s}$ | Normalized space domain receiver |
| $\bar{\mathbf{w}}^{\mathrm{p}}_{k,s}$ | Normalized beam domain precoder |
| $\bar{\mathbf{c}}^{\mathrm{p}}_{k,s}$ | Normalized beam domain receiver |
| $\mathcal{R}^{\mathrm{p}}_{k,s}$ | Ergodic rate |

propose the DL/UL BBCM and present the DL/UL signal models and statistical properties of the satellite channel.

### A. System Configuration

Consider a multiple LEO satellite communication system, where $S$ satellites provide services to $K$ multi-antenna ground UTs simultaneously. As illustrated in Fig. 1, each satellite $s$ is equipped with a uniform planar array (UPA) with $M_{\mathrm{x}}$ antennas in the x-axis and $M_{\mathrm{y}}$ antennas in the y-axis, where $s \in \mathcal{S}$ with $\mathcal{S} = \{1, 2, \ldots, S\}$. The number of antenna elements at each satellite is $M = M_{\mathrm{x}} M_{\mathrm{y}}$. Meanwhile, each UT $k$ is equipped with a UPA with $N_{\mathrm{x}'}$ antennas in the x′-axis and $N_{\mathrm{y}'}$ antennas in the y′-axis, where $k \in \mathcal{K}$ with $\mathcal{K} = \{1, 2, \ldots, K\}$. The number of antenna elements at each UT is $N = N_{\mathrm{x}'} N_{\mathrm{y}'}$. The set of UTs served by satellite $s$ is denoted as $\mathcal{K}_s$ with $|\mathcal{K}_s| = K_s$. The set of satellites visible to UT $k$ is denoted as $\mathcal{S}_k$ with $|\mathcal{S}_k| = S_k$. We assume that each UT's UPA lies on the horizontal x′-y′ plane, and its rotation about the local vertical $z'$- axis is modeled as random and independent across UTs. For each satellite, the $z$- axis of the UPA points toward Earth's center, while the x- and y- axes are oriented toward the geographic South Pole and West, respectively.

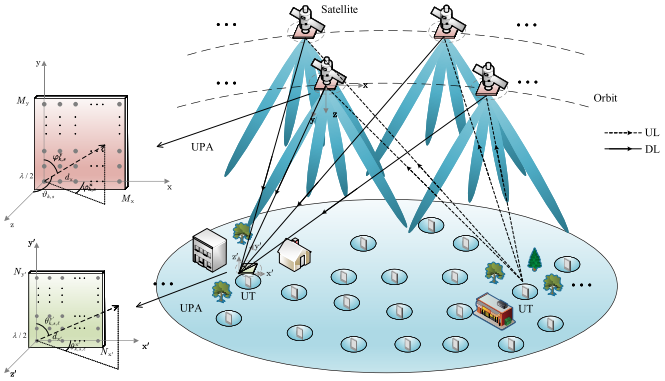



Fig. 1. Massive MIMO multi-LEO SATCOM system.

We adopt orthogonal frequency division multiplexing (OFDM) modulation in the multi-satellite massive MIMO system, with $N_{\mathrm{sc}}$ subcarriers and $N_{\mathrm{cp}}$ cyclic prefix (CP) samples. Let $N_{\mathrm{t}} = N_{\mathrm{cp}} + N_{\mathrm{sc}}$. The symbol duration with and without CP can be given by $T = T_{\mathrm{cp}} + T_{\mathrm{sc}}$ and $T_{\mathrm{sc}} = T_{\mathrm{s}} N_{\mathrm{sc}}$, respectively, where $T_{\mathrm{cp}} = T_{\mathrm{s}} N_{\mathrm{cp}}$, and $T_{\mathrm{s}}$ is the system sampling period.

### B. DL/UL BBCM

After appropriate time and frequency synchronizations at the UT side, the DL channel matrix between satellite $s$ and UT $k$ over a specific subcarrier exhibits the spatial single path property and is given by [31]

$$\mathbf{H}^{\mathrm{dl}}_{k,s} = \mathbf{d}^{\mathrm{dl}}_{k,s} (\mathbf{g}^{\mathrm{dl}}_{k,s})^{H}, \tag{1}$$

where $\mathbf{d}^{\mathrm{dl}}_{k,s} = \sum_{\ell=0}^{L_{k,s}-1} \hat{a}^{\mathrm{dl}}_{k,s,\ell} \mathbf{d}^{\mathrm{dl}}_{k,s,\ell}$, $L_{k,s}$ denotes the number of multi-paths and $\hat{a}^{\mathrm{dl}}_{k,s,\ell}$ is the effective channel gain related to path $\ell$ of the channel between satellite $s$ and UT $k$. $\mathbf{g}^{\mathrm{dl}}_{k,s}$ and $\mathbf{d}^{\mathrm{dl}}_{k,s,\ell}$ refer to the DL steering vector at satellite $s$ for UT $k$ and the DL steering vector at UT $k$ for satellite $s$ associated with the $\ell$-th path, respectively, and are expressed as

$$\mathbf{g}^{\mathrm{dl}}_{k,s} = \mathbf{a}^{\mathrm{dl}}_{\mathrm{x}}(g^{\mathrm{x}}_{k,s}) \otimes \mathbf{a}^{\mathrm{dl}}_{\mathrm{y}}(g^{\mathrm{y}}_{k,s}), \tag{2a}$$

$$\mathbf{d}^{\mathrm{dl}}_{k,s,\ell} = \mathbf{a}^{\mathrm{dl}}_{\mathrm{x}'}(d^{\mathrm{x}'}_{k,s,\ell}) \otimes \mathbf{a}^{\mathrm{dl}}_{\mathrm{y}'}(d^{\mathrm{y}'}_{k,s,\ell}), \tag{2b}$$

where $g^{\mathrm{x}}_{k,s} = \cos\varphi^{\mathrm{x}}_{k,s} \sin\varphi^{\mathrm{y}}_{k,s}$ and $g^{\mathrm{y}}_{k,s} = \cos\varphi^{\mathrm{y}}_{k,s}$ are the directional cosines of satellite $s$ for UT $k$ with respect to (w.r.t.) x-axis and y-axis, respectively, and $d^{\mathrm{x}'}_{k,s,\ell} = \cos\theta^{\mathrm{x}'}_{k,s,\ell} \sin\theta^{\mathrm{y}'}_{k,s,\ell}$ and $d^{\mathrm{y}'}_{k,s,\ell} = \cos\theta^{\mathrm{y}'}_{k,s,\ell}$ are the directional cosines of UT $k$ for satellite $s$ w.r.t. x′-axis and y′-axis related to the $\ell$-th path of the channel, respectively. $\varphi^{\mathrm{x}}_{k,s}$, $\varphi^{\mathrm{y}}_{k,s}$, $\theta^{\mathrm{x}'}_{k,s,\ell}$, and $\theta^{\mathrm{y}'}_{k,s,\ell}$ are the corresponding angles w.r.t. the x-axis, y-axis, x′-axis and y′-axis, respectively. Let $\mathrm{v} \in \{\mathrm{x}, \mathrm{y}, \mathrm{x}', \mathrm{y}'\}$, then $\mathbf{a}^{\mathrm{dl}}_{\mathrm{v}}(x)$ in (2) is defined as

$$\mathbf{a}^{\mathrm{dl}}_{\mathrm{v}}(x) = \frac{\omega^{\mathrm{dl}}_{\mathrm{v}}(x)}{\sqrt{A_{\mathrm{v}}}} \left[1, e^{-\jmath\pi\chi^{\mathrm{dl}}_{\mathrm{v}} x}, \ldots, e^{-\jmath\pi(A_{\mathrm{v}}-1)\chi^{\mathrm{dl}}_{\mathrm{v}} x}\right]^{T}, \tag{3}$$

where $\omega^{\mathrm{dl}}_{\mathrm{v}}(x) = e^{\jmath\pi(A_{\mathrm{v}}-1)\frac{\chi^{\mathrm{dl}}_{\mathrm{v}}}{2} x}$ is chosen to make $[\mathbf{a}^{\mathrm{dl}}_{\mathrm{v}}(x)]^{*}_{n} = [\mathbf{a}^{\mathrm{dl}}_{\mathrm{v}}(x)]_{A_{\mathrm{v}}-n+1}$, $A_{\mathrm{v}}$ is the antenna number in the v-axis, and $\chi^{\mathrm{dl}}_{\mathrm{v}} = \frac{2d_{\mathrm{v}}}{\lambda_{\mathrm{dl}}}$ with $\lambda_{\mathrm{dl}} = \frac{c}{f^{\mathrm{dl}}_{\mathrm{c}}}$. $d_{\mathrm{v}}$, $f^{\mathrm{dl}}_{\mathrm{c}}$, and $c$ are the antenna spacing in the v-axis, the DL central frequency, and the light speed, respectively. Note that $\omega^{\mathrm{dl}}_{\mathrm{v}}(x)$ will be utilized to reduce

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the complexity of the subsequently proposed DL BSP and UL BSR at each satellite side. We refer to $\chi_{\mathrm{v}}^{\mathrm{dl}}x$ in (3) as the scaled directional cosine.

In a frequency division duplex (FDD) massive MIMO-OFDM multi-LEO satellite communication system, where each satellite employs the same antenna array for both DL transmission and UL reception, the antenna array can not realize half-wavelength spacing simultaneously for both links due to the inherent carrier frequency difference. To obtain a BBCM, uniform sampling for scaled directional cosines $\chi_{\mathrm{t}}^{\mathrm{dl}}g_{k,s}^{\mathrm{t}}$ and $\chi_{\mathrm{u}}^{\mathrm{dl}}d_{k,s,\ell}^{\mathrm{u}}$ is adopted, where $\mathrm{t}\in\{\mathrm{x},\mathrm{y}\}$ and $\mathrm{u}\in\{\mathrm{x}',\mathrm{y}'\}$. Let $\vartheta_{\max}\in\left[0,\frac{\pi}{2}\right)$ be the satellite maximum nadir angle, then the angle w.r.t. the t-axis i.e., $\varphi_{k,s}^{\mathrm{t}}$ should satisfy $\varphi_{k,s}^{\mathrm{t}}\in\left[\frac{\pi}{2}-\vartheta_{\max},\frac{\pi}{2}+\vartheta_{\max}\right]$. The corresponding scaled directional cosines $\chi_{\mathrm{t}}^{\mathrm{dl}}g_{k,s}^{\mathrm{t}}$ should satisfy $\chi_{\mathrm{t}}^{\mathrm{dl}}g_{k,s}^{\mathrm{t}}\in[-g_{\max}^{\mathrm{dl}},g_{\max}^{\mathrm{dl}}]$, where $g_{\max}^{\mathrm{dl}}=\chi_{\mathrm{t}}^{\mathrm{dl}}\cos\left(\frac{\pi}{2}-\vartheta_{\max}\right)$. Let $\mathcal{C}_{\breve{m}_{\mathrm{t}}}^{\mathrm{sat,dl}}=[2\breve{m}_{\mathrm{t}}/\bar{M}_{\mathrm{t}}-g_{\max}^{\mathrm{dl}},2(\breve{m}_{\mathrm{t}}+1)/\bar{M}_{\mathrm{t}}-g_{\max}^{\mathrm{dl}})$, where $\bar{M}_{\mathrm{t}}=F_{\mathrm{t}}^{\mathrm{sat}}M_{\mathrm{t}}$, $0\le\breve{m}_{\mathrm{t}}\le\breve{M}_{\mathrm{t}}^{\mathrm{dl}}-1$, $\breve{M}_{\mathrm{t}}^{\mathrm{dl}}=\lceil g_{\max}^{\mathrm{dl}}\bar{M}_{\mathrm{t}}\rceil$ and $F_{\mathrm{t}}^{\mathrm{sat}}$ are the sampling number and the fine factor related to the t-axis, respectively. Let $\mathcal{C}_{\breve{n}_{\mathrm{u}}}^{\mathrm{ut,dl}}=[2\breve{n}_{\mathrm{u}}/\bar{N}_{\mathrm{u}}-\chi_{\mathrm{u}}^{\mathrm{dl}},2(\breve{n}_{\mathrm{u}}+1)/\bar{N}_{\mathrm{u}}-\chi_{\mathrm{u}}^{\mathrm{dl}})$, where $\bar{N}_{\mathrm{u}}=F_{\mathrm{u}}^{\mathrm{ut}}N_{\mathrm{u}}$, $0\le\breve{n}_{\mathrm{u}}\le\breve{N}_{\mathrm{u}}^{\mathrm{dl}}-1$, $\breve{N}_{\mathrm{u}}^{\mathrm{dl}}=\lceil\chi_{\mathrm{u}}^{\mathrm{dl}}\bar{N}_{\mathrm{u}}\rceil$ and $F_{\mathrm{u}}^{\mathrm{ut}}$ are the sampling number and fine factor related to the u-axis, respectively. Define the DL beam matrices of t-axis and u-axis as

$$\mathbf{V}_{\mathrm{t}}^{\mathrm{dl}}=\left[\breve{\mathbf{a}}_{\mathrm{t}}^{\mathrm{dl}}\left(b_{0}^{\mathrm{sat,dl}}\right),\breve{\mathbf{a}}_{\mathrm{t}}^{\mathrm{dl}}\left(b_{1}^{\mathrm{sat,dl}}\right),\cdots,\breve{\mathbf{a}}_{\mathrm{t}}^{\mathrm{dl}}\left(b_{\breve{M}_{\mathrm{t}}^{\mathrm{dl}}-1}^{\mathrm{sat,dl}}\right)\right],\tag{4a}$$

$$\mathbf{U}_{\mathrm{u}}^{\mathrm{dl}}=\left[\breve{\mathbf{a}}_{\mathrm{u}}^{\mathrm{dl}}\left(b_{0}^{\mathrm{ut,dl}}\right),\breve{\mathbf{a}}_{\mathrm{u}}^{\mathrm{dl}}\left(b_{1}^{\mathrm{ut,dl}}\right),\cdots,\breve{\mathbf{a}}_{\mathrm{u}}^{\mathrm{dl}}\left(b_{\breve{N}_{\mathrm{u}}^{\mathrm{dl}}-1}^{\mathrm{ut,dl}}\right)\right],\tag{4b}$$

where $b_{\breve{m}_{\mathrm{t}}}^{\mathrm{sat,dl}}=\frac{2\breve{m}_{\mathrm{t}}}{\bar{M}_{\mathrm{t}}}-g_{\max}^{\mathrm{dl}}$ and $b_{\breve{n}_{\mathrm{u}}}^{\mathrm{ut,dl}}=\frac{2\breve{n}_{\mathrm{u}}}{\bar{N}_{\mathrm{u}}}-\chi_{\mathrm{u}}^{\mathrm{dl}}$ are the approximated scaled directional cosines in sets $\mathcal{C}_{\breve{m}_{\mathrm{t}}}^{\mathrm{sat,dl}}$ and $\mathcal{C}_{\breve{n}_{\mathrm{u}}}^{\mathrm{ut,dl}}$, respectively, and $\breve{\mathbf{a}}_{\mathrm{v}}^{\mathrm{dl}}(x)$ is defined as

$$\breve{\mathbf{a}}_{\mathrm{v}}^{\mathrm{dl}}(x)=\frac{\breve{\omega}_{\mathrm{v}}^{\mathrm{dl}}(x)}{\sqrt{A_{\mathrm{v}}}}\left[1,e^{-\jmath\pi x},\ldots,e^{-\jmath\pi(A_{\mathrm{v}}-1)x}\right]^{T},\tag{5}$$

with $\breve{\omega}_{\mathrm{v}}^{\mathrm{dl}}(x)=e^{\jmath\pi(A_{\mathrm{v}}-1)\frac{x}{2}}$. Note that the beam matrices are constructed with sampled steering vectors, and each column of the beam matrices corresponds to a specific beam in the space domain. We can express $\mathbf{a}_{\mathrm{t}}^{\mathrm{dl}}(g_{k,s}^{\mathrm{t}})$ in (2a) as

$$\mathbf{a}_{\mathrm{t}}^{\mathrm{dl}}(g_{k,s}^{\mathrm{t}})=\mathbf{V}_{\mathrm{t}}^{\mathrm{dl}}\tilde{\mathbf{g}}_{\mathrm{t}}^{\mathrm{dl}}(g_{k,s}^{\mathrm{t}}),\tag{6}$$

where $\tilde{\mathbf{g}}_{\mathrm{t}}^{\mathrm{dl}}(g_{k,s}^{\mathrm{t}})$ denotes the diffusion vector, representing how the energy related to the actual UT direction is distributed across multiple adjacent beams and can be obtained by using a sparse signal recovery algorithm, e.g. multiple measurement vector focal underdetermined system solver (MFOCUSS) [33]. Notice that when $\chi_{\mathrm{t}}^{\mathrm{dl}}g_{k,s}^{\mathrm{t}}=b_{\breve{m}_{\mathrm{t}}}^{\mathrm{sat,dl}},\forall\breve{m}_{\mathrm{t}}=0,1,\ldots,\breve{M}_{\mathrm{t}}^{\mathrm{dl}}-1$, we have $\tilde{\mathbf{g}}_{\mathrm{t}}^{\mathrm{dl}}(g_{k,s}^{\mathrm{t}})=\mathbf{e}_{\breve{m}_{\mathrm{t}},\breve{M}_{\mathrm{t}}^{\mathrm{dl}}}$ where $\mathbf{e}_{m,\breve{M}_{\mathrm{t}}^{\mathrm{dl}}}$ is an $\breve{M}_{\mathrm{t}}^{\mathrm{dl}}$-length vector, with the $\breve{m}_{\mathrm{t}}$-th element being 1 and all the other elements being 0, indicating that all the energy is concentrated in the $\breve{m}_{\mathrm{t}}$-th beam. For simplicity, we denote $\tilde{\mathbf{g}}_{k,s}^{\mathrm{dl,t}}=\tilde{\mathbf{g}}_{\mathrm{t}}^{\mathrm{dl}}(g_{k,s}^{\mathrm{t}})$. The steering vector at each satellite side in (2a) can be rewritten as

$$\mathbf{g}_{k,s}^{\mathrm{dl}}=(\mathbf{V}_{\mathrm{x}}^{\mathrm{dl}}\otimes\mathbf{V}_{\mathrm{y}}^{\mathrm{dl}})(\tilde{\mathbf{g}}_{k,s}^{\mathrm{dl,x}}\otimes\tilde{\mathbf{g}}_{k,s}^{\mathrm{dl,y}})=\mathbf{V}^{\mathrm{dl}}\tilde{\mathbf{g}}_{k,s}^{\mathrm{dl}},\tag{7}$$

where $\mathbf{V}^{\mathrm{dl}}=\mathbf{V}_{\mathrm{x}}^{\mathrm{dl}}\otimes\mathbf{V}_{\mathrm{y}}^{\mathrm{dl}}\in\mathbb{C}^{M\times\breve{M}^{\mathrm{dl}}}$, $\tilde{\mathbf{g}}_{k,s}^{\mathrm{dl}}=\tilde{\mathbf{g}}_{k,s}^{\mathrm{dl,x}}\otimes\tilde{\mathbf{g}}_{k,s}^{\mathrm{dl,y}}\in\mathbb{C}^{\breve{M}^{\mathrm{dl}}\times1}$, and $\breve{M}^{\mathrm{dl}}=\breve{M}_{\mathrm{x}}^{\mathrm{dl}}\breve{M}_{\mathrm{y}}^{\mathrm{dl}}$. For the steering vector at each UT side, $\mathbf{a}_{\mathrm{u}}^{\mathrm{dl}}(d_{k,s,\ell}^{\mathrm{u}})$ can be expressed as

$$\mathbf{a}_{\mathrm{u}}^{\mathrm{dl}}(d_{k,s,\ell}^{\mathrm{u}})=\mathbf{U}_{\mathrm{u}}^{\mathrm{dl}}\tilde{\mathbf{d}}_{\mathrm{u}}^{\mathrm{dl}}(d_{k,s,\ell}^{\mathrm{u}}),\tag{8}$$

where $\tilde{\mathbf{d}}_{\mathrm{u}}^{\mathrm{dl}}(d_{k,s,\ell}^{\mathrm{u}})$ denotes the corresponding diffusion vector. For simplicity, we denote $\tilde{\mathbf{d}}_{k,s,\ell}^{\mathrm{dl,u}}=\tilde{\mathbf{d}}_{\mathrm{u}}^{\mathrm{dl}}(d_{k,s,\ell}^{\mathrm{u}})$. The steering vector at each UT side in (2b) can be rewritten as

$$\mathbf{d}_{k,s,\ell}^{\mathrm{dl}}=(\mathbf{U}_{\mathrm{x}'}^{\mathrm{dl}}\otimes\mathbf{U}_{\mathrm{y}'}^{\mathrm{dl}})(\tilde{\mathbf{d}}_{k,s,\ell}^{\mathrm{dl,x}'}\otimes\tilde{\mathbf{d}}_{k,s,\ell}^{\mathrm{dl,y}'})=\mathbf{U}^{\mathrm{dl}}\tilde{\mathbf{d}}_{k,s,\ell}^{\mathrm{dl}},\tag{9}$$

where $\mathbf{U}^{\mathrm{dl}}=\mathbf{U}_{\mathrm{x}'}^{\mathrm{dl}}\otimes\mathbf{U}_{\mathrm{y}'}^{\mathrm{dl}}\in\mathbb{C}^{N\times\breve{N}^{\mathrm{dl}}}$, $\tilde{\mathbf{d}}_{k,s,\ell}^{\mathrm{dl}}=\tilde{\mathbf{d}}_{k,s,\ell}^{\mathrm{dl,x}'}\otimes\tilde{\mathbf{d}}_{k,s,\ell}^{\mathrm{dl,y}'}\in\mathbb{C}^{\breve{N}^{\mathrm{dl}}\times1}$, and $\breve{N}^{\mathrm{dl}}=\breve{N}_{\mathrm{x}'}^{\mathrm{dl}}\breve{N}_{\mathrm{y}'}^{\mathrm{dl}}$. Then, we can rewrite the DL channel matrix in (1) as

$$\mathbf{H}_{k,s}^{\mathrm{dl}}=\mathbf{U}^{\mathrm{dl}}\tilde{\mathbf{d}}_{k,s}^{\mathrm{dl}}(\tilde{\mathbf{g}}_{k,s}^{\mathrm{dl}})^{H}(\mathbf{V}^{\mathrm{dl}})^{H}=\mathbf{U}^{\mathrm{dl}}\widetilde{\mathbf{H}}_{k,s}^{\mathrm{dl}}(\mathbf{V}^{\mathrm{dl}})^{H},\tag{10}$$

where $\tilde{\mathbf{d}}_{k,s}^{\mathrm{dl}}=\sum_{\ell=0}^{L_{k,s}-1}\hat{a}_{k,s,\ell}^{\mathrm{dl}}\tilde{\mathbf{d}}_{k,s,\ell}^{\mathrm{dl}}$, and $\widetilde{\mathbf{H}}_{k,s}^{\mathrm{dl}}=\tilde{\mathbf{d}}_{k,s}^{\mathrm{dl}}(\tilde{\mathbf{g}}_{k,s}^{\mathrm{dl}})^{H}$ denotes the channel model in the beam domain. We refer to the channel representation in (10) as the DL BBCM.

For the UL, after appropriate frequency and time synchronizations at the UT side, the UL channel matrix from UT $k$ to satellite $s$ over a particular subcarrier can be expressed as [32]

$$\mathbf{H}_{k,s}^{\mathrm{ul}}=\mathbf{g}_{k,s}^{\mathrm{ul}}(\mathbf{d}_{k,s}^{\mathrm{ul}})^{H},\tag{11}$$

where $\mathbf{d}_{k,s}^{\mathrm{ul}}=\sum_{\ell=0}^{L_{k,s}-1}\hat{a}_{k,s,\ell}^{\mathrm{ul}}\mathbf{d}_{k,s,\ell}^{\mathrm{ul}}$, and $\hat{a}_{k,s,\ell}^{\mathrm{ul}}$ is the effective channel gain related to path $\ell$ of the UL channel between UT $k$ and satellite $s$. $\mathbf{g}_{k,s}^{\mathrm{ul}}$ and $\mathbf{d}_{k,s,\ell}^{\mathrm{ul}}$ refer to the UL steering vector at satellite $s$ for UT $k$ and the UL steering vector at UT $k$ for satellite $s$ associated with the $\ell$-th path, respectively, and are expressed as

$$\mathbf{g}_{k,s}^{\mathrm{ul}}=\mathbf{a}_{\mathrm{x}}^{\mathrm{ul}}(g_{k,s}^{\mathrm{x}})\otimes\mathbf{a}_{\mathrm{y}}^{\mathrm{ul}}(g_{k,s}^{\mathrm{y}}),\tag{12a}$$

$$\mathbf{d}_{k,s,\ell}^{\mathrm{ul}}=\mathbf{a}_{\mathrm{x}'}^{\mathrm{ul}}(d_{k,s,\ell}^{\mathrm{x}'})\otimes\mathbf{a}_{\mathrm{y}'}^{\mathrm{ul}}(d_{k,s,\ell}^{\mathrm{y}'}),\tag{12b}$$

where

$$\mathbf{a}_{\mathrm{v}}^{\mathrm{ul}}(x)=\frac{\omega_{\mathrm{v}}^{\mathrm{ul}}(x)}{\sqrt{A_{\mathrm{v}}}}\left[1,e^{-\jmath\pi\chi_{\mathrm{v}}^{\mathrm{ul}}x},\ldots,e^{-\jmath\pi(A_{\mathrm{v}}-1)\chi_{\mathrm{v}}^{\mathrm{ul}}x}\right]^{T},\tag{13}$$

$\omega_{\mathrm{v}}^{\mathrm{ul}}(x)=e^{\jmath\pi(A_{\mathrm{v}}-1)\frac{\chi_{\mathrm{v}}^{\mathrm{ul}}}{2}x}$, $\chi_{\mathrm{v}}^{\mathrm{ul}}=\frac{2d_{\mathrm{v}}}{\lambda_{\mathrm{ul}}}$, $\lambda_{\mathrm{ul}}=\frac{c}{f_{\mathrm{c}}^{\mathrm{ul}}}$ and $f_{\mathrm{c}}^{\mathrm{ul}}$ is the UL central frequency. Let $\mathbf{V}_{\mathrm{t}}^{\mathrm{ul}}\in\mathbb{C}^{M_{\mathrm{t}}\times\breve{M}_{\mathrm{t}}^{\mathrm{ul}}}$ and $\mathbf{U}_{\mathrm{u}}^{\mathrm{ul}}\in\mathbb{C}^{N_{\mathrm{u}}\times\breve{N}_{\mathrm{u}}^{\mathrm{ul}}}$ be the UL beam matrices that are defined similarly as DL beam matrices in (4). The UL steering vectors at satellite sides and UT sides can be rewritten as

$$\mathbf{g}_{k,s}^{\mathrm{ul}}=(\mathbf{V}_{\mathrm{x}}^{\mathrm{ul}}\otimes\mathbf{V}_{\mathrm{y}}^{\mathrm{ul}})(\tilde{\mathbf{g}}_{k,s}^{\mathrm{ul,x}}\otimes\tilde{\mathbf{g}}_{k,s}^{\mathrm{ul,y}})=\mathbf{V}^{\mathrm{ul}}\tilde{\mathbf{g}}_{k,s}^{\mathrm{ul}},\tag{14a}$$

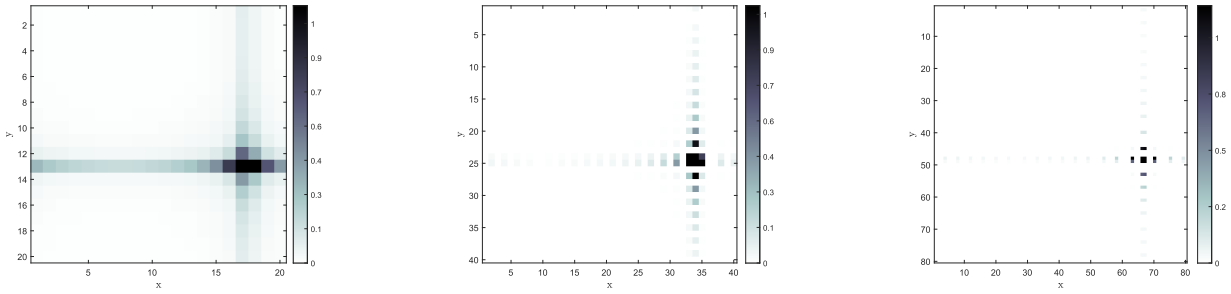
$$\mathbf{d}_{k,s,\ell}^{\mathrm{ul}}=(\mathbf{U}_{\mathrm{x}'}^{\mathrm{ul}}\otimes\mathbf{U}_{\mathrm{y}'}^{\mathrm{ul}})(\tilde{\mathbf{d}}_{k,s,\ell}^{\mathrm{ul,x}'}\otimes\tilde{\mathbf{d}}_{k,s,\ell}^{\mathrm{dl,y}'})=\mathbf{U}^{\mathrm{ul}}\tilde{\mathbf{d}}_{k,s,\ell}^{\mathrm{ul}},\tag{14b}$$

where $\mathbf{V}^{\mathrm{ul}}=\mathbf{V}_{\mathrm{x}}^{\mathrm{ul}}\otimes\mathbf{V}_{\mathrm{y}}^{\mathrm{ul}}\in\mathbb{C}^{M\times\breve{M}^{\mathrm{ul}}}$, $\breve{M}^{\mathrm{ul}}=\breve{M}_{\mathrm{x}}^{\mathrm{ul}}\breve{M}_{\mathrm{y}}^{\mathrm{ul}}$, $\mathbf{U}^{\mathrm{ul}}=\mathbf{U}_{\mathrm{x}'}^{\mathrm{ul}}\otimes\mathbf{U}_{\mathrm{y}'}^{\mathrm{ul}}\in\mathbb{C}^{N\times\breve{N}^{\mathrm{ul}}}$, $\breve{N}^{\mathrm{ul}}=\breve{N}_{\mathrm{x}'}^{\mathrm{ul}}\breve{N}_{\mathrm{y}'}^{\mathrm{ul}}$, $\tilde{\mathbf{g}}_{k,s}^{\mathrm{ul}}=\tilde{\mathbf{g}}_{k,s}^{\mathrm{ul,x}}\otimes\tilde{\mathbf{g}}_{k,s}^{\mathrm{ul,y}}$, $\tilde{\mathbf{d}}_{k,s,\ell}^{\mathrm{ul}}=\tilde{\mathbf{d}}_{k,s,\ell}^{\mathrm{ul,x}'}\otimes\tilde{\mathbf{d}}_{k,s,\ell}^{\mathrm{ul,y}'}$, and $\tilde{\mathbf{g}}_{k,s}^{\mathrm{ul,t}}$ and $\tilde{\mathbf{d}}_{k,s,\ell}^{\mathrm{ul,u}}$ denote the corresponding diffusion vectors. Then, we can rewrite the UL channel matrix in (11) as

$$\mathbf{H}_{k,s}^{\mathrm{ul}}=\mathbf{V}^{\mathrm{ul}}\tilde{\mathbf{g}}_{k,s}^{\mathrm{ul}}(\tilde{\mathbf{d}}_{k,s}^{\mathrm{ul}})^{H}(\mathbf{U}^{\mathrm{ul}})^{H}=\mathbf{V}^{\mathrm{ul}}\widetilde{\mathbf{H}}_{k,s}^{\mathrm{ul}}(\mathbf{U}^{\mathrm{ul}})^{H},\tag{15}$$

where $\widetilde{\mathbf{H}}_{k,s}^{\mathrm{ul}}=\tilde{\mathbf{g}}_{k,s}^{\mathrm{ul}}(\tilde{\mathbf{d}}_{k,s}^{\mathrm{ul}})^{H}$ denotes the UL channel model in the beam domain, and $\tilde{\mathbf{d}}_{k,s}^{\mathrm{ul}}=\sum_{\ell=0}^{L_{k,s}-1}\hat{a}_{k,s,\ell}^{\mathrm{ul}}\tilde{\mathbf{d}}_{k,s,\ell}^{\mathrm{ul}}$. We

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(a) $F_{\mathrm{x}}^{\mathrm{sat}} = F_{\mathrm{y}}^{\mathrm{sat}} = 1$ (b) $F_{\mathrm{x}}^{\mathrm{sat}} = F_{\mathrm{y}}^{\mathrm{sat}} = 2$ (c) $F_{\mathrm{x}}^{\mathrm{sat}} = F_{\mathrm{y}}^{\mathrm{sat}} = 4$

Fig. 2. Magnitude of $|\tilde{\mathbf{g}}_{k,s}^{\mathrm{p}}|$ with different fine factors.

refer to the channel representation in (15) as the UL BBCM. To ensure effective sampling and avoid spatial aliasing, the proposed DL/UL BBCM assumes that the scaling factor $\chi_{\mathrm{v}}^{\mathrm{p}}$ satisfies $\chi_{\mathrm{v}}^{\mathrm{p}} \leq 1$ with $\mathrm{p} \in \{\mathrm{dl}, \mathrm{ul}\}$.

Note that beam matrices $\mathbf{V}^{\mathrm{p}}$ and $\mathbf{U}^{\mathrm{p}}$ are independent from OFDM subcarriers and common for the UTs. Define $\breve{\boldsymbol{\omega}}_{\mathrm{t}}^{\mathrm{sat,p}} = \frac{1}{\sqrt{M_{\mathrm{t}}}}[\breve{\omega}_{\mathrm{t}}^{\mathrm{p}}(b_0^{\mathrm{sat,p}}), \ldots, \breve{\omega}_{\mathrm{t}}^{\mathrm{p}}(b_{\breve{M}_{\mathrm{t}}^{\mathrm{p}}-1}^{\mathrm{sat,p}})]^T$, $\breve{\boldsymbol{\omega}}_{\mathrm{u}}^{\mathrm{ut,p}} = \frac{1}{\sqrt{N_{\mathrm{u}}}}[\breve{\omega}_{\mathrm{u}}^{\mathrm{p}}(b_0^{\mathrm{ut,p}}), \ldots, \breve{\omega}_{\mathrm{u}}^{\mathrm{p}}(b_{\breve{N}_{\mathrm{u}}^{\mathrm{p}}-1}^{\mathrm{ut,p}})]^T$, $\hat{\boldsymbol{\omega}}_{\mathrm{t}}^{\mathrm{sat,p}} = [1, e^{\jmath\pi g_{\max}^{\mathrm{p}}}, \ldots, e^{\jmath\pi(M_{\mathrm{t}}-1)g_{\max}^{\mathrm{p}}}]^T$, $\hat{\boldsymbol{\omega}}_{\mathrm{u}}^{\mathrm{ut,p}} = [1, e^{\jmath\pi\chi_{\mathrm{u}}^{\mathrm{p}}}, \ldots, e^{\jmath\pi(N_{\mathrm{u}}-1)\chi_{\mathrm{u}}^{\mathrm{p}}}]^T$, $\hat{\boldsymbol{\omega}}^{\mathrm{sat,p}} = \hat{\boldsymbol{\omega}}_{\mathrm{x}}^{\mathrm{sat,p}} \otimes \hat{\boldsymbol{\omega}}_{\mathrm{y}}^{\mathrm{sat,p}}$, $\hat{\boldsymbol{\omega}}^{\mathrm{ut,p}} = \hat{\boldsymbol{\omega}}_{\mathrm{x}'}^{\mathrm{ut,p}} \otimes \hat{\boldsymbol{\omega}}_{\mathrm{y}'}^{\mathrm{ut,p}}$, $\breve{\boldsymbol{\omega}}^{\mathrm{sat,p}} = \breve{\boldsymbol{\omega}}_{\mathrm{x}}^{\mathrm{sat,p}} \otimes \breve{\boldsymbol{\omega}}_{\mathrm{y}}^{\mathrm{sat,p}}$, and $\breve{\boldsymbol{\omega}}^{\mathrm{ut,p}} = \breve{\boldsymbol{\omega}}_{\mathrm{x}'}^{\mathrm{ut,p}} \otimes \breve{\boldsymbol{\omega}}_{\mathrm{y}'}^{\mathrm{ut,p}}$, then $\mathbf{V}^{\mathrm{p}}$ and $\mathbf{U}^{\mathrm{p}}$ can be rewritten as

$$\begin{aligned}\mathbf{V}^{\mathrm{p}} &= \mathbf{V}_{\mathrm{x}}^{\mathrm{p}} \otimes \mathbf{V}_{\mathrm{y}}^{\mathrm{p}} \\ &= \mathrm{diag}\big(\hat{\boldsymbol{\omega}}_{\mathrm{x}}^{\mathrm{sat,p}}\big)[\mathbf{F}_{\bar{M}_{\mathrm{x}}}]_{1:M_{\mathrm{x}},1:\breve{M}_{\mathrm{x}}^{\mathrm{p}}}\mathrm{diag}\big(\breve{\boldsymbol{\omega}}_{\mathrm{x}}^{\mathrm{sat,p}}\big) \\ &\otimes \mathrm{diag}\big(\hat{\boldsymbol{\omega}}_{\mathrm{y}}^{\mathrm{sat,p}}\big)[\mathbf{F}_{\bar{M}_{\mathrm{y}}}]_{1:M_{\mathrm{y}},1:\breve{M}_{\mathrm{y}}^{\mathrm{p}}}\mathrm{diag}\big(\breve{\boldsymbol{\omega}}_{\mathrm{y}}^{\mathrm{sat,p}}\big) \\ &= \mathrm{diag}\big(\hat{\boldsymbol{\omega}}^{\mathrm{sat,p}}\big)[\mathbf{F}_{\bar{M}}]_{\mathcal{R}_{\mathrm{sat}},\mathcal{C}_{\mathrm{sat}}^{\mathrm{p}}}\mathrm{diag}\big(\breve{\boldsymbol{\omega}}^{\mathrm{sat,p}}\big), \end{aligned} \tag{16a}$$

$$\begin{aligned}\mathbf{U}^{\mathrm{p}} &= \mathbf{U}_{\mathrm{x}'}^{\mathrm{p}} \otimes \mathbf{U}_{\mathrm{y}'}^{\mathrm{p}} \\ &= \mathrm{diag}\big(\hat{\boldsymbol{\omega}}_{\mathrm{x}'}^{\mathrm{ut,p}}\big)[\mathbf{F}_{\bar{N}_{\mathrm{x}'}}]_{1:N_{\mathrm{x}'},1:\breve{N}_{\mathrm{x}'}^{\mathrm{p}}}\mathrm{diag}\big(\breve{\boldsymbol{\omega}}_{\mathrm{x}'}^{\mathrm{ut,p}}\big) \\ &\otimes \mathrm{diag}\big(\hat{\boldsymbol{\omega}}_{\mathrm{y}'}^{\mathrm{ut,p}}\big)[\mathbf{F}_{\bar{N}_{\mathrm{y}'}}]_{1:N_{\mathrm{y}'},1:\breve{N}_{\mathrm{y}'}^{\mathrm{p}}}\mathrm{diag}\big(\breve{\boldsymbol{\omega}}_{\mathrm{y}'}^{\mathrm{ut,p}}\big) \\ &= \mathrm{diag}\big(\hat{\boldsymbol{\omega}}^{\mathrm{ut,p}}\big)[\mathbf{F}_{\bar{N}}]_{\mathcal{R}_{\mathrm{ut}},\mathcal{C}_{\mathrm{ut}}^{\mathrm{p}}}\mathrm{diag}\big(\breve{\boldsymbol{\omega}}^{\mathrm{ut,p}}\big), \end{aligned} \tag{16b}$$

where $\mathbf{V}_{\mathrm{t}}^{\mathrm{p}} = \mathrm{diag}(\hat{\boldsymbol{\omega}}_{\mathrm{t}}^{\mathrm{sat,p}})[\mathbf{F}_{\bar{M}_{\mathrm{t}}}]_{1:M_{\mathrm{t}},1:\breve{M}_{\mathrm{t}}^{\mathrm{p}}}\mathrm{diag}(\breve{\boldsymbol{\omega}}_{\mathrm{t}}^{\mathrm{sat,p}})$, $\mathbf{U}_{\mathrm{u}}^{\mathrm{p}} = \mathrm{diag}(\hat{\boldsymbol{\omega}}_{\mathrm{u}}^{\mathrm{ut,p}})[\mathbf{F}_{\bar{N}_{\mathrm{u}}}]_{1:N_{\mathrm{u}},1:\breve{N}_{\mathrm{u}}^{\mathrm{p}}}\mathrm{diag}(\breve{\boldsymbol{\omega}}_{\mathrm{u}}^{\mathrm{ut,p}})$, and $\mathbf{F}_{\bar{A}_{\mathrm{v}}}$ is the $\bar{A}_{\mathrm{v}}$-point DFT matrix. Moreover, $[\mathbf{F}_{\bar{M}}]_{\mathcal{R}_{\mathrm{sat}},\mathcal{C}_{\mathrm{sat}}^{\mathrm{p}}} = [\mathbf{F}_{\bar{M}_{\mathrm{x}}}]_{1:M_{\mathrm{x}},1:\breve{M}_{\mathrm{x}}^{\mathrm{p}}} \otimes [\mathbf{F}_{\bar{M}_{\mathrm{y}}}]_{1:M_{\mathrm{y}},1:\breve{M}_{\mathrm{y}}^{\mathrm{p}}}$ and $[\mathbf{F}_{\bar{N}}]_{\mathcal{R}_{\mathrm{ut}},\mathcal{C}_{\mathrm{ut}}^{\mathrm{p}}} = [\mathbf{F}_{\bar{N}_{\mathrm{x}'}}]_{1:N_{\mathrm{x}'},1:\breve{N}_{\mathrm{x}'}^{\mathrm{p}}} \otimes [\mathbf{F}_{\bar{N}_{\mathrm{y}'}}]_{1:N_{\mathrm{y}'},1:\breve{N}_{\mathrm{y}'}^{\mathrm{p}}}$, with row and column sets $\mathcal{R}_{\mathrm{sat}}, \mathcal{C}_{\mathrm{sat}}^{\mathrm{p}}, \mathcal{R}_{\mathrm{ut}}$, and $\mathcal{C}_{\mathrm{ut}}^{\mathrm{p}}$ given by $\mathcal{R}_{\mathrm{sat}} = \{(i_{\mathrm{x}}-1)\bar{M}_{\mathrm{y}}+i_{\mathrm{y}} \big| 1 \leq i_{\mathrm{x}} \leq \bar{M}_{\mathrm{x}}, 1 \leq i_{\mathrm{y}} \leq \bar{M}_{\mathrm{y}}\}$, $\mathcal{C}_{\mathrm{sat}}^{\mathrm{p}} = \{(j_{\mathrm{x}}-1)\bar{M}_{\mathrm{y}}+j_{\mathrm{y}} \big| 1 \leq j_{\mathrm{x}} \leq \breve{M}_{\mathrm{x}}^{\mathrm{p}}, 1 \leq j_{\mathrm{y}} \leq \breve{M}_{\mathrm{y}}^{\mathrm{p}}\}$, $\mathcal{R}_{\mathrm{ut}} = \{(i_{\mathrm{x}'}-1)\bar{N}_{\mathrm{y}'}+i_{\mathrm{y}'} \big| 1 \leq i_{\mathrm{x}'} \leq N_{\mathrm{x}'}, 1 \leq i_{\mathrm{y}'} \leq N_{\mathrm{y}'}\}$, and $\mathcal{C}_{\mathrm{ut}}^{\mathrm{p}} = \{(j_{\mathrm{x}'}-1)\bar{N}_{\mathrm{y}'}+j_{\mathrm{y}'} \big| 1 \leq j_{\mathrm{x}'} \leq \breve{N}_{\mathrm{x}'}^{\mathrm{p}}, 1 \leq j_{\mathrm{y}'} \leq \breve{N}_{\mathrm{y}'}^{\mathrm{p}}\}$, respectively. Note that the beam matrices can be constructed with the phase-shifted partial DFT matrix, which will be utilized to reduce the complexity of the subsequently proposed DL BSP and UL BSR. Even though not all the columns of the DFT matrix are included, all practically relevant directional cosines within the satellite's angular visibility are covered.

Owing to the limited number of satellite antennas, the channel energy is spread across different beams at each satellite side. As shown in Fig. 2, when fine factors are set to $F_{\mathrm{x}}^{\mathrm{sat}} = F_{\mathrm{y}}^{\mathrm{sat}} = 1$, the energy disperses widely among various beams, leading to pronounced leakage into neighboring beams. When these fine factors are increased to 2, the energy distribution becomes more concentrated. When $F_{\mathrm{x}}^{\mathrm{sat}} = F_{\mathrm{y}}^{\mathrm{sat}} = 4$, the diffusion is markedly diminished, yielding a sharply defined and localized energy peak. This indicates that higher fine factors effectively limit energy leakage while confining most of the energy within the diamond-shaped region. Then, we define the following beam index set $\mathcal{B}_{k,s}^{\mathrm{dia,p}}$ for the chosen elements of $\tilde{\mathbf{g}}_{k,s}$ to capture this observed energy concentration.

$$\begin{aligned}\mathcal{B}_{k,s}^{\mathrm{dia,p}} = \big\{&(b_{\mathrm{x}}^{\mathrm{p}}-1)\breve{M}_{\mathrm{y}}^{\mathrm{p}} + b_{\mathrm{y}}^{\mathrm{p}} \big| b_{\mathrm{x}}^{\mathrm{p}} = ((\delta_{\mathrm{x}} - 1 + g_{\max}^{\mathrm{x,p}}))_{\breve{M}_{\mathrm{x}}^{\mathrm{p}}} + 1, \\ &b_{\mathrm{y}}^{\mathrm{p}} = ((\delta_{\mathrm{y}} - 1 + g_{\max}^{\mathrm{y,p}}))_{\breve{M}_{\mathrm{y}}^{\mathrm{p}}} + 1, \delta_{\mathrm{x}} \in \mathbb{Z}, \delta_{\mathrm{y}} \in \mathbb{Z}, \\ &|\delta_{\mathrm{x}}| \leq \gamma, |\delta_{\mathrm{y}}| \leq \gamma, |\delta_{\mathrm{x}}| + |\delta_{\mathrm{y}}| \leq \gamma, \\ &g_{\max}^{\mathrm{x,p}} = \arg\max_{g_{\mathrm{x}}^{\mathrm{p}}} [\tilde{\mathbf{g}}_{k,s}^{\mathrm{p,x}}]_{g_{\mathrm{x}}^{\mathrm{p}}}, g_{\max}^{\mathrm{y,p}} = \arg\max_{g_{\mathrm{y}}^{\mathrm{p}}} [\tilde{\mathbf{g}}_{k,s}^{\mathrm{p,y}}]_{g_{\mathrm{y}}^{\mathrm{p}}}\big\}, \end{aligned} \tag{17}$$

with $|\mathcal{B}_{k,s}^{\mathrm{dia,p}}| = B_{k,s}^{\mathrm{dia,p}}$ and $((\cdot))_M$ denoting the modulo operation with $M$. Note that $\mathcal{B}_{k,s}^{\mathrm{dia,p}}$ denotes a beam index set associated with a diamond-shaped beam arrangement, whose geometric center is positioned at a distance parameter $\gamma$ from each vertex. Then, the DL and UL BBCMs in (10) and (15) can be approximated as

$$\mathbf{H}_{k,s}^{\mathrm{dl}} \approx \mathbf{U}^{\mathrm{dl}}\check{\mathbf{H}}_{k,s}^{\mathrm{dl}}(\boldsymbol{\Omega}_{k,s}^{\mathrm{dl}})^H(\mathbf{V}^{\mathrm{dl}})^H, \tag{18}$$

and

$$\mathbf{H}_{k,s}^{\mathrm{ul}} \approx \mathbf{V}^{\mathrm{ul}}\boldsymbol{\Omega}_{k,s}^{\mathrm{ul}}\check{\mathbf{H}}_{k,s}^{\mathrm{ul}}(\mathbf{U}^{\mathrm{ul}})^H, \tag{19}$$

respectively, where $\check{\mathbf{H}}_{k,s}^{\mathrm{dl}} = \tilde{\mathbf{d}}_{k,s}^{\mathrm{dl}}(\check{\mathbf{g}}_{k,s}^{\mathrm{dl}})^H$, $\check{\mathbf{H}}_{k,s}^{\mathrm{ul}} = \check{\mathbf{g}}_{k,s}^{\mathrm{ul}}(\tilde{\mathbf{d}}_{k,s}^{\mathrm{ul}})^H$, $\check{\mathbf{g}}_{k,s}^{\mathrm{p}} = [\tilde{\mathbf{g}}_{k,s}^{\mathrm{p}}]_{\mathcal{B}_{k,s}^{\mathrm{dia,p}}}$, and $\boldsymbol{\Omega}_{k,s}^{\mathrm{p}} \in \mathbb{R}^{\breve{M}^{\mathrm{p}} \times B_{k,s}^{\mathrm{dia,p}}}$ denotes the beam selection matrix, given by

$$[\boldsymbol{\Omega}_{k,s}^{\mathrm{p}}]_{m,n} = \begin{cases} 1, & m = \big[\mathcal{B}_{k,s}^{\mathrm{dia,p}}\big]_n, \\ 0, & \text{otherwise.} \end{cases} \tag{20}$$

### C. Statistical Properties of the Satellite Channel

As discussed in Subsection II-B, the steering vector $\mathbf{g}_{k,s}^{\mathrm{p}}$ only depends on the relative locations of satellite $s$ and UT $k$ and changes relatively slowly, then we can reasonably assume that $\{\mathbf{g}_{k,s}^{\mathrm{p}}\}_{\forall k}$ can be known at each satellite $s$ [20], [21], [31], [32]. For $\mathrm{p} \in \{\mathrm{dl}, \mathrm{ul}\}$, we model $\mathbf{d}_{k,s}^{\mathrm{p}}$ by using the Rician distribution, which is expressed as

$$\mathbf{d}_{k,s}^{\mathrm{p}} = \sqrt{\frac{\kappa_{k,s}\beta_{k,s}^{\mathrm{p}}}{\kappa_{k,s}+1}}\mathbf{d}_{k,s,0}^{\mathrm{p}} + \sqrt{\frac{\beta_{k,s}^{\mathrm{p}}}{\kappa_{k,s}+1}}\breve{\mathbf{d}}_{k,s}^{\mathrm{p}}, \tag{21}$$

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where $\kappa_{k,s}$ is the Rician factor, $\beta^{\mathrm{p}}_{k,s} = \mathbb{E}\{||\mathbf{d}^{\mathrm{p}}_{k,s}||^2\}$ is the channel power, $\mathbf{d}^{\mathrm{p}}_{k,s,0}$ is the steering vector of the line-of-sight (LoS) path at UT side, and the vector $\breve{\mathbf{d}}^{\mathrm{p}}_{k,s}$ refers to the random non-line-of-sight (NLoS) scattering part at UT side. The channel correlation matrices at satellites' and UTs' sides are given by

$$\begin{aligned}\mathbf{R}^{\mathrm{sat,dl}}_{k,s} &= \mathbb{E}\{(\mathbf{H}^{\mathrm{dl}}_{k,s})^H\mathbf{H}^{\mathrm{dl}}_{k,s}\} = \beta^{\mathrm{dl}}_{k,s}\mathbf{g}^{\mathrm{dl}}_{k,s}(\mathbf{g}^{\mathrm{dl}}_{k,s})^H \\ &= \mathbf{V}^{\mathrm{dl}}\widetilde{\mathbf{R}}^{\mathrm{sat,dl}}_{k,s}(\mathbf{V}^{\mathrm{dl}})^H, \quad (22a)\\ \mathbf{R}^{\mathrm{sat,ul}}_{k,s} &= \mathbb{E}\{\mathbf{H}^{\mathrm{ul}}_{k,s}(\mathbf{H}^{\mathrm{ul}}_{k,s})^H\} = \beta^{\mathrm{ul}}_{k,s}\mathbf{g}^{\mathrm{ul}}_{k,s}(\mathbf{g}^{\mathrm{ul}}_{k,s})^H \\ &= \mathbf{V}^{\mathrm{ul}}\widetilde{\mathbf{R}}^{\mathrm{sat,ul}}_{k,s}(\mathbf{V}^{\mathrm{ul}})^H, \quad (22b)\\ \mathbf{R}^{\mathrm{ut,dl}}_{k,s} &= \mathbb{E}\{\mathbf{H}^{\mathrm{dl}}_{k,s}(\mathbf{H}^{\mathrm{dl}}_{k,s})^H\} = \mathbb{E}\{\mathbf{d}^{\mathrm{dl}}_{k,s}(\mathbf{d}^{\mathrm{dl}}_{k,s})^H\} \\ &= \mathbf{U}^{\mathrm{dl}}\widetilde{\mathbf{R}}^{\mathrm{ut,dl}}_{k,s}(\mathbf{U}^{\mathrm{dl}})^H, \quad (22c)\\ \mathbf{R}^{\mathrm{ut,ul}}_{k,s} &= \mathbb{E}\{(\mathbf{H}^{\mathrm{ul}}_{k,s})^H\mathbf{H}^{\mathrm{ul}}_{k,s}\} = \mathbb{E}\{\mathbf{d}^{\mathrm{ul}}_{k,s}(\mathbf{d}^{\mathrm{ul}}_{k,s})^H\} \\ &= \mathbf{U}^{\mathrm{ul}}\widetilde{\mathbf{R}}^{\mathrm{ut,ul}}_{k,s}(\mathbf{U}^{\mathrm{ul}})^H, \quad (22d)\end{aligned}$$

respectively, where $\widetilde{\mathbf{R}}^{\mathrm{sat,p}}_{k,s} = \tilde{\mathbf{g}}^{\mathrm{p}}_{k,s}(\tilde{\mathbf{g}}^{\mathrm{p}}_{k,s})^H$ and $\widetilde{\mathbf{R}}^{\mathrm{ut,p}}_{k,s} = \mathbb{E}\{\tilde{\mathbf{d}}^{\mathrm{p}}_{k,s}(\tilde{\mathbf{d}}^{\mathrm{p}}_{k,s})^H\}$ refer to the beam domain channel correlation matrices at satellites' and UTs' sides, respectively. Note that $\mathrm{rank}(\mathbf{R}^{\mathrm{sat,p}}_{k,s}) = 1$, which indicates that the satellite channels are highly correlated in the space domain at the satellite side. On the other hand, the rank of $\mathbf{R}^{\mathrm{ut,p}}_{k,s}$ depends on the propagation environment around UT $k$, which can be acquired at each UT $k$ through DL channel estimation. We define the set $\mathcal{H}_{k,s}$ that involves the long-term sCSI between satellite $s$ and UT $k$ as $\mathcal{H}_{k,s} = \{g^{\mathrm{t}}_{k,s}, d^{\mathrm{u}}_{k,s,0}, \kappa_{k,s}, \beta^{\mathrm{p}}_{k,s}, \mathbf{R}^{\mathrm{ut,p}}_{k,s}\}$, where $\mathrm{t} \in \{\mathrm{x}, \mathrm{y}\}$ and $\mathrm{u} \in \{\mathrm{x}', \mathrm{y}'\}$. We denote $\mathcal{H}^{\mathrm{sat}}_s$ and $\mathcal{H}^{\mathrm{ut}}_k$ as the local sCSI at satellite $s$ and UT $k$, respectively, where $\mathcal{H}^{\mathrm{sat}}_s = \bigcup_{k\in\mathcal{K}}\mathcal{H}_{k,s}$ and $\mathcal{H}^{\mathrm{ut}}_k = \bigcup_{s\in\mathcal{S}}\mathcal{H}_{k,s}$.

The calculation of directional cosines relies on the location information of each UT and each satellite, which can be acquired from the ephemeris and global navigation satellite system (GNSS) positioning [20], [21]. With modern precise orbit determination (POD) techniques providing centimeter-level satellite-position errors [34] and GNSS achieving meter-level horizontal accuracy for the UT [35], the resulting angular uncertainty is below the milliradian level when the satellite altitude is 600 km. Each satellite $s$ can acquire its local sCSI in $\mathcal{H}^{\mathrm{sat}}_s$ through UT feedback or UL channel estimation. Thanks to the statistical reciprocity between the UL and DL channels [20], each satellite can first estimate its sCSI from the UL pilot signals, and then use the estimated sCSI for the subsequent DL transmission. The sCSI set $\{\mathcal{H}_{k,s}\}_{\forall k,s}$ is assumed quasi-static while satellites and UTs remain within a certain range. When any satellite or UT moves beyond this range, the corresponding $\mathcal{H}_{k,s}$ should be updated accordingly.

### D. DL/UL Signal Model

For the DL, each satellite adopts linear precoders for generating transmit signals. To mitigate the effect of time and frequency asynchronizations, a linear receive processing is adopted at each UT for each satellite to extract the desired data stream, and time and frequency compensations are performed for each stream individually. Specifically, the received signal at UT $k$ of satellite $s$ over a specific subcarrier is given by [31]

$$\begin{aligned}y^{\mathrm{dl}}_{k,s} &= (\mathbf{c}^{\mathrm{dl}}_{k,s})^H\mathbf{H}^{\mathrm{dl}}_{k,s}\sqrt{p^{\mathrm{dl}}_{k,s}}\mathbf{w}^{\mathrm{dl}}_{k,s}s^{\mathrm{dl}}_{k,s} \\ &\quad + (\mathbf{c}^{\mathrm{dl}}_{k,s})^H\mathbf{H}^{\mathrm{dl}}_{k,s}\sum_{i\in\mathcal{K}_s\setminus\{k\}}\sqrt{p^{\mathrm{dl}}_{i,s}}\mathbf{w}^{\mathrm{dl}}_{i,s}s^{\mathrm{dl}}_{i,s} + z^{\mathrm{dl}}_{k,s}, \quad (23)\end{aligned}$$

where $\mathbf{c}^{\mathrm{dl}}_{k,s}$ is the DL linear receiver at UT $k$ for satellite $s$, $p^{\mathrm{dl}}_{k,s}$ is the transmit power of satellite $s$ for UT $k$, $\mathbf{w}^{\mathrm{dl}}_{k,s}$ is the DL normalized precoder at satellite $s$ for UT $k$ with $||\mathbf{w}^{\mathrm{dl}}_{k,s}|| = 1$, $s^{\mathrm{dl}}_{k,s}$ is the transmit data stream of satellite $s$ for UT $k$ with 0 mean and unit variance, and the total transmit power of satellite $s$ is $P^{\mathrm{dl}}_s = \sum_{k\in\mathcal{K}_s}p^{\mathrm{dl}}_{k,s}$. $z^{\mathrm{dl}}_{k,s}$ is the noise signal combined with asynchronous interference signal with variance $\mathbb{E}\{z^{\mathrm{dl}}_{k,s}(z^{\mathrm{dl}}_{k,s})^*\} = (\mathbf{c}^{\mathrm{dl}}_{k,s})^H\mathbf{R}^{\mathrm{dl}}_{k,s}\mathbf{c}^{\mathrm{dl}}_{k,s}$. Here, $\mathbf{R}^{\mathrm{dl}}_{k,s}$ is given by

$$\mathbf{R}^{\mathrm{dl}}_{k,s} = \sum_{j\in\mathcal{S}\setminus\{s\}}\sum_{i\in\mathcal{K}_j}p^{\mathrm{dl}}_{i,j}\left|(\mathbf{g}^{\mathrm{dl}}_{k,j})^H\mathbf{w}^{\mathrm{dl}}_{i,j}\right|^2\mathbf{R}^{\mathrm{ut,dl}}_{k,j} + (\sigma^{\mathrm{dl}}_k)^2\mathbf{I}_N, \quad (24)$$

where $\sigma^{\mathrm{dl}}_k$ is the variance of the noise signal at UT $k$.

For the UL, each UT adopts a linear precoder with time and frequency pre-compensations for each satellite to synchronize the signals intended for a specific satellite. Each satellite adopts a linear receiver for receiving the signals. Specifically, the received signal of satellite $s$ from UT $k$ over a specific subcarrier is given by [32]

$$\begin{aligned}y^{\mathrm{ul}}_{k,s} &= (\mathbf{c}^{\mathrm{ul}}_{k,s})^H\mathbf{H}^{\mathrm{ul}}_{k,s}\sqrt{p^{\mathrm{ul}}_{k,s}}\mathbf{w}^{\mathrm{ul}}_{k,s}s^{\mathrm{ul}}_{k,s} \\ &\quad + (\mathbf{c}^{\mathrm{ul}}_{k,s})^H\sum_{i\in\mathcal{K}_s\setminus\{k\}}\mathbf{H}^{\mathrm{ul}}_{i,s}\sqrt{p^{\mathrm{ul}}_{i,s}}\mathbf{w}^{\mathrm{ul}}_{i,s}s^{\mathrm{ul}}_{i,s} + (\mathbf{c}^{\mathrm{ul}}_{k,s})^H\mathbf{z}^{\mathrm{ul}}_s, \quad (25)\end{aligned}$$

where $\mathbf{c}^{\mathrm{ul}}_{k,s}$ is the UL linear receiver at satellite $s$ for UT $k$, $p^{\mathrm{ul}}_{k,s}$ is the transmit power of UT $k$ for satellite $s$, $\mathbf{w}^{\mathrm{ul}}_{k,s}$ is the UL normalized precoder at UT $k$ for satellite $s$ with $||\mathbf{w}^{\mathrm{ul}}_{k,s}|| = 1$, $s^{\mathrm{ul}}_{k,s}$ is the transmit data stream of UT $k$ for satellite $s$ with 0 mean and unit variance, and the total transmit power of UT $k$ is $P^{\mathrm{ul}}_k = \sum_{s\in\mathcal{S}_k}p^{\mathrm{ul}}_{k,s}$. $\mathbf{z}^{\mathrm{ul}}_s$ represents the noise signal combined with asynchronous interference signals with variance $\mathbb{E}\{\mathbf{z}^{\mathrm{ul}}_s(\mathbf{z}^{\mathrm{ul}}_s)^H\} = \mathbf{R}^{\mathrm{ul}}_s$. Here, $\mathbf{R}^{\mathrm{ul}}_s$ is given by

$$\mathbf{R}^{\mathrm{ul}}_s = \sum_{i\in\mathcal{K}}\sum_{j\in\mathcal{S}_i\setminus\{s\}}p^{\mathrm{ul}}_{i,j}(\mathbf{w}^{\mathrm{ul}}_{i,j})^H\mathbf{R}^{\mathrm{ut,ul}}_{i,s}\mathbf{w}^{\mathrm{ul}}_{i,j}\mathbf{g}^{\mathrm{ul}}_{i,s}(\mathbf{g}^{\mathrm{ul}}_{i,s})^H + (\sigma^{\mathrm{ul}}_s)^2\mathbf{I}_M. \quad (26)$$

where $\sigma^{\mathrm{ul}}_s$ is the variance of the noise signal at satellite $s$.

## III. DL/UL DPR Design

In this section, we investigate the decoupled designs of the DL/UL precoders and receivers with only slow-varying sCSI for massive MIMO multi-satellite systems. The proposed designs at each satellite and each UT only require their local sCSI knowledge.

### A. DL DPR Design

Consider the efficient average signal-to-leakage-plus-noise ratio (ASLNR) design metric [19] for the DL precoder design with given DL receiver for massive MIMO multi-satellite

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communications. Specifically, for the DL transmission, the ASLNR of satellite $s$ for UT $k$ is calculated as

$$\mathrm{ASLNR}_{k,s}^{\mathrm{dl}}(\mathbf{w}_{k,s}^{\mathrm{dl}})=\frac{(\mathbf{c}_{k,s}^{\mathrm{dl}})^H\mathbf{R}_{k,s}^{\mathrm{ut,dl}}\mathbf{c}_{k,s}^{\mathrm{dl}}\left|(\mathbf{g}_{k,s}^{\mathrm{dl}})^H\mathbf{w}_{k,s}^{\mathrm{dl}}\right|^2}{(\mathbf{w}_{k,s}^{\mathrm{dl}})^H\mathbf{D}_{k,s}^{\mathrm{dl}}\mathbf{w}_{k,s}^{\mathrm{dl}}}, \tag{27}$$

where $\mathbf{D}_{k,s}^{\mathrm{dl}}=\frac{1}{\rho_{k,s}^{\mathrm{dl}}}\mathbf{I}+\sum_{j\neq s}^{S}\sum_{i\in\mathcal{K}_j}(\mathbf{c}_{i,j}^{\mathrm{dl}})^H\mathbf{R}_{i,s}^{\mathrm{ut,dl}}\mathbf{c}_{i,j}^{\mathrm{dl}}\mathbf{g}_{i,s}^{\mathrm{dl}}(\mathbf{g}_{i,s}^{\mathrm{dl}})^H+\sum_{i\in\mathcal{K}_s\setminus\{k\}}(\mathbf{c}_{i,s}^{\mathrm{dl}})^H\mathbf{R}_{i,s}^{\mathrm{ut,dl}}\mathbf{c}_{i,s}^{\mathrm{dl}}\mathbf{g}_{i,s}^{\mathrm{dl}}(\mathbf{g}_{i,s}^{\mathrm{dl}})^H$, and $\rho_{k,s}^{\mathrm{dl}}=\frac{p_{k,s}^{\mathrm{dl}}}{(\sigma_k^{\mathrm{dl}})^2}$ denotes the signal-to-noise ratio (SNR) in the DL. The DL precoder that maximizes the ASLNR in (27) is given by

$$\mathbf{w}_{k,s}^{\mathrm{aslnr,dl}}=\frac{1}{\eta_{k,s}^{\mathrm{aslnr,dl}}}\left(\bar{\mathbf{D}}_{k,s}^{\mathrm{dl}}\right)^{-1}\mathbf{g}_{k,s}^{\mathrm{dl}}, \tag{28}$$

where $\eta_{k,s}^{\mathrm{aslnr,dl}}$ denotes the power normalization factor that makes $\|\mathbf{w}_{k,s}^{\mathrm{aslnr,dl}}\|=1$, and $\bar{\mathbf{D}}_{k,s}^{\mathrm{dl}}=\frac{1}{\rho_{k,s}^{\mathrm{dl}}}\mathbf{I}+\sum_{i\in\mathcal{K}}\sum_{j\in\mathcal{S}_i}(\mathbf{c}_{i,j}^{\mathrm{dl}})^H\mathbf{R}_{i,s}^{\mathrm{ut,dl}}\mathbf{c}_{i,j}^{\mathrm{dl}}\mathbf{g}_{i,s}^{\mathrm{dl}}(\mathbf{g}_{i,s}^{\mathrm{dl}})^H$. Next, we consider the average signal-to-interference-plus-noise ratio (ASINR) design metric for the DL receiver design with given DL precoder. Specifically, the ASINR of UT $k$ for satellite $s$ is given by

$$\mathrm{ASINR}_{k,s}^{\mathrm{dl}}(\mathbf{c}_{k,s}^{\mathrm{dl}})=\frac{(\mathbf{c}_{k,s}^{\mathrm{dl}})^H\mathbf{R}_{k,s}^{\mathrm{ut,dl}}\mathbf{c}_{k,s}^{\mathrm{dl}}\left|(\mathbf{g}_{k,s}^{\mathrm{dl}})^H\mathbf{w}_{k,s}^{\mathrm{dl}}\right|^2p_{k,s}^{\mathrm{dl}}}{(\mathbf{c}_{k,s}^{\mathrm{dl}})^H\mathbf{E}_{k,s}^{\mathrm{dl}}\mathbf{c}_{k,s}^{\mathrm{dl}}}, \tag{29}$$

where $\mathbf{E}_{k,s}^{\mathrm{dl}}=\mathbf{R}_{k,s}^{\mathrm{dl}}+\sum_{i\in\mathcal{K}_s\setminus\{k\}}p_{i,s}^{\mathrm{dl}}|(\mathbf{g}_{k,s}^{\mathrm{dl}})^H\mathbf{w}_{i,s}^{\mathrm{dl}}|^2\mathbf{R}_{k,s}^{\mathrm{ut,dl}}$. The DL receiver that maximizes the ASINR in (29) is given by

$$\mathbf{c}_{k,s}^{\mathrm{asinr,dl}}=\boldsymbol{\xi}_{\max}\left(\left(\bar{\mathbf{E}}_k^{\mathrm{dl}}\right)^{-1}\mathbf{R}_{k,s}^{\mathrm{ut,dl}}\right), \tag{30}$$

where $\boldsymbol{\xi}_{\max}(\mathbf{A})$ denotes the normalized eigenvector related to the maximum eigenvalue of matrix $\mathbf{A}$, and $\bar{\mathbf{E}}_k^{\mathrm{dl}}=\sum_{j=1}^{S}\sum_{i\in\mathcal{K}_j}p_{i,j}^{\mathrm{dl}}|(\mathbf{g}_{k,j}^{\mathrm{dl}})^H\mathbf{w}_{i,j}^{\mathrm{dl}}|^2\mathbf{R}_{k,j}^{\mathrm{ut,dl}}+(\sigma_k^{\mathrm{dl}})^2\mathbf{I}$.

From (28) and (30), we can observe that the optimal DL precoders and receivers are interdependent. Specifically, the design of DL precoders at each satellite requires the knowledge of all receivers at the UTs, while the design of DL receivers at each UT depends on all precoders at the satellites. Such dependency is generally infeasible in practice, and the convergence of the alternating optimization between the DL precoders and receivers cannot be guaranteed. To obtain the decoupled designs of the DL precoders and receivers, we first derive the following theorem.

*Theorem 1:* The designs of DL precoders and receivers can be decoupled under the following three cases: i) $\rho_{k,s}^{\mathrm{dl}}\to\infty,\mathbf{R}_{k,s}^{\mathrm{ut,dl}}\mathbf{R}_{k,j}^{\mathrm{ut,dl}}=\mathbf{0},\forall k,s,j,s\neq j$, ii) $\rho_{k,s}^{\mathrm{dl}}\to 0,\forall k,s$, and iii) $M\to\infty$. In these cases, the decoupled solutions of DL precoders $\{\mathbf{w}_{k,s}^{\mathrm{dl}}\}_{\forall k\in\mathcal{K}_s,s\in\mathcal{S}}$, and receivers $\{\mathbf{c}_{k,s}^{\mathrm{dl}}\}_{\forall k\in\mathcal{K}_s,s\in\mathcal{S}}$, are given by

$$\mathbf{w}_{k,s}^{\mathrm{aslnr,dl,de}}=\frac{1}{\eta_{k,s}^{\mathrm{aslnr,dl,de}}}\left(\frac{1}{\rho_{k,s}^{\mathrm{dl}}}\mathbf{I}+\sum_{i\in\mathcal{K}}\xi_{i,s}^{\mathrm{dl}}\mathbf{g}_{i,s}^{\mathrm{dl}}(\mathbf{g}_{i,s}^{\mathrm{dl}})^H\right)^{-1}\mathbf{g}_{k,s}, \tag{31a}$$

$$\mathbf{c}_{k,s}^{\mathrm{asinr,dl,de}}=\boldsymbol{\xi}_{\max}\left(\left(\sum_{j\in\mathcal{S}_k}\rho_{k,j}^{\mathrm{dl}}\mathbf{R}_{k,j}^{\mathrm{ut,dl}}+\mathbf{I}\right)^{-1}\mathbf{R}_{k,s}^{\mathrm{ut,dl}}\right), \tag{31b}$$

where $\eta_{k,s}^{\mathrm{aslnr,dl,de}}$ is the power normalization factor of $\mathbf{w}_{k,s}^{\mathrm{aslnr,dl,de}}$, $\xi_{i,s}^{\mathrm{dl}}=\zeta_{\max}(\mathbf{R}_{i,s}^{\mathrm{ut,dl}})$, and $\zeta_{\max}(\mathbf{A})$ denotes the maximum eigenvalue of matrix $\mathbf{A}$.

*Proof:* See Appendix A. ■

Theorem 1 reveals that the designs of the DL precoders and receivers can be independent and only depend on the local sCSI. Specifically, the decoupled DL precoder at satellite $s$ for UT $k$ requires local sCSI $\{\mathbf{g}_{i,s}^{\mathrm{dl}},\xi_{i,s}^{\mathrm{dl}}\}_{\forall i\in\mathcal{K}}$, and the decoupled DL receiver at UT $k$ for satellite $s$ requires $\{\mathbf{R}_{k,s}^{\mathrm{ut,dl}}\}_{\forall s\in\mathcal{S}_k}$, which represent partial information of $\mathcal{H}_s^{\mathrm{sat}}$ and $\mathcal{H}_k^{\mathrm{ut}}$, respectively. Notice that since the satellite channels are typically LoS dominated, which corresponds to a relatively larger Rician factor, the maximum eigenvalue of the channel correlation matrix $\mathbf{R}_{i,s}^{\mathrm{ut,dl}}$, i.e., $\xi_{i,s}^{\mathrm{dl}}$, approaches the link gain of the LoS component of the channel between satellite $s$ and UT $i$.

Next, we discuss the feasibility of the orthogonality condition in the high SNR case. First, the channel correlation matrix at UTs' sides in (22c) can be rewritten as

$$\mathbf{R}_{k,s}^{\mathrm{ut,dl}}=\mathbb{E}\{\mathbf{d}_{k,s}^{\mathrm{dl}}(\mathbf{d}_{k,s}^{\mathrm{dl}})^H\}=\mathbf{U}^{\mathrm{dl}}\boldsymbol{\Gamma}_{k,s}^{\mathrm{dl}}\widetilde{\mathbf{R}}_{k,s}^{\mathrm{ut,dl,nz}}(\boldsymbol{\Gamma}_{k,s}^{\mathrm{dl}})^H(\mathbf{U}^{\mathrm{dl}})^H, \tag{32}$$

where $\widetilde{\mathbf{R}}_{k,s}^{\mathrm{ut,dl,nz}}$ denotes the beam domain channel correlation matrix at UTs' sides, and $\boldsymbol{\Gamma}_{k,s}^{\mathrm{dl}}$ is the beam selection matrix of UT $k$ for satellite $s$, defined as

$$[\boldsymbol{\Gamma}_{k,s}^{\mathrm{dl}}]_{m,n}\begin{cases}1, & m=\left[\mathcal{B}_{k,s}^{\mathrm{dl,nz}}\right]_n,\\ 0, & \text{otherwise}\end{cases} \tag{33}$$

with $\mathcal{B}_{k,s}^{\mathrm{dl,nz}}$ being the non-zero element index set of $\tilde{\mathbf{d}}_{k,s}^{\mathrm{dl}}$. Then, $\mathbf{R}_{k,s}^{\mathrm{ut,dl}}\mathbf{R}_{k,j}^{\mathrm{ut,dl}}$ can be expressed as

$$\mathbf{R}_{k,s}^{\mathrm{ut,dl}}\mathbf{R}_{k,j}^{\mathrm{ut,dl}}=\mathbf{U}^{\mathrm{dl}}\boldsymbol{\Gamma}_{k,s}^{\mathrm{dl}}\widetilde{\mathbf{R}}_{k,s}^{\mathrm{ut,dl,nz}}(\boldsymbol{\Gamma}_{k,s}^{\mathrm{dl}})^H(\mathbf{U}^{\mathrm{dl}})^H\mathbf{U}^{\mathrm{dl}}\boldsymbol{\Gamma}_{k,j}^{\mathrm{dl}}\widetilde{\mathbf{R}}_{k,j}^{\mathrm{ut,dl,nz}}(\boldsymbol{\Gamma}_{k,j}^{\mathrm{dl}})^H(\mathbf{U}^{\mathrm{dl}})^H. \tag{34}$$

Notice that $(\mathbf{U}_{\mathrm{u}}^{\mathrm{p}})^H\mathbf{U}_{\mathrm{u}}^{\mathrm{p}}$ can be expressed as

$$\left[(\mathbf{U}_{\mathrm{u}}^{\mathrm{p}})^H(\mathbf{U}_{\mathrm{u}}^{\mathrm{p}})\right]_{i_{\mathrm{u}},j_{\mathrm{u}}}=\Upsilon_{\mathrm{u}}\left(\frac{i_{\mathrm{u}}-j_{\mathrm{u}}}{N_{\mathrm{u}}}\right), \tag{35}$$

where $\Upsilon_{\mathrm{u}}(x)=\frac{\sin(\pi N_{\mathrm{u}}x)}{N_{\mathrm{u}}\sin(\pi x)}$. As $|i_{\mathrm{u}}-j_{\mathrm{u}}|$ increases, $|[(\mathbf{U}_{\mathrm{u}}^{\mathrm{p}})^H\mathbf{U}_{\mathrm{u}}^{\mathrm{p}}]_{i_{\mathrm{u}},j_{\mathrm{u}}}|$ decreases. Then, we can obtain that $(\boldsymbol{\Gamma}_{k,s}^{\mathrm{p}})^H(\mathbf{U}^{\mathrm{p}})^H\mathbf{U}^{\mathrm{p}}\boldsymbol{\Gamma}_{k,j}^{\mathrm{p}}\to\mathbf{0}$ and $\mathbf{R}_{k,s}^{\mathrm{ut,p}}\mathbf{R}_{k,j}^{\mathrm{ut,p}}\to\mathbf{0}$ when the number of antennas, $N$, is large enough, and $\mathcal{B}_{k,s}^{\mathrm{p,nz}}\cap\mathcal{B}_{k,j}^{\mathrm{p,nz}}=\emptyset,\forall s,j,s\neq j$, which implies the asymptotic orthogonality among the beams. Note that even though the number of antennas is finite in practice, each element in $(\boldsymbol{\Gamma}_{k,s}^{\mathrm{p}})^H(\mathbf{U}^{\mathrm{p}})^H\mathbf{U}^{\mathrm{p}}\boldsymbol{\Gamma}_{k,j}^{\mathrm{p}}$ can be relatively small, due to the large space angle difference between any two satellites and a UT. Therefore, the orthogonality condition in the high SNR case can be satisfied exactly in the asymptotic regime and approximately for finite arrays.

### B. UL DPR Design

In this subsection, we investigate the decoupled designs of UL receivers and precoders by maximizing the ASINR and ASLNR, respectively. For given UL precoders, the ASINR of

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satellite $s$ for UT $k$ is a function of $\mathbf{c}_{k,s}^{\mathrm{ul}}$, which is expressed as

$$\mathrm{ASINR}_{k,s}^{\mathrm{ul}}(\mathbf{c}_{k,s}^{\mathrm{ul}}) = \frac{p_{k,s}^{\mathrm{ul}}(\mathbf{w}_{k,s}^{\mathrm{ul}})^H \mathbf{R}_{k,s}^{\mathrm{ut,ul}} \mathbf{w}_{k,s}^{\mathrm{ul}} \left|(\mathbf{g}_{k,s}^{\mathrm{ul}})^H \mathbf{c}_{k,s}^{\mathrm{ul}}\right|^2}{(\mathbf{c}_{k,s}^{\mathrm{ul}})^H \mathbf{E}_{k,s}^{\mathrm{ul}} \mathbf{c}_{k,s}^{\mathrm{ul}}}, \tag{36}$$

where $\mathbf{E}_{k,s}^{\mathrm{ul}} = \sum_{i\in\mathcal{K}_s\setminus\{k\}} p_{i,s}^{\mathrm{ul}}(\mathbf{w}_{i,s}^{\mathrm{ul}})^H \mathbf{R}_{i,s}^{\mathrm{ut,ul}} \mathbf{w}_{i,s}^{\mathrm{ul}} \mathbf{g}_{i,s}^{\mathrm{ul}} (\mathbf{g}_{i,s}^{\mathrm{ul}})^H + \mathbf{R}_s^{\mathrm{ul}}$.The UL receiver maximizing the ASINR in (36) is given by

$$\mathbf{c}_{k,s}^{\mathrm{asinr,ul}} = \left(\bar{\mathbf{E}}_{k,s}^{\mathrm{ul}}\right)^{-1} \mathbf{g}_{k,s}^{\mathrm{ul}}, \tag{37}$$

where $\bar{\mathbf{E}}_{k,s}^{\mathrm{ul}} = \sum_{j=1}^{S} \sum_{i\in\mathcal{K}_j} p_{i,j}^{\mathrm{ul}}(\mathbf{w}_{i,j}^{\mathrm{ul}})^H \mathbf{R}_{i,s}^{\mathrm{ut,ul}} \mathbf{w}_{i,j}^{\mathrm{ul}} \mathbf{g}_{i,s}^{\mathrm{ul}} (\mathbf{g}_{i,s}^{\mathrm{ul}})^H + (\sigma_s^{\mathrm{ul}})^2 \mathbf{I}$. With given UL receivers, the ASLNR of UT $k$ for satellite $s$ is a function of $\mathbf{w}_{k,s}^{\mathrm{ul}}$, calculated as

$$\mathrm{ASLNR}_{k,s}^{\mathrm{ul}}(\mathbf{w}_{k,s}^{\mathrm{ul}}) = \frac{(\mathbf{w}_{k,s}^{\mathrm{ul}})^H \mathbf{R}_{k,s}^{\mathrm{ut,ul}} \mathbf{w}_{k,s}^{\mathrm{ul}} \left|(\mathbf{g}_{k,s}^{\mathrm{ul}})^H \mathbf{c}_{k,s}^{\mathrm{ul}}\right|^2}{(\mathbf{w}_{k,s}^{\mathrm{ul}})^H \mathbf{D}_{k,s}^{\mathrm{ul}} \mathbf{w}_{k,s}^{\mathrm{ul}}}, \tag{38}$$

where $\mathbf{D}_{k,s}^{\mathrm{ul}} = \frac{1}{\rho_{k,s}^{\mathrm{ul}}}\mathbf{I} + \sum_{j\neq s}^{S} \sum_{i\in\mathcal{K}_s\setminus\{k\}} |(\mathbf{g}_{k,j}^{\mathrm{ul}})^H \mathbf{c}_{i,j}^{\mathrm{ul}}|^2 \mathbf{R}_{k,j}^{\mathrm{ut,ul}} + \sum_{i\in\mathcal{K}_s\setminus\{k\}} |(\mathbf{g}_{k,s}^{\mathrm{ul}})^H \mathbf{c}_{i,s}^{\mathrm{ul}}|^2 \mathbf{R}_{k,s}^{\mathrm{ut,ul}}$, and $\rho_{k,s}^{\mathrm{ul}} = \frac{p_{k,s}^{\mathrm{ul}}}{(\sigma_s^{\mathrm{ul}})^2}$ denotes the UL SNR. The UL precoder maximizing the ASLNR in (38) is given by

$$\mathbf{w}_{k,s}^{\mathrm{aslnr,ul}} = \boldsymbol{\xi}_{\max}\left(\left(\bar{\mathbf{D}}_{k,s}^{\mathrm{ul}}\right)^{-1} \mathbf{R}_{k,s}^{\mathrm{ut,ul}}\right), \tag{39}$$

where $\bar{\mathbf{D}}_{k,s}^{\mathrm{ul}} = \sum_{j=1}^{S} \sum_{i\in\mathcal{K}_j} |(\mathbf{g}_{k,j}^{\mathrm{ul}})^H \mathbf{c}_{i,j}^{\mathrm{ul}}|^2 \mathbf{R}_{k,j}^{\mathrm{ut,ul}} + \frac{1}{\rho_{k,s}^{\mathrm{ul}}}\mathbf{I}$. From (37) and (39), it can be observed that the designs of UL receivers and precoders are coupled. To obtain the decoupled designs of the UL precoders and receivers, we derive the following theorem.

*Theorem 2:* The designs of UL precoders and receivers can be decoupled under the following three cases: i) $\rho_{k,s}^{\mathrm{ul}} \to \infty, \mathbf{R}_{k,s}^{\mathrm{ut,ul}} \mathbf{R}_{k,j}^{\mathrm{ut,ul}} = \mathbf{0}, \forall k, s, j, s \neq j$, ii) $\rho_{k,s}^{\mathrm{ul}} \to 0, \forall k, s$, and iii) $M \to \infty$. In these cases, the decoupled solutions of UL precoders $\{\mathbf{w}_{k,s}^{\mathrm{ul}}\}_{\forall k\in\mathcal{K}_s, s\in\mathcal{S}}$, and receivers $\{\mathbf{c}_{k,s}^{\mathrm{ul}}\}_{\forall k\in\mathcal{K}_s, s\in\mathcal{S}}$, are given by

$$\mathbf{w}_{k,s}^{\mathrm{aslnr,ul,de}} = \boldsymbol{\xi}_{\max}\left(\left(\sum_{j\in\mathcal{S}} \mathbf{R}_{k,j}^{\mathrm{ut,ul}} + \frac{1}{\rho_{k,s}^{\mathrm{ul}}}\mathbf{I}\right)^{-1} \mathbf{R}_{k,s}^{\mathrm{ut,ul}}\right), \tag{40a}$$

$$\mathbf{c}_{k,s}^{\mathrm{asinr,ul,de}} = \left(\mathbf{I} + \sum_{i\in\mathcal{K}_s} \rho_{i,s}^{\mathrm{ul}} \xi_{i,s}^{\mathrm{ul}} \mathbf{g}_{i,s}^{\mathrm{ul}} (\mathbf{g}_{i,s}^{\mathrm{ul}})^H\right)^{-1} \mathbf{g}_{k,s}^{\mathrm{ul}}. \tag{40b}$$

*Proof:* The proof is similar to that of Theorem 1, and is omitted here. ■

Theorem 2 demonstrates that the UL receivers at each satellite and UL precoders at each UT can be designed separately with only their local sCSI. Specifically, the decoupled UL precoder at UT $k$ for satellite $s$ and the decoupled UL receiver at satellite $s$ for UT $k$ rely on the local sCSI $\{\mathbf{R}_{k,s}^{\mathrm{ut,ul}}\}_{\forall s\in\mathcal{S}}$ and $\{\mathbf{g}_{i,s}^{\mathrm{ul}}, \xi_{i,s}^{\mathrm{ul}}\}_{\forall i\in\mathcal{K}_s}$, respectively, which represent partial information of $\mathcal{H}_k^{\mathrm{ut}}$ and $\mathcal{H}_s^{\mathrm{sat}}$.

## IV. DL BSP and UL BSR Design

In this section, with the established BBCM, we demonstrate that the designs of DL precoder and UL receiver in the space domain can be transformed into that of the lower-dimensional vectors in the beam domain, and the resulting space domain precoder and receiver are beam structured.

### A. DL BSP Design

In this subsection, for given space domain DL receivers, we propose the DL BSP design to maximize the ASLNR in (27). We first obtain the following theorem.

*Theorem 3:* When $M \to \infty$, and $\mathcal{B}_{k,s}^{\mathrm{dia,dl}} \cap \mathcal{B}_{i,s}^{\mathrm{dia,dl}} = \emptyset, \forall k, i, s$, the optimal space domain DL precoder maximizing the ASLNR in (27) has the following structure:

$$\mathbf{w}_{k,s}^{\mathrm{aslnr,dl}} = \check{\mathbf{V}}_{k,s}^{\mathrm{dl}} \check{\mathbf{w}}_{k,s}^{\mathrm{aslnr,dl}}, \forall k \in \mathcal{K}_s, s \in \mathcal{S}, \tag{41}$$

where $\check{\mathbf{V}}_{k,s}^{\mathrm{dl}} = \mathbf{V}^{\mathrm{dl}} \boldsymbol{\Omega}_{k,s}^{\mathrm{dl}}$, and

$$\check{\mathbf{w}}_{k,s}^{\mathrm{aslnr,dl}} = \arg\max_{\check{\mathbf{w}}_{k,s}^{\mathrm{dl}}} \mathrm{ASLNR}_{k,s}^{\mathrm{dl,b}}\left(\check{\mathbf{w}}_{k,s}^{\mathrm{dl}}, \mathcal{B}_{k,s}^{\mathrm{dia,dl}}\right) \mathrm{s.t.} ||\check{\mathbf{w}}_{k,s}^{\mathrm{dl}}||^2 = 1, \tag{42}$$

with

$$\mathrm{ASLNR}_{k,s}^{\mathrm{dl,b}}\left(\check{\mathbf{w}}_{k,s}^{\mathrm{dl}}, \mathcal{B}_{k,s}^{\mathrm{dia,dl}}\right) = \frac{(\mathbf{c}_{k,s}^{\mathrm{dl}})^H \mathbf{R}_{k,s}^{\mathrm{ut,dl}} \mathbf{c}_{k,s}^{\mathrm{dl}} \left|(\check{\mathbf{g}}_{k,s}^{\mathrm{dl}})^H \check{\mathbf{w}}_{k,s}^{\mathrm{dl}}\right|^2}{(\check{\mathbf{w}}_{k,s}^{\mathrm{dl}})^H \check{\mathbf{D}}_{k,s}^{\mathrm{dl}} \check{\mathbf{w}}_{k,s}^{\mathrm{dl}}}, \tag{43}$$

and

$$\begin{aligned}\check{\mathbf{D}}_{k,s}^{\mathrm{dl}} = &\sum_{j\neq s}^{S} \sum_{i\in\mathcal{K}_j} (\mathbf{c}_{i,j}^{\mathrm{dl}})^H \mathbf{R}_{i,s}^{\mathrm{ut,dl}} \mathbf{c}_{i,j}^{\mathrm{dl}} (\check{\mathbf{V}}_{k,s}^{\mathrm{dl}})^H \check{\mathbf{V}}_{i,s}^{\mathrm{dl}} \check{\mathbf{g}}_{i,s}^{\mathrm{dl}} (\check{\mathbf{g}}_{i,s}^{\mathrm{dl}})^H \\ &(\check{\mathbf{V}}_{i,s}^{\mathrm{dl}})^H \check{\mathbf{V}}_{k,s}^{\mathrm{dl}} + \sum_{i\in\mathcal{K}_s\setminus\{k\}} (\mathbf{c}_{i,s}^{\mathrm{dl}})^H \mathbf{R}_{i,s}^{\mathrm{ut,dl}} \mathbf{c}_{i,s}^{\mathrm{dl}} (\check{\mathbf{V}}_{k,s}^{\mathrm{dl}})^H \\ &\check{\mathbf{V}}_{i,s}^{\mathrm{dl}} \check{\mathbf{g}}_{i,s}^{\mathrm{dl}} (\check{\mathbf{g}}_{i,s}^{\mathrm{dl}})^H (\check{\mathbf{V}}_{i,s}^{\mathrm{dl}})^H \check{\mathbf{V}}_{k,s}^{\mathrm{dl}} + \frac{(\sigma_k^{\mathrm{dl}})^2}{p_{k,s}^{\mathrm{dl}}}\mathbf{I}.\end{aligned} \tag{44}$$

*Proof:* See Appendix B. ■

$\mathrm{ASLNR}_{k,s}^{\mathrm{dl,b}}(\check{\mathbf{w}}_{k,s}^{\mathrm{dl}}, \mathcal{B}_{k,s}^{\mathrm{dia,dl}})$ in (43) refers to the value of ASLNR achieved by $\check{\mathbf{w}}_{k,s}^{\mathrm{dl}}$ with beam index set $\mathcal{B}_{k,s}^{\mathrm{dia,dl}}$. From Theorem 3, when the number of satellite antennas tends to infinity, and beam index sets of each satellite for any two UTs are non-overlapping, we can convert the design of space domain DL precoders $\{\mathbf{w}_{k,s}^{\mathrm{dl}}\}_{k\in\mathcal{K}_s, s\in\mathcal{S}}$ into that of beam domain vectors $\{\check{\mathbf{w}}_{k,s}^{\mathrm{dl}}\}_{k\in\mathcal{K}_s, s\in\mathcal{S}}$. Typically, the dimension of $\check{\mathbf{w}}_{k,s}^{\mathrm{dl}}$ is much lower than that of $\mathbf{w}_{k,s}^{\mathrm{dl}}$, due to the sparsity of the beam domain satellite channel. Therefore, the complexity of the DL precoder design can be significantly reduced. We refer to the space domain precoder $\mathbf{w}_{k,s}^{\mathrm{aslnr,dl}}$ in (41) as the DL BSP and $\check{\mathbf{w}}_{k,s}^{\mathrm{aslnr,dl}}$ as the beam domain DL precoder.

In massive MIMO multi-satellite systems, the number of UTs might be huge, and the beam index sets of each satellite for UTs may overlap. Therefore, more beams should be incorporated in the design of beam domain DL precoder to improve the system performance. Let $\bar{\mathcal{B}}_{k,s}^{\mathrm{dia,dl}}$ be the beam index set, which is defined similarly to $\mathcal{B}_{k,s}^{\mathrm{dia,dl}}$ in (17), except that $\gamma$ is replaced with $\bar{\gamma}$, and $\bar{\gamma} > \gamma$. Then, with beam index sets $\mathcal{B}_{k,s}^{\mathrm{dia,dl}}$ and $\bar{\mathcal{B}}_{k,s}^{\mathrm{dia,dl}}$, we provide a beam selection criterion for the design of DL BSP in the following theorem.

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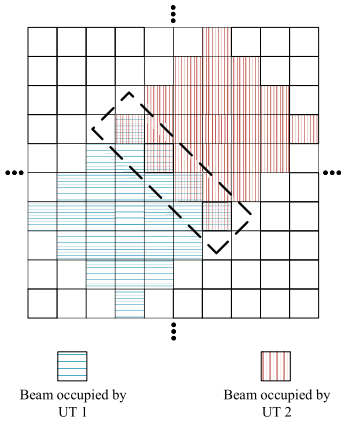



Fig. 3. Illustration of the beam selection criteria.

*Theorem 4:* When $M \to \infty$, we have

$$\mathrm{ASLNR}_{k,s}^{\mathrm{dl,b}}\left(\check{\mathbf{w}}_{k,s}^{\mathrm{aslnr,dl}},\mathcal{B}_{k,s}^{\mathrm{dia,dl}}\right) \le \mathrm{ASLNR}_{k,s}^{\mathrm{dl,b}}\left(\bar{\mathbf{w}}_{k,s}^{\mathrm{aslnr,dl}},\bar{\mathcal{B}}_{k,s}^{\mathrm{dia,dl}}\right), \tag{45}$$

where the equality holds when $\{\bar{b}_{k,s,m}\}_{m=1}^{\bar{B}_{k,s}^{\mathrm{dia,dl}}-B_{k,s}^{\mathrm{dia,dl}}}$ in the newly selected beam index set $\bar{\mathcal{B}}_{k,s}^{\mathrm{dia,dl}} \setminus \mathcal{B}_{k,s}^{\mathrm{dia,dl}}$ satisfy

$$\bar{b}_{k,s,m} \notin \bigcup_{i\in\mathcal{K}_j, j\in\mathcal{S}, \mathcal{B}_{i,s}^{\mathrm{dia,dl}}\cap\mathcal{B}_{k,s}^{\mathrm{dia,dl}}\neq\emptyset} \mathcal{B}_{i,s}^{\mathrm{dia,dl}},$$
$$\forall s\in\mathcal{S}, k\in\mathcal{K}_s, m=1,2,\ldots,\bar{B}_{k,s}^{\mathrm{dia,dl}}-B_{k,s}^{\mathrm{dia,dl}}. \tag{46}$$

*Proof:* See Appendix C. ∎

As demonstrated in Theorem 4, the performance of the DL BSP can be improved by incorporating more beams into the design. Specifically, (46) specifies the criteria for beam selection, indicating that beams satisfying this condition should be excluded. Moreover, the beams satisfying $\bar{b}_{k,s,m} \in \bigcup_{i\in\mathcal{K}_j, j\in\mathcal{S}, \mathcal{B}_{i,s}^{\mathrm{dia,dl}}\cap\mathcal{B}_{k,s}^{\mathrm{dia,dl}}\neq\emptyset} \mathcal{B}_{i,s}^{\mathrm{dia,dl}}$ can be incorporated to improve the performance of the beam structured designs. As illustrated in Fig. 3, the beam index set of UT 1 overlaps with that of UT 2. According to the beam selection criteria in Theorem 4, four additional beams, highlighted by the dotted rectangle, can be included in UT 1's beam structured precoder design to enhance the system performance.

Although the number of satellite antennas remains constrained in practice, the proposed beam structured DL precoder design can still achieve near-optimal performance with a reasonably large number of satellite antennas. Therefore, we constrain the DL precoder in the space domain to satisfy

$$\mathbf{w}_{k,s}^{\mathrm{dl}} = \bar{\mathbf{V}}_{k,s}^{\mathrm{dl}}\bar{\mathbf{w}}_{k,s}^{\mathrm{dl}}, \tag{47}$$

where $\bar{\mathbf{V}}_{k,s}^{\mathrm{dl}} = \mathbf{V}^{\mathrm{dl}}\bar{\mathbf{\Omega}}_{k,s}^{\mathrm{dl}}$, and $\bar{\mathbf{\Omega}}_{k,s}^{\mathrm{dl}} \in \mathbb{R}^{\check{M}^{\mathrm{dl}}\times\bar{B}_{k,s}^{\mathrm{dia,dl}}}$ is defined similarly to $\mathbf{\Omega}_{k,s}^{\mathrm{dl}}$ in (20), except that $\mathcal{B}_{k,s}^{\mathrm{dia,dl}}$ is replaced with $\bar{\mathcal{B}}_{k,s}^{\mathrm{dia,dl}}$, and $|\bar{\mathcal{B}}_{k,s}^{\mathrm{dia,dl}}| = \bar{B}_{k,s}^{\mathrm{dia,dl}}$. Then we can rewrite the ASLNR as

$$\mathrm{ASLNR}_{k,s}^{\mathrm{dl,b}}(\bar{\mathbf{w}}_{k,s}^{\mathrm{dl}}) = \frac{(\mathbf{c}_{k,s}^{\mathrm{dl}})^H\mathbf{R}_{k,s}^{\mathrm{ut,dl}}\mathbf{c}_{k,s}^{\mathrm{dl}}\left|(\check{\mathbf{g}}_{k,s}^{\mathrm{dl}})^H(\tilde{\mathbf{J}}_{k,k,s}^{\mathrm{dl}})^H\bar{\mathbf{w}}_{k,s}^{\mathrm{dl}}\right|^2}{(\bar{\mathbf{w}}_{k,s}^{\mathrm{dl}})^H\bar{\mathbf{D}}_{k,s}^{\mathrm{dl,b}}\bar{\mathbf{w}}_{k,s}^{\mathrm{dl}}}, \tag{48}$$

where $\bar{\mathbf{D}}_{k,s}^{\mathrm{dl,b}} = \sum_{i\in\mathcal{K}_s\setminus\{k\}}(\mathbf{c}_{i,s}^{\mathrm{dl}})^H\mathbf{R}_{i,s}^{\mathrm{ut,dl}}\mathbf{c}_{i,s}^{\mathrm{dl}}\tilde{\mathbf{J}}_{k,i,s}^{\mathrm{dl}}\check{\mathbf{g}}_{i,s}^{\mathrm{dl}}(\check{\mathbf{g}}_{i,s}^{\mathrm{dl}})^H(\tilde{\mathbf{J}}_{k,i,s}^{\mathrm{dl}})^H + \sum_{j\neq s}^{S}\sum_{i\in\mathcal{K}_j}(\mathbf{c}_{i,j}^{\mathrm{dl}})^H\mathbf{R}_{i,s}^{\mathrm{ut,dl}}\mathbf{c}_{i,j}^{\mathrm{dl}}\tilde{\mathbf{J}}_{k,i,s}^{\mathrm{dl}}\check{\mathbf{g}}_{i,s}^{\mathrm{dl}}(\check{\mathbf{g}}_{i,s}^{\mathrm{dl}})^H(\tilde{\mathbf{J}}_{k,i,s}^{\mathrm{dl}})^H + \frac{1}{\rho_{k,s}^{\mathrm{dl}}}\tilde{\mathbf{J}}_{k,s}^{\mathrm{dl}}$, $\tilde{\mathbf{J}}_{k,i,s}^{\mathrm{p}} = (\bar{\mathbf{V}}_{k,s}^{\mathrm{p}})^H\check{\mathbf{V}}_{i,s}^{\mathrm{p}}$, and $\bar{\mathbf{J}}_{k,s}^{\mathrm{p}} = (\bar{\mathbf{V}}_{k,s}^{\mathrm{p}})^H\bar{\mathbf{V}}_{k,s}^{\mathrm{p}}$. Then, the beam domain precoder maximizing $\mathrm{ASLNR}_{k,s}^{\mathrm{dl,b}}$ in (48) is given by

$$\bar{\mathbf{w}}_{k,s}^{\mathrm{aslnr,dl}} = \frac{1}{\bar{\eta}_{k,s}^{\mathrm{aslnr,dl}}}\left(\sum_{i\in\mathcal{K}}\sum_{j\in\mathcal{S}_i}(\mathbf{c}_{i,j}^{\mathrm{dl}})^H\mathbf{R}_{i,s}^{\mathrm{ut,dl}}\mathbf{c}_{i,j}^{\mathrm{dl}}\tilde{\mathbf{J}}_{k,i,s}^{\mathrm{dl}}\check{\mathbf{g}}_{i,s}^{\mathrm{dl}}(\check{\mathbf{g}}_{i,s}^{\mathrm{dl}})^H(\tilde{\mathbf{J}}_{k,i,s}^{\mathrm{dl}})^H + \frac{1}{\rho_{k,s}^{\mathrm{dl}}}\bar{\mathbf{J}}_{k,s}^{\mathrm{dl}}\right)^{-1}\tilde{\mathbf{J}}_{k,k,s}^{\mathrm{dl}}\check{\mathbf{g}}_{k,s}^{\mathrm{dl}},\ \forall k\in\mathcal{K}_s, s\in\mathcal{S}, \tag{49}$$

where $\bar{\eta}_{k,s}^{\mathrm{aslnr,dl}}$ denotes the normalization factor that makes $(\bar{\mathbf{w}}_k^{\mathrm{aslnr,dl}})^H\bar{\mathbf{J}}_{k,s}^{\mathrm{dl}}\bar{\mathbf{w}}_k^{\mathrm{aslnr,dl}} = 1$. By utilizing the results provided in Theorem 1, we can obtain the decoupled beam domain DL precoder as (50), shown at the bottom of the page, where $\bar{\eta}_{k,s}^{\mathrm{aslnr,dl,de}}$ is the corresponding normalization factor making $(\bar{\mathbf{w}}_k^{\mathrm{aslnr,dl,de}})^H(\bar{\mathbf{\Omega}}_{k,s}^{\mathrm{dl}})^H(\mathbf{V}^{\mathrm{dl}})^H\mathbf{V}^{\mathrm{dl}}\bar{\mathbf{\Omega}}_{k,s}^{\mathrm{dl}}\bar{\mathbf{w}}_k^{\mathrm{aslnr,dl,de}} = 1$, and $\tilde{\mathbf{g}}_{i,s}^{\mathrm{dl}} = \mathbf{\Omega}_{i,s}^{\mathrm{dl}}\check{\mathbf{g}}_{i,s}^{\mathrm{dl}}$. The resulting space domain DL precoder is given by

$$\mathbf{w}_{k,s}^{\mathrm{aslnr,dl,de}} = \bar{\mathbf{V}}_{k,s}^{\mathrm{dl}}\bar{\mathbf{w}}_{k,s}^{\mathrm{aslnr,dl,de}}. \tag{51}$$

Notice that (50) provides a decoupled closed-form beam domain DL precoder with only local sCSI, whose dimension is typically much smaller than that of the space domain precoder in (31a), then the computational overhead of the DL satellite payload can be significantly reduced.

### B. UL BSR Design

In this subsection, for fixed space domain UL precoders, we propose the UL BSR design to maximize the ASINR in (36). We first derive the following theorem.

*Theorem 5:* When $M \to \infty$, and $\mathcal{B}_{k,s}^{\mathrm{dia,ul}} \cap \mathcal{B}_{i,s}^{\mathrm{dia,ul}} = \emptyset, \forall k,i,s$, the optimal space domain UL receiver maximizing the ASINR in (36) has the following structure:

$$\mathbf{c}_{k,s}^{\mathrm{asinr,ul}} = \check{\mathbf{V}}_{k,s}^{\mathrm{ul}}\check{\mathbf{c}}_{k,s}^{\mathrm{asinr,ul}}, \forall k\in\mathcal{K}_s, s\in\mathcal{S}, \tag{52}$$

where

$$\check{\mathbf{c}}_{k,s}^{\mathrm{asinr,ul}} = \arg\max_{\check{\mathbf{c}}_{k,s}^{\mathrm{ul}}} \mathrm{ASINR}_{k,s}^{\mathrm{ul,b}}\left(\check{\mathbf{c}}_{k,s}^{\mathrm{ul}}, \mathcal{B}_{k,s}^{\mathrm{dia,ul}}\right) \mathrm{s.t.} ||\check{\mathbf{c}}_{k,s}^{\mathrm{ul}}||^2 = 1, \tag{53}$$

$$\bar{\mathbf{w}}_{k,s}^{\mathrm{aslnr,dl,de}} = \frac{1}{\bar{\eta}_{k,s}^{\mathrm{aslnr,dl,de}}}\left(\sum_{i\in\mathcal{K}}\xi_{i,s}^{\mathrm{dl}}(\bar{\mathbf{\Omega}}_{k,s}^{\mathrm{dl}})^H(\mathbf{V}^{\mathrm{dl}})^H\mathbf{V}^{\mathrm{dl}}\tilde{\mathbf{g}}_{i,s}^{\mathrm{dl}}(\tilde{\mathbf{g}}_{i,s}^{\mathrm{dl}})^H(\mathbf{V}^{\mathrm{dl}})^H\mathbf{V}^{\mathrm{dl}}\bar{\mathbf{\Omega}}_{k,s}^{\mathrm{dl}} + \frac{1}{\rho_{k,s}^{\mathrm{dl}}}(\bar{\mathbf{\Omega}}_{k,s}^{\mathrm{dl}})^H(\mathbf{V}^{\mathrm{dl}})^H\mathbf{V}^{\mathrm{dl}}\bar{\mathbf{\Omega}}_{k,s}^{\mathrm{dl}}\right)^{-1}(\bar{\mathbf{\Omega}}_{k,s}^{\mathrm{dl}})^H(\mathbf{V}^{\mathrm{dl}})^H\mathbf{V}^{\mathrm{dl}}\tilde{\mathbf{g}}_{k,s}^{\mathrm{dl}}, \tag{50}$$

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with

$$\mathrm{ASINR}_{k,s}^{\mathrm{ul,b}}(\check{\mathbf{c}}_{k,s}^{\mathrm{ul}},\mathcal{B}_{k,s}^{\mathrm{dia}})=\frac{p_{k,s}^{\mathrm{ul}}(\mathbf{w}_{k,s}^{\mathrm{ul}})^H\mathbf{R}_{k,s}^{\mathrm{ut,ul}}\mathbf{w}_{k,s}^{\mathrm{ul}}\left|(\check{\mathbf{g}}_{k,s}^{\mathrm{ul}})^H\check{\mathbf{c}}_{k,s}^{\mathrm{ul}}\right|^2}{(\check{\mathbf{c}}_{k,s}^{\mathrm{ul}})^H\check{\mathbf{E}}_{k,s}^{\mathrm{ul}}\check{\mathbf{c}}_{k,s}^{\mathrm{ul}}},\tag{54}$$

and

$$\begin{aligned}\check{\mathbf{E}}_{k,s}^{\mathrm{ul}}=&\sum_{j\neq s}^{S}\sum_{i\in\mathcal{K}_j}p_{i,j}^{\mathrm{ul}}(\mathbf{w}_{i,j}^{\mathrm{ul}})^H\mathbf{R}_{i,s}^{\mathrm{ut,ul}}\mathbf{w}_{i,j}^{\mathrm{ul}}(\check{\mathbf{V}}_{k,s}^{\mathrm{ul}})^H\check{\mathbf{V}}_{i,s}^{\mathrm{ul}}\check{\mathbf{g}}_{i,s}^{\mathrm{ul}}\\&(\check{\mathbf{g}}_{i,s}^{\mathrm{ul}})^H(\check{\mathbf{V}}_{i,s}^{\mathrm{ul}})^H\check{\mathbf{V}}_{k,s}^{\mathrm{ul}}+\sum_{i\in\mathcal{K}_s\setminus\{k\}}p_{i,s}^{\mathrm{ul}}(\mathbf{w}_{i,s}^{\mathrm{ul}})^H\mathbf{R}_{i,s}^{\mathrm{ut,ul}}\mathbf{w}_{i,s}^{\mathrm{ul}}\\&(\check{\mathbf{V}}_{k,s}^{\mathrm{ul}})^H\check{\mathbf{V}}_{i,s}^{\mathrm{ul}}\check{\mathbf{g}}_{i,s}^{\mathrm{ul}}(\check{\mathbf{g}}_{i,s}^{\mathrm{ul}})^H(\check{\mathbf{V}}_{i,s}^{\mathrm{ul}})^H\check{\mathbf{V}}_{k,s}^{\mathrm{ul}}+(\sigma_s^{\mathrm{ul}})^2\mathbf{I}.\end{aligned}\tag{55}$$

*Proof:* The proof is similar to that of Theorem 3, and is omitted here. ■

$\mathrm{ASINR}_{k,s}^{\mathrm{ul,b}}(\check{\mathbf{c}}_{k,s}^{\mathrm{ul}},\mathcal{B}_{k,s}^{\mathrm{dia,ul}})$ in (54) denotes the ASINR of $\check{\mathbf{c}}_{k,s}^{\mathrm{ul}}$ with beam index set $\mathcal{B}_{k,s}^{\mathrm{dia,ul}}$. Theorem 5 indicates that the design of space domain UL receivers $\{\mathbf{c}_{k,s}^{\mathrm{ul}}\}_{k\in\mathcal{K}_s,s\in\mathcal{S}}$ can be transformed into that of beam domain vectors $\{\check{\mathbf{c}}_{k,s}^{\mathrm{ul}}\}_{k\in\mathcal{K}_s,s\in\mathcal{S}}$, which are referred to as the beam domain UL receivers. Then, the complexity of the UL receiver design can be significantly reduced. We refer to the space domain receiver $\mathbf{c}_{k,s}^{\mathrm{asinr,ul}}$ in (52) as the beam structured UL receiver.

The following theorem provides a beam selection criteria for incorporating additional beams into the beam structured UL receiver design, when the beam index sets of the satellite for different UTs are overlapping.

*Theorem 6:* When $M\to\infty$, for given beam index sets $\bar{\mathcal{B}}_{k,s}^{\mathrm{dia,ul}}$ and $\mathcal{B}_{k,s}^{\mathrm{dia,ul}}$, we have

$$\mathrm{ASINR}_{k,s}^{\mathrm{ul,b}}\left(\check{\mathbf{c}}_{k,s}^{\mathrm{asinr,ul}},\mathcal{B}_{k,s}^{\mathrm{dia,ul}}\right)\leq\mathrm{ASINR}_{k,s}^{\mathrm{ul,b}}\left(\check{\mathbf{c}}_{k,s}^{\mathrm{asinr,ul}},\bar{\mathcal{B}}_{k,s}^{\mathrm{dia,ul}}\right),\tag{56}$$

where the equality holds when $\{\bar{b}_{k,s,m}\}_{m=1}^{\bar{B}_{k,s}^{\mathrm{dia,ul}}-B_{k,s}^{\mathrm{dia,ul}}}$ in the newly selected beam index set $\bar{\mathcal{B}}_{k,s}^{\mathrm{dia,ul}}\setminus\mathcal{B}_{k,s}^{\mathrm{dia,ul}}$ satisfy

$$\begin{aligned}\bar{b}_{k,s,m}\notin&\bigcup_{i\in\mathcal{K}_j,j\in\mathcal{S},\mathcal{B}_{i,s}^{\mathrm{dia,ul}}\cap\mathcal{B}_{k,s}^{\mathrm{dia,ul}}\neq\emptyset}\mathcal{B}_{i,s}^{\mathrm{dia,ul}},\\&\forall s\in\mathcal{S},k\in\mathcal{K}_s,m=1,2,\ldots,\bar{B}_{k,s}^{\mathrm{dia,ul}}-B_{k,s}^{\mathrm{dia,ul}}.\end{aligned}\tag{57}$$

*Proof:* The proof is similar to that of Theorem 4, and is omitted here. ■

Hereafter, we restrict the space domain UL receiver to satisfy the following structure:

$$\mathbf{c}_{k,s}^{\mathrm{ul}}=\bar{\mathbf{V}}_{k,s}^{\mathrm{ul}}\bar{\mathbf{c}}_{k,s}^{\mathrm{ul}},\tag{58}$$

where $\bar{\mathbf{V}}_{k,s}^{\mathrm{ul}}$ is defined similarly to $\bar{\mathbf{V}}_{k,s}^{\mathrm{dl}}$. The ASINR can be rewritten as

$$\mathrm{ASINR}_{k,s}^{\mathrm{ul,b}}(\bar{\mathbf{c}}_{k,s}^{\mathrm{ul}})=\frac{p_{k,s}^{\mathrm{ul}}(\mathbf{w}_{k,s}^{\mathrm{ul}})^H\mathbf{R}_{k,s}^{\mathrm{ut,ul}}\mathbf{w}_{k,s}^{\mathrm{ul}}\left|(\check{\mathbf{g}}_{k,s}^{\mathrm{ul}})^H\bar{\mathbf{c}}_{k,s}^{\mathrm{ul}}\right|^2}{(\bar{\mathbf{c}}_{k,s}^{\mathrm{ul}})^H\bar{\mathbf{E}}_{k,s}^{\mathrm{ul,b}}\bar{\mathbf{c}}_{k,s}^{\mathrm{ul}}},\tag{59}$$

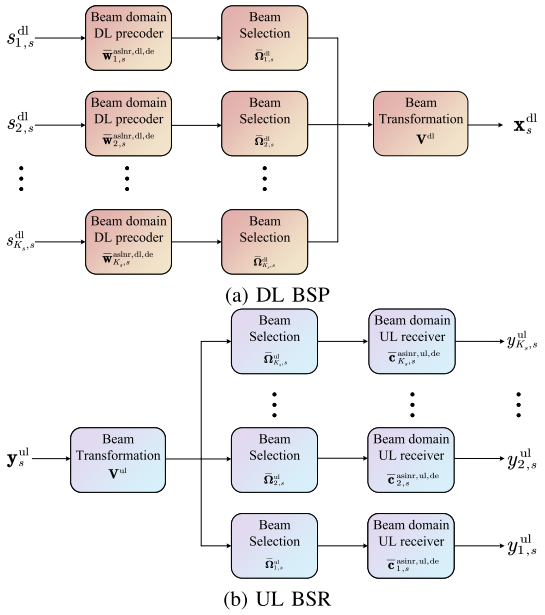



Fig. 4. Illustration of DL BSP and UL BSR.

where $\bar{\mathbf{E}}_{k,s}^{\mathrm{ul,b}}=\sum_{i\in\mathcal{K}_s\setminus\{k\}}p_{i,s}^{\mathrm{ul}}(\mathbf{w}_{i,s}^{\mathrm{ul}})^H\mathbf{R}_{i,s}^{\mathrm{ut,ul}}\mathbf{w}_{i,s}^{\mathrm{ul}}\tilde{\mathbf{J}}_{k,i,s}^{\mathrm{ul}}\check{\mathbf{g}}_{i,s}^{\mathrm{ul}}(\check{\mathbf{g}}_{i,s}^{\mathrm{ul}})^H(\tilde{\mathbf{J}}_{k,i,s}^{\mathrm{ul}})^H+\sum_{j\neq s}^{S}\sum_{i\in\mathcal{K}_j}p_{i,j}^{\mathrm{ul}}(\mathbf{w}_{i,j}^{\mathrm{ul}})^H\mathbf{R}_{i,s}^{\mathrm{ut,ul}}\mathbf{w}_{i,j}^{\mathrm{ul}}\tilde{\mathbf{J}}_{k,i,s}^{\mathrm{ul}}\check{\mathbf{g}}_{i,s}^{\mathrm{ul}}(\check{\mathbf{g}}_{i,s}^{\mathrm{ul}})^H(\tilde{\mathbf{J}}_{k,i,s}^{\mathrm{ul}})^H+(\sigma_s^{\mathrm{ul}})^2\bar{\mathbf{J}}_{k,s}^{\mathrm{ul}}$. Then, the beam domain receiver maximizing $\mathrm{ASINR}_{k,s}^{\mathrm{ul,b}}$ in (59) is given by

$$\begin{aligned}\bar{\mathbf{c}}_{k,s}^{\mathrm{asinr,ul}}=&\left(\sum_{j=1}^{S}\sum_{i\in\mathcal{K}_j}p_{i,j}^{\mathrm{ul}}(\mathbf{w}_{i,j}^{\mathrm{ul}})^H\mathbf{R}_{i,s}^{\mathrm{ut,ul}}\mathbf{w}_{i,j}^{\mathrm{ul}}\tilde{\mathbf{J}}_{k,i,s}^{\mathrm{ul}}\check{\mathbf{g}}_{i,s}^{\mathrm{ul}}(\check{\mathbf{g}}_{i,s}^{\mathrm{ul}})^H(\tilde{\mathbf{J}}_{k,i,s}^{\mathrm{ul}})^H\right.\\&\left.+(\sigma_s^{\mathrm{ul}})^2\bar{\mathbf{J}}_{k,s}^{\mathrm{ul}}\right)^{-1}\tilde{\mathbf{J}}_{k,k,s}^{\mathrm{ul}}\check{\mathbf{g}}_{k,s}^{\mathrm{ul}},\forall k\in\mathcal{K}_s,s\in\mathcal{S}.\end{aligned}\tag{60}$$

By utilizing the results in Theorem 2, we can obtain the decoupled beam domain UL receiver as (61), shown at the bottom of the page, where $\tilde{\mathbf{g}}_{i,s}^{\mathrm{ul}}=\mathbf{\Omega}_{i,s}^{\mathrm{ul}}\check{\mathbf{g}}_{i,s}^{\mathrm{ul}}$.

The resulting space domain UL receiver is expressed as

$$\mathbf{c}_{k,s}^{\mathrm{asinr,ul,de}}=\bar{\mathbf{V}}_{k,s}^{\mathrm{ul}}\bar{\mathbf{c}}_{k,s}^{\mathrm{asinr,ul,de}}.\tag{62}$$

Note that the calculation of the closed-form lower-dimensional beam domain UL receiver in (61) requires only the local sCSI, thereby reducing the computational overhead of the UL satellite payload. The proposed DL BSP and UL BSR are illustrated in Fig. 4. We assume that all UTs and satellites in the system are active on the same time-frequency resource.

## V. Low-Complexity Design and Implementation for DL BSP and UL BSR

In this section, we show that the computation of lower-dimensional beam domain precoder and receiver only involves real-valued matrices, and develop a low-complexity design and implementation for the proposed decoupled DL BSP and UL BSR by exploiting the properties of the beam matrix.

$$\bar{\mathbf{c}}_{k,s}^{\mathrm{asinr,ul,de}}=\left(\sum_{i\in\mathcal{K}_s}p_{i,s}^{\mathrm{ul}}\xi_{i,s}^{\mathrm{ul}}(\bar{\mathbf{\Omega}}_{k,s}^{\mathrm{ul}})^H(\mathbf{V}^{\mathrm{ul}})^H\mathbf{V}^{\mathrm{ul}}\tilde{\mathbf{g}}_{i,s}^{\mathrm{ul}}(\tilde{\mathbf{g}}_{i,s}^{\mathrm{ul}})^H(\mathbf{V}^{\mathrm{ul}})^H\mathbf{V}^{\mathrm{ul}}\bar{\mathbf{\Omega}}_{k,s}^{\mathrm{ul}}+(\sigma_s^{\mathrm{ul}})^2(\bar{\mathbf{\Omega}}_{k,s}^{\mathrm{ul}})^H(\mathbf{V}^{\mathrm{ul}})^H\mathbf{V}^{\mathrm{ul}}\bar{\mathbf{\Omega}}_{k,s}^{\mathrm{ul}}\right)^{-1}(\bar{\mathbf{\Omega}}_{k,s}^{\mathrm{ul}})^H(\mathbf{V}^{\mathrm{ul}})^H\mathbf{V}^{\mathrm{ul}}\tilde{\mathbf{g}}_{k,s}^{\mathrm{ul}}.\tag{61}$$

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### A. Low-Complexity Design

The design complexity of the decoupled beam structured DL precoder and UL receiver comes from (50) and (61), respectively. First, we show that computing $\{\bar{\mathbf{w}}_{k,s}^{\text{aslnr,dl,de}}\}_{k\in\mathcal{K}_s,s\in\mathcal{S}}$ and $\{\bar{\mathbf{c}}_{k,s}^{\text{asinr,ul,de}}\}_{k\in\mathcal{K}_s,s\in\mathcal{S}}$ relies solely on real-valued matrix. Let $\mathbf{R}_N$ be the reverse permutation matrix with $\mathbf{R}_N\mathbf{R}_N=\mathbf{I}$. For $\mathrm{t}\in\{\mathrm{x},\mathrm{y}\}$, since each column of beam matrices $\mathbf{V}_{\mathrm{t}}^{\mathrm{p}}$ is conjugate centrosymmetric, then we have $\mathbf{R}_{M_{\mathrm{t}}}\mathbf{V}_{\mathrm{t}}^{\mathrm{p}}=(\mathbf{V}_{\mathrm{t}}^{\mathrm{p}})^*$, and $(\mathbf{V}^{\mathrm{p}})^H\mathbf{V}^{\mathrm{p}}$ in (50) and (61) can be rewritten as

$$(\mathbf{V}^{\mathrm{p}})^H\mathbf{V}^{\mathrm{p}}=(\mathbf{V}_{\mathrm{x}}^{\mathrm{p}})^T(\mathbf{V}_{\mathrm{x}}^{\mathrm{p}})^*\otimes(\mathbf{V}_{\mathrm{y}}^{\mathrm{p}})^T(\mathbf{V}_{\mathrm{y}}^{\mathrm{p}})^*=(\mathbf{V}^{\mathrm{p}})^T(\mathbf{V}^{\mathrm{p}})^*, \tag{63}$$

which implies that $(\mathbf{V}^{\mathrm{p}})^H\mathbf{V}^{\mathrm{p}}$ is real-valued. Next, since the steering vector, as defined in (3), is also conjugate centrosymmetric, and $\mathbf{g}_{k,s}^{\mathrm{p}}=\mathbf{V}^{\mathrm{p}}\tilde{\mathbf{g}}_{k,s}^{\mathrm{p}}$, then $\{\tilde{\mathbf{g}}_{k,s}^{\mathrm{p}}\}_{k\in\mathcal{K}_s,s\in\mathcal{S}}$ can be real-valued vectors. Moreover, the beam selection matrix is also real-valued. Then, the computations of $\{\bar{\mathbf{w}}_{k,s}^{\text{aslnr,dl,de}}\}_{k\in\mathcal{K}_s,s\in\mathcal{S}}$ and $\{\bar{\mathbf{c}}_{k,s}^{\text{asinr,ul,de}}\}_{k\in\mathcal{K}_s,s\in\mathcal{S}}$ rely solely on real-valued matrix.

Further, because $(\mathbf{V}^{\mathrm{p}})^H\mathbf{V}^{\mathrm{p}}$ does not depend on the specific signal, then it can be pre-calculated and stored. Specifically, we have

$$\begin{aligned}&\left[(\mathbf{V}^{\mathrm{p}})^H\mathbf{V}^{\mathrm{p}}\right]_{m,n}\\&=[\mathbf{\Xi}_{\mathrm{x}}^{\mathrm{p}}]_{\lfloor\frac{m-1}{\breve{M}_{\mathrm{y}}^{\mathrm{p}}}\rfloor+1,\lfloor\frac{n-1}{\breve{M}_{\mathrm{y}}^{\mathrm{p}}}\rfloor+1}[\mathbf{\Xi}_{\mathrm{y}}^{\mathrm{p}}]_{((m-1))_{\breve{M}_{\mathrm{y}}^{\mathrm{p}}}+1,((n-1))_{\breve{M}_{\mathrm{y}}^{\mathrm{p}}}+1},\end{aligned} \tag{64}$$

where $\lfloor\cdot\rfloor$ denotes the floor division, and $\mathbf{\Xi}_{\mathrm{t}}^{\mathrm{p}}=(\mathbf{V}_{\mathrm{t}}^{\mathrm{p}})^H\mathbf{V}_{\mathrm{t}}^{\mathrm{p}}$ can be obtained with its first column $[\mathbf{\Xi}_{\mathrm{t}}^{\mathrm{p}}]_{:,1}=(\mathbf{V}_{\mathrm{t}}^{\mathrm{p}})^H[\mathbf{V}_{\mathrm{t}}^{\mathrm{p}}]_{:,1}$, i.e.,

$$[\mathbf{\Xi}_{\mathrm{t}}^{\mathrm{p}}]_{n,n'}=[\mathbf{\Xi}_{\mathrm{t}}^{\mathrm{p}}]_{|n-n'|+1,1}\,. \tag{65}$$

Therefore, we can obtain $(\mathbf{V}^{\mathrm{p}})^H\mathbf{V}^{\mathrm{p}}$ with $[\mathbf{\Xi}_{\mathrm{x}}^{\mathrm{p}}]_{:,1}$ and $[\mathbf{\Xi}_{\mathrm{y}}^{\mathrm{p}}]_{:,1}$, whose computational complexity and store complexity are $\mathcal{O}(\bar{M}_{\mathrm{x}}\log_2\bar{M}_{\mathrm{x}}+\bar{M}_{\mathrm{y}}\log_2\bar{M}_{\mathrm{y}})$ from fast Fourier transform (FFT) and $\mathcal{O}(\breve{M}_{\mathrm{x}}^{\mathrm{p}}+\breve{M}_{\mathrm{y}}^{\mathrm{p}})$, respectively.

Hereafter, we can obtain an easy-handling expression of the steering vectors at satellite sides by introducing additional $\bar{M}_{\mathrm{t}}'$ sampling points $\{b_{i,m}^{\mathrm{sat}}\}_{m=0}^{\bar{M}_{\mathrm{t}}'-1}$ within $[b_i^{\mathrm{sat}},b_{i+1}^{\mathrm{sat}})$ with $b_{i,0}^{\mathrm{sat}}=b_i^{\mathrm{sat}}$. Let $\bar{M}'=\bar{M}_{\mathrm{x}}'\bar{M}_{\mathrm{y}}'$ and $b_{k,s}^{\mathrm{sat,t,min}}$ denote the sampling point closest to the scaled directional cosine $\chi_{\mathrm{t}}g_{k,s}^{\mathrm{t}}$, which is expressed as

$$b_{k,s}^{\mathrm{sat,t,min}}=\arg\min_{b_{i,m}^{\mathrm{sat}}}\left|\chi_{\mathrm{t}}g_{k,s}^{\mathrm{t}}-b_{i,m}^{\mathrm{sat}}\right|. \tag{66}$$

Then, the steering vector $\mathbf{a}_{\mathrm{t}}^{\mathrm{p}}(g_{k,s}^{\mathrm{t}})$ can be approximated as

$$\mathbf{a}_{\mathrm{t}}^{\mathrm{p}}(g_{k,s}^{\mathrm{t}})\approx\breve{\mathbf{a}}_{\mathrm{t}}^{\mathrm{p}}(b_{k,s}^{\mathrm{sat,t,min}})=\mathbf{V}_{\mathrm{t}}^{\mathrm{p}}\tilde{\mathbf{g}}_{\mathrm{t}}^{\mathrm{p}}(b_{k,s}^{\mathrm{sat,t,min}}). \tag{67}$$

Notice that $\tilde{\mathbf{g}}_{\mathrm{t}}(b_{i,m}^{\mathrm{sat}})$ can be derived from $\tilde{\mathbf{g}}_{\mathrm{t}}(b_{0,m}^{\mathrm{sat}})$ as

$$\begin{aligned}&[\tilde{\mathbf{g}}_{\mathrm{t}}^{\mathrm{p}}(b_{i,m}^{\mathrm{sat}})]_l=\\&e^{\jmath\pi\frac{(M_{\mathrm{t}}-1)i}{\bar{M}_{\mathrm{t}}}}\left([\breve{\boldsymbol{\omega}}_{\mathrm{t}}^{\mathrm{sat,p}}]_l\right)^{-1}\left[\mathrm{diag}\left(\breve{\boldsymbol{\omega}}_{\mathrm{t}}^{\mathrm{sat,p}}\right)\tilde{\mathbf{g}}_{\mathrm{t}}^{\mathrm{p}}(b_{0,m}^{\mathrm{sat}})\right]_{((l-i-1))_{\bar{M}_{\mathrm{t}}}+1},\end{aligned} \tag{68}$$

when $\breve{M}_{\mathrm{t}}^{\mathrm{p}}=\bar{M}_{\mathrm{t}}$. Note that $\tilde{\mathbf{g}}_{\mathrm{t}}^{\mathrm{p}}(b_{i,m}^{\mathrm{sat}}),\forall i,m$, constitute the diffusion vector codebook, which depends only on the sampled directional cosines and can be generated offline and stored on-board, and the induced approximation error becomes negligible under a sufficiently fine sampling grid. The vector $(\mathbf{V}^{\mathrm{p}})^H\mathbf{V}^{\mathrm{p}}\tilde{\mathbf{g}}_{k,s}^{\mathrm{p}}$ in (50) and (61) can be expressed as

$$\begin{aligned}&\left[(\mathbf{V}^{\mathrm{p}})^H\mathbf{V}^{\mathrm{p}}\tilde{\mathbf{g}}_{k,s}^{\mathrm{p}}\right]_i=\\&\left[\mathbf{\Xi}_{\mathrm{x}}^{\mathrm{p}}\tilde{\mathbf{g}}_{\mathrm{x}}^{\mathrm{p}}(b_{k,s}^{\mathrm{sat,x,min}})\right]_{\lfloor\frac{i-1}{\breve{M}_{\mathrm{y}}^{\mathrm{p}}}\rfloor+1}\left[\mathbf{\Xi}_{\mathrm{y}}^{\mathrm{p}}\tilde{\mathbf{g}}_{\mathrm{y}}^{\mathrm{p}}(b_{k,s}^{\mathrm{sat,y,min}})\right]_{((i-1))_{\breve{M}_{\mathrm{y}}^{\mathrm{p}}}+1},\end{aligned} \tag{69}$$

and $\left[\mathbf{\Xi}_{\mathrm{t}}^{\mathrm{p}}\tilde{\mathbf{g}}_{\mathrm{t}}^{\mathrm{p}}(b_{k,s}^{\mathrm{sat,t,min}})\right]_m$ is calculated as

$$\left[\mathbf{\Xi}_{\mathrm{t}}^{\mathrm{p}}\tilde{\mathbf{g}}_{\mathrm{t}}^{\mathrm{p}}(b_{k,s}^{\mathrm{sat,t,min}})\right]_m=\sum_{m'=1}^{\breve{M}_{\mathrm{t}}^{\mathrm{p}}}[\mathbf{\Xi}_{\mathrm{t}}^{\mathrm{p}}]_{m,m'}\left[\tilde{\mathbf{g}}_{\mathrm{t}}^{\mathrm{p}}(b_{k,s}^{\mathrm{sat,t,min}})\right]_{((m'-1))_{\breve{M}_{\mathrm{t}}^{\mathrm{p}}}+1}. \tag{70}$$

Then, with (65), (68), (69), and (70), $(\mathbf{V}^{\mathrm{p}})^H\mathbf{V}^{\mathrm{p}}\tilde{\mathbf{g}}_{k,s}^{\mathrm{p}},\forall k\in\mathcal{K}_s,s\in\mathcal{S}$, can be obtained with a computational complexity $\mathcal{O}((\breve{M}_{\mathrm{x}}^{\mathrm{p}})^2\bar{M}_{\mathrm{x}}'+(\breve{M}_{\mathrm{y}}^{\mathrm{p}})^2\bar{M}_{\mathrm{y}}'+\breve{M}^{\mathrm{p}}\bar{M}')$ and a store complexity $\mathcal{O}(\bar{M}'\breve{M}^{\mathrm{p}})$. Therefore, the design complexity of the beam domain DL precoders $\{\bar{\mathbf{w}}_{k,s}^{\text{aslnr,dl,de}}\}_{k\in\mathcal{K}_s,s\in\mathcal{S}}$ and beam domain UL receivers $\{\bar{\mathbf{c}}_{k,s}^{\text{asinr,ul,de}}\}_{k\in\mathcal{K}_s,s\in\mathcal{S}}$ are given by $\mathcal{O}\left(\sum_{s=1}^{S}K_s\left(\frac{K}{2}(\bar{B}_{\mathrm{dia}}^{\mathrm{dl}})^2+(\bar{B}_{\mathrm{dia}}^{\mathrm{dl}})^3+(\bar{B}_{\mathrm{dia}}^{\mathrm{dl}})^2\right)\right)$, and $\mathcal{O}\left(\sum_{s=1}^{S}K_s\left(\frac{K_s}{2}(\bar{B}_{\mathrm{dia}}^{\mathrm{ul}})^2+(\bar{B}_{\mathrm{dia}}^{\mathrm{ul}})^3+(\bar{B}_{\mathrm{dia}}^{\mathrm{ul}})^2\right)\right)$, respectively, where $\bar{B}_{\mathrm{dia}}^{\mathrm{p}}=\bar{B}_{k,s}^{\mathrm{dia,p}},\forall k\in\mathcal{K}_s,s\in\mathcal{S}$. For comparison, when the DL SNRs of each satellite for different UTs are identical, the design complexity of the space domain DL precoder and UL receiver in (31a) and (40b) are given by $\mathcal{O}\left(\sum_{s=1}^{S}M^3+K_sM^2+\frac{K}{2}M^2\right)$, and $\mathcal{O}\left(\sum_{s=1}^{S}M^3+\frac{3}{2}K_sM^2\right)$, respectively.

### B. Low-Complexity Implementation

The implementation involves each satellite applying the designed precoders to its transmit data streams for signal generation, and utilizing the designed receivers to process the received signals for desired data stream recovery. With the designed beam structured DL precoder in (51), the DL transmit signal of satellite $s$ can be expressed as

$$\mathbf{x}_s^{\mathrm{dl}}=\mathbf{V}^{\mathrm{dl}}\mathbf{s}_s^{\mathrm{dl}}=\mathrm{vec}\left(\mathbf{V}_{\mathrm{y}}\mathbf{S}_s^{\mathrm{dl}}\mathbf{V}_{\mathrm{x}}^T\right), \tag{71}$$

where $\mathbf{s}_s^{\mathrm{dl}}=\sum_{k=1}^{K_s}\bar{\mathbf{\Omega}}_{k,s}^{\mathrm{dl}}\sqrt{p_{k,s}^{\mathrm{dl}}}\bar{\mathbf{w}}_{k,s}^{\text{aslnr,dl,de}}s_{k,s}^{\mathrm{dl}}$, and $\mathbf{S}_s^{\mathrm{dl}}=\mathrm{unvec}(\mathbf{s}_s^{\mathrm{dl}})\in\mathbb{C}^{\breve{M}_{\mathrm{y}}^{\mathrm{dl}}\times\breve{M}_{\mathrm{x}}^{\mathrm{dl}}}$. Then the complexity of generating the transmit signal with the beam structured DL precoder is given by $\mathcal{O}\left(\sum_{s=1}^{S}\bar{M}\log_2\bar{M}_{\mathrm{y}}+M_{\mathrm{y}}\bar{M}_{\mathrm{x}}\log_2\bar{M}_{\mathrm{x}}+K_s\bar{B}_{\mathrm{dia}}^{\mathrm{dl}}\right)$, while the complexity of generating the transmit signal with the space domain DL precoder is given by $\mathcal{O}\left(\sum_{s=1}^{S}K_sM\right)$.

By using the designed beam structured UL receiver in (62), the signal at satellite $s$ of UT $k$ after the receiver is given by

$$\begin{aligned}y_{k,s}^{\mathrm{ul}}&=(\bar{\mathbf{c}}_{k,s}^{\text{asinr,ul,de}})^H(\bar{\mathbf{\Omega}}_{k,s}^{\mathrm{ul}})^H(\mathbf{V}^{\mathrm{ul}})^H\mathbf{y}_s^{\mathrm{ul}}\\&=(\bar{\mathbf{c}}_{k,s}^{\text{asinr,ul,de}})^H(\bar{\mathbf{\Omega}}_{k,s}^{\mathrm{ul}})^H\mathrm{vec}\left((\mathbf{V}_{\mathrm{y}}^{\mathrm{ul}})^H\mathbf{Y}_s^{\mathrm{ul}}(\mathbf{V}_{\mathrm{x}}^{\mathrm{ul}})^*\right),\end{aligned} \tag{72}$$

where $\mathbf{Y}_s^{\mathrm{ul}}=\mathrm{unvec}(\mathbf{y}_s^{\mathrm{ul}})\in\mathbb{C}^{M_{\mathrm{y}}\times M_{\mathrm{x}}}$, and $\mathbf{y}_s^{\mathrm{ul}}=\mathbf{z}_s^{\mathrm{ul}}+\sum_{i\in\mathcal{K}_s}\mathbf{H}_{i,s}^{\mathrm{ul}}\sqrt{p_{i,s}^{\mathrm{ul}}}\mathbf{w}_{i,s}^{\mathrm{ul}}s_{i,s}^{\mathrm{ul}}$. Then the complexity of recovering the received signal with

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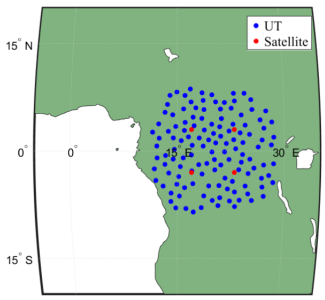



Fig. 5. Distributions of satellites and UTs.

TABLE II
SIMULATION PARAMETERS

| Parameter | Value |
|---|---|
| Number of satellites $S$ | 4 |
| Number of UTs $K$ | 128 |
| Earth Radius $R_{\mathrm{e}}$ | 6378 km |
| Orbit altitude $H$ | 600 km |
| Bandwidth $B$ | 10 MHz |
| Number of antennas at each UT $N_{\mathrm{x}}, N_{\mathrm{y}}$ | $8, 8$ |
| Number of antennas at each satellite $M_{\mathrm{x}}, M_{\mathrm{y}}$ | $20, 20$ |
| Nadir angle of satellite $s$ $\vartheta_s$, $\forall s$ | $45°$ |
| Central frequency $f_{\mathrm{c}}^{\mathrm{dl}}, f_{\mathrm{c}}^{\mathrm{ul}}$ | 2 GHz, 1.9 GHz |
| Noise Temperature $T_n$ | 273 K |
| Transmit Power of UT $k$ $P_k^{\mathrm{ul}}$, $\forall k$ | 10 dBm $\sim$ 30 dBm |
| Transmit Power of satellite $s$ $P_s^{\mathrm{dl}}$, $\forall s$ | 5 dBW $\sim$ 25 dBW |

the beam structured UL receiver is given by $\mathcal{O}\left(\sum_{s=1}^{S} \bar{M} \log_2 \bar{M}_{\mathrm{x}} + M_{\mathrm{x}} \bar{M}_{\mathrm{y}} \log_2 \bar{M}_{\mathrm{y}} + K_s \bar{B}_{\mathrm{dia}}^{\mathrm{ul}}\right)$, and the complexity of recovering the received signal with the space domain UL receiver is given by $\mathcal{O}\left(\sum_{s=1}^{S} K_s M\right)$.

For the total computational complexity, we assume that the DL precoders and UL receivers at each satellite for UTs are designed once and implemented within 100 milliseconds for 4 resource blocks, and there are one subframe per millisecond, 14 OFDM symbols per subframe, and 12 subcarriers per resource block [36]. At each satellite side, the total computational complexities of the beam structured DL precoder and UL receiver are given by $\mathcal{O}\left(\left(\sum_{s=1}^{S} K_s \left(\frac{K}{2}(\bar{B}_{\mathrm{dia}}^{\mathrm{dl}})^2 + (\bar{B}_{\mathrm{dia}}^{\mathrm{dl}})^3 + (\bar{B}_{\mathrm{dia}}^{\mathrm{dl}})^2\right)\right) + F\left(\sum_{s=1}^{S} \bar{M} \log_2 \bar{M}_{\mathrm{y}} + M_{\mathrm{y}} \bar{M}_{\mathrm{x}} \log_2 \bar{M}_{\mathrm{x}} + K_s \bar{B}_{\mathrm{dia}}^{\mathrm{dl}}\right)\right)$ and $\mathcal{O}\left(\left(\sum_{s=1}^{S} K_s \left(\frac{K_s}{2}(\bar{B}_{\mathrm{dia}}^{\mathrm{ul}})^2 + (\bar{B}_{\mathrm{dia}}^{\mathrm{ul}})^3 + (\bar{B}_{\mathrm{dia}}^{\mathrm{ul}})^2\right)\right) + F\left(\sum_{s=1}^{S} \bar{M} \log_2 \bar{M}_{\mathrm{x}} + M_{\mathrm{x}} \bar{M}_{\mathrm{y}} \log_2 \bar{M}_{\mathrm{y}} + K_s \bar{B}_{\mathrm{dia}}^{\mathrm{ul}}\right)\right)$, respectively, where $F = 12 \times 4 \times 100 \times 14$. For comparison, the total computational complexity of both space domain DL precoder and UL receiver are $\mathcal{O}\left(\left(\sum_{s=1}^{S} M^3 + K_s M^2 + \frac{K}{2} M^2\right) + F\left(\sum_{s=1}^{S} K_s M\right)\right)$ and $\mathcal{O}\left(\left(\sum_{s=1}^{S} M^3 + \frac{3}{2} K_s M^2\right) + F\left(\sum_{s=1}^{S} K_s M\right)\right)$, respectively.

## VI. SIMULATION RESULTS

In this section, we present numerical results to evaluate the effectiveness of the proposed decoupled designs for massive MIMO multi-satellite systems. Simulation parameters are outlined in Table II. Let $R_{\mathrm{e}}$ represent Earth's radius and $H$ the orbital altitude. Then, the satellite orbit radius is $R_{\mathrm{sat}} = H + R_{\mathrm{e}}$. The elevation angle $\alpha_{k,s}$ for UT $k$ and satellite $s$ follows $\alpha_{k,s} = \cos^{-1}\left(\frac{R_{\mathrm{sat}} \sin(\vartheta_{k,s})}{R_{\mathrm{e}}}\right)$ [12]. The nadir angle for satellite $s$ is denoted as $\vartheta_s$, which is set as $\vartheta_s = 45°, \forall s \in \mathcal{S}$. Then, the set of UT served by satellite $s$ is expressed as $\mathcal{K}_s = \{k | \vartheta_{k,s} \leq \vartheta_s, k \in \mathcal{K}\}$, and the set of satellites visible to UT $k$ is expressed as $\mathcal{S}_k = \{s | \vartheta_{k,s} \leq \vartheta_s, s \in \mathcal{S}\}$. As shown in Fig. 5, satellites are located at the latitude/longitude pairs $(3°, 17°)$, $(3°, 23°)$, $(-3°, 17°)$, $(-3°, 23°)$, and ground UTs are uniformly distributed in the satellites' coverage area. The distance between UT $k$ and satellite $s$ is $D_{k,s} = -\sin(\alpha_{k,s}) R_{\mathrm{e}} + \sqrt{2R_{\mathrm{e}}H + (\sin(\alpha_{k,s}) R_{\mathrm{e}})^2 + H^2}$ [37]. We denote the antenna gain of satellite $s$ for UT $k$ as $G_{\mathrm{sat}}(\varphi_{k,s}^{\mathrm{x}}, \varphi_{k,s}^{\mathrm{y}})$ with the consideration of radiation power pattern [38] and denote $G_{\mathrm{ut}}$ as the UT antenna gain. The steering vectors $\mathbf{g}_{k,s}$ and $\mathbf{d}_{k,s,0}$ can be generated by using $(\varphi_{k,s}^{\mathrm{x}}, \varphi_{k,s}^{\mathrm{y}})$, and $(\theta_{k,s,0}^{\mathrm{x}'}, \theta_{k,s,0}^{\mathrm{y}'})$, respectively. We assume that the LoS angles at UTs' sides satisfy $\sin(\theta_{k,s,0}^{\mathrm{x}'}) \sin(\theta_{k,s,0}^{\mathrm{y}'}) = \sin(\alpha_{k,s})$, ensuring that UT arrays are placed horizontally. To generate $\breve{\mathbf{d}}_{k,s}^{\mathrm{p}}$, the complex channel gains and the remaining angles at the UT side related to the NLoS path $\{\theta_{k,s,\ell}^{\mathrm{x}'}, \theta_{k,s,\ell}^{\mathrm{y}'}\}_{\ell=1}^{L_{k,s}-1}$ are simulated by using the exponential power delay profile and the wrapped Gaussian distribution, respectively [21], [37]. Moreover, the values of shadow fading and Rician factors for different elevation angles are adopted from [37] for urban scenario. Then, the channel correlation matrix at the UT side, $\mathbf{R}_{k,s}^{\mathrm{ut,p}}$, can be calculated through sample average operations by using the results of $\mathbf{d}_{k,s,0}^{\mathrm{p}}$ and $\breve{\mathbf{d}}_{k,s}^{\mathrm{p}}$ over multiple channel samples.

In the following, we set $|\mathcal{B}_{k,s}^{\mathrm{dia,dl}}| = |\mathcal{B}_{k,s}^{\mathrm{dia,ul}}| = B_{k,s}^{\mathrm{dia}}$ and $|\bar{\mathcal{B}}_{k,s}^{\mathrm{dia,dl}}| = |\bar{\mathcal{B}}_{k,s}^{\mathrm{dia,ul}}| = \bar{B}_{k,s}^{\mathrm{dia}}$. We first evaluate the accuracy of the proposed diffusion vector for different fine factors and numbers of beams by using the normalized mean-squared error (NMSE), which is defined as

$$\mathrm{NMSE} = \frac{\sum_{s=1}^{S} \sum_{k=1}^{K} \left\| \mathbf{g}_{k,s}^{\mathrm{p}} - \mathbf{V}_{k,s}^{\mathrm{p}} \mathbf{\Omega}_{k,s}^{\mathrm{p}} \breve{\mathbf{g}}_{k,s}^{\mathrm{p}} \right\|^2}{\sum_{s=1}^{S} \sum_{k=1}^{K} \left\| \mathbf{g}_{k,s}^{\mathrm{p}} \right\|^2}. \tag{73}$$

In Fig. 6, $B_{k,s}^{\mathrm{squ}} = |\mathcal{B}_{k,s}^{\mathrm{squ,dl}}| = |\mathcal{B}_{k,s}^{\mathrm{squ,ul}}|$, and $\mathcal{B}_{k,s}^{\mathrm{squ,p}}$ is defined as

$$\mathcal{B}_{k,s}^{\mathrm{squ,p}} = \left\{ (b_{\mathrm{x}}^{\mathrm{p}} - 1) \breve{M}_{\mathrm{y}}^{\mathrm{p}} + b_{\mathrm{y}}^{\mathrm{p}} | b_{\mathrm{x}}^{\mathrm{p}} = ((\delta_{\mathrm{x}} - 1 + g_{\max}^{\mathrm{x,p}}))_{\breve{M}_{\mathrm{x}}^{\mathrm{p}}} + 1, \right.$$
$$\left. b_{\mathrm{y}}^{\mathrm{p}} = ((\delta_{\mathrm{y}} - 1 + g_{\max}^{\mathrm{y,p}}))_{\breve{M}_{\mathrm{y}}^{\mathrm{p}}} + 1, \delta_{\mathrm{x}} \in \mathbb{Z}, \delta_{\mathrm{y}} \in \mathbb{Z}, \right.$$

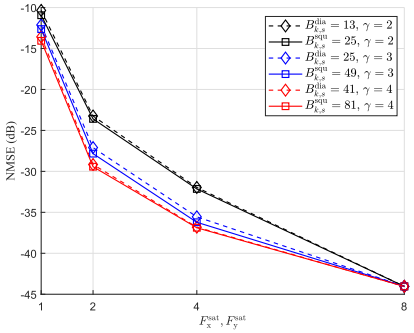
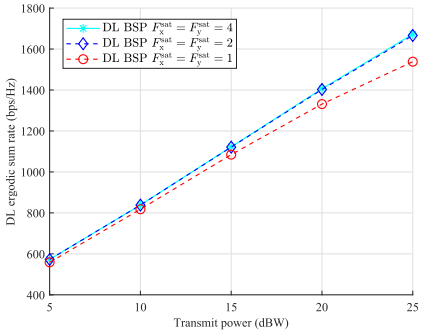
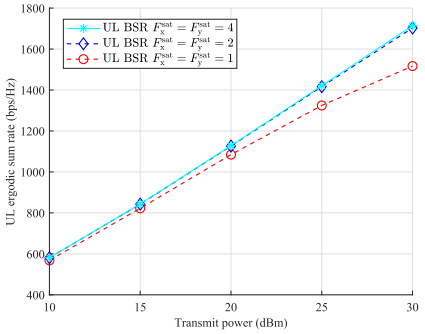



Fig. 6. NMSE performance versus the fine factor for different numbers of beams.

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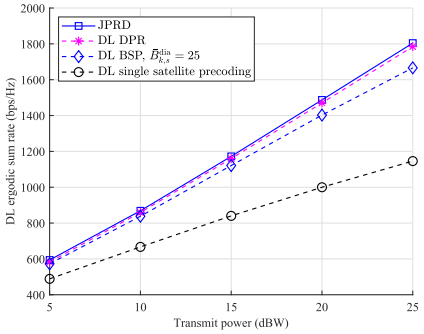



(a) DL

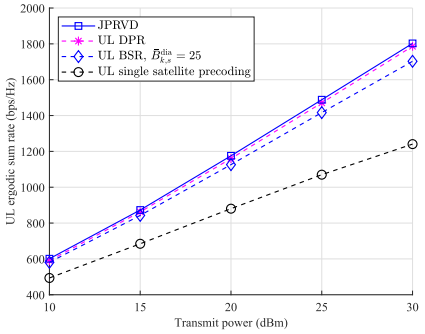



(b) UL

Fig. 7. Ergodic sum rate performance of the proposed DL BSP and UL BSR ($\bar{B}_{k,s}^{\text{dia}} = 25$) for different fine factors.

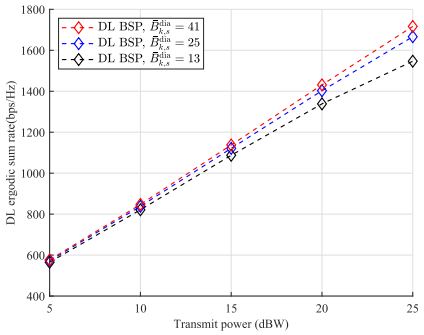



(a) DL

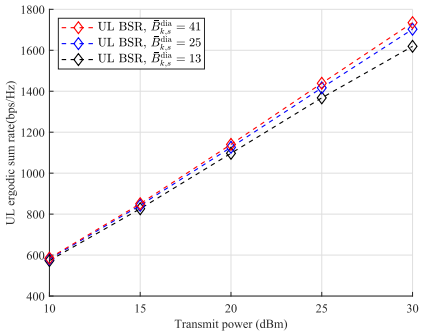



(b) UL

Fig. 8. Ergodic sum rate performance of the proposed DL/UL DPR, DL BSP, and UL BSR compared to the single satellite precoding strategy and the ergodic sum rate based optimal designs.




(a) DL




(b) UL

Fig. 9. Ergodic sum rate performance of the proposed DL BSP and UL BSR for different numbers of beams.

$$g_{\max}^{\text{x,p}} = \arg\max_{g_{\text{x}}^{\text{p}}} [\tilde{\mathbf{g}}_{k,s}^{\text{p,x}}]_{g_{\text{x}}^{\text{p}}}, g_{\max}^{\text{y,p}} = \arg\max_{g_{\text{y}}^{\text{p}}} [\tilde{\mathbf{g}}_{k,s}^{\text{p,y}}]_{g_{\text{y}}^{\text{p}}},$$
$$|\delta_{\text{x}}| \le \gamma, |\delta_{\text{y}}| \le \gamma\big\}, \tag{74}$$

which is similar to $\mathcal{B}_{k,s}^{\text{dia,p}}$ as defined in (17). $\mathcal{B}_{k,s}^{\text{squ,p}}$ represents the beam index set corresponding to the beams arranged in a square shape, with the center located at a distance of $\gamma$ from each of its side. Comparing to beam set $\mathcal{B}_{k,s}^{\text{squ,p}}$, which contains $(2\gamma+1)^2$ beams, the beam set $\mathcal{B}_{k,s}^{\text{dia,p}}$ effectively removes the four corners of the square-shaped beam index set $\mathcal{B}_{k,s}^{\text{squ,p}}$, and involves only $\frac{(2\gamma+1)^2+1}{2}$ beams. From the figure, we can observe that the number of selected beams can be significantly reduced by using a diamond-shaped beam index without sacrificing much accuracy of the proposed BBCM.

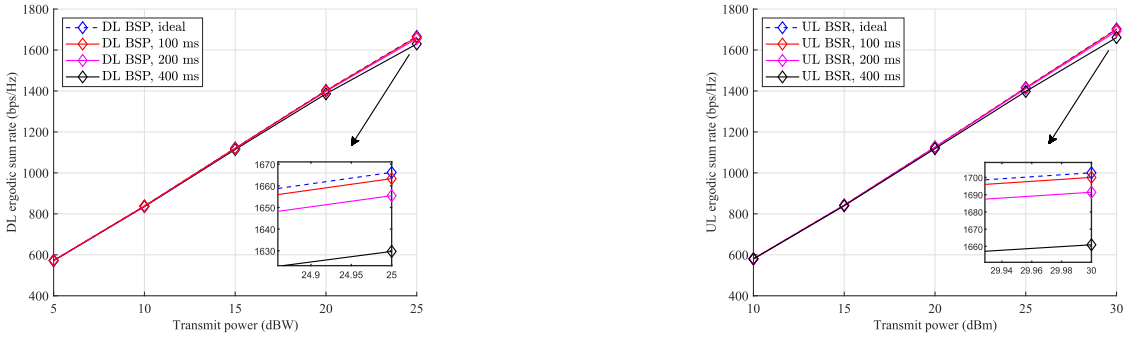
We evaluate the system performance by using the DL and UL ergodic sum rates with Monte-Carlo simulations, which are given by $\mathcal{R}_{\text{sum}}^{\text{dl}} = \sum_{s=1}^{S} \sum_{k\in\mathcal{K}_s} \mathcal{R}_{k,s}^{\text{dl}}$, and $\mathcal{R}_{\text{sum}}^{\text{ul}} = \sum_{s=1}^{S} \sum_{k\in\mathcal{K}_s} \mathcal{R}_{k,s}^{\text{ul}}$, respectively, where

$$\mathcal{R}_{k,s}^{\text{dl}} = \mathbb{E}\left\{\log\left(1+(v_{k,s}^{\text{dl}})^{-1} p_{k,s}^{\text{dl}} \left|(\mathbf{g}_{k,s}^{\text{dl}})^H \mathbf{w}_{k,s}^{\text{dl}}\right|^2 \left|(\mathbf{d}_{k,s}^{\text{dl}})^H \mathbf{c}_{k,s}^{\text{dl}}\right|^2\right)\right\}, \tag{75}$$

and

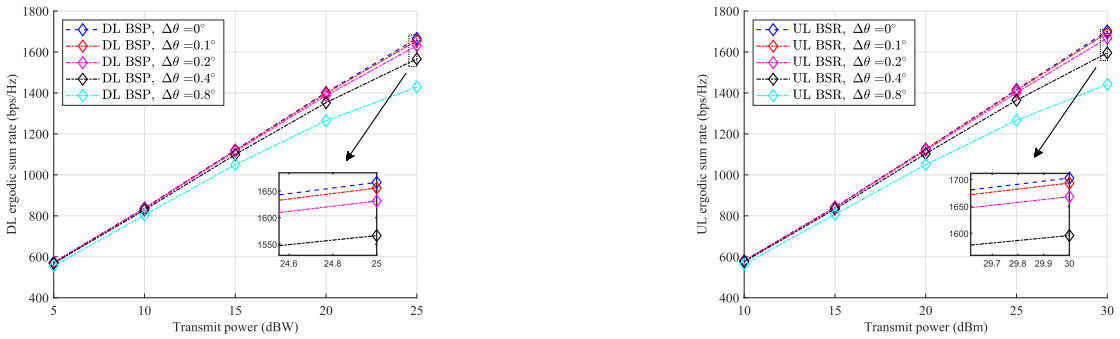
$$\mathcal{R}_{k,s}^{\text{ul}} = \mathbb{E}\left\{\log\left(1+(v_{k,s}^{\text{ul}})^{-1} p_{k,s}^{\text{ul}} \left|(\mathbf{g}_{k,s}^{\text{ul}})^H \mathbf{c}_{k,s}^{\text{ul}}\right|^2 \left|(\mathbf{d}_{k,s}^{\text{ul}})^H \mathbf{w}_{k,s}^{\text{ul}}\right|^2\right)\right\}, \tag{76}$$

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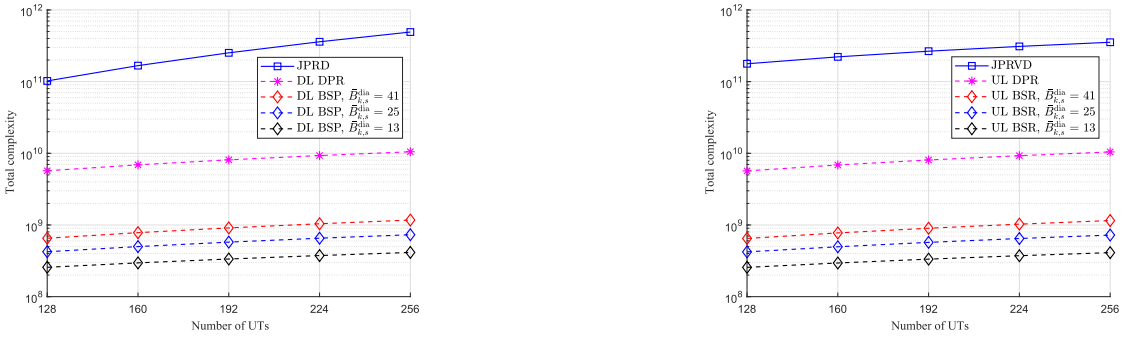

(a) DL (b) UL

Fig. 10. Ergodic sum rate performance of proposed DL BSP and UL BSR ($\bar{B}_{k,s}^{\mathrm{dia}} = 25$) for different update periods.




(a) DL (b) UL

Fig. 11. Ergodic sum rate performance of proposed DL BSP and UL BSR ($\bar{B}_{k,s}^{\mathrm{dia}} = 25$) for different satellite array inclination angles.




(a) DL (b) UL

Fig. 12. Total complexity of proposed DL/UL DPR, DL BSP, and UL BSR compared to their optimal counterparts (JPRD/JPRVD) for different numbers of beams at satellite sides.

with $v_{k,s}^{\mathrm{dl}} = \sum_{i\in\mathcal{K}_s\setminus\{k\}} p_{i,s}^{\mathrm{dl}} |(\mathbf{g}_{k,s}^{\mathrm{dl}})^H \mathbf{w}_{i,s}^{\mathrm{dl}}|^2 |(\mathbf{d}_{k,s}^{\mathrm{dl}})^H \mathbf{c}_{k,s}^{\mathrm{dl}}|^2 + (\mathbf{c}_{k,s}^{\mathrm{dl}})^H \mathbf{R}_{k,s}^{\mathrm{dl}} \mathbf{c}_{k,s}^{\mathrm{dl}}$, and $v_{k,s}^{\mathrm{ul}} = (\mathbf{c}_{k,s}^{\mathrm{ul}})^H \mathbf{R}_s^{\mathrm{ul}} \mathbf{c}_{k,s}^{\mathrm{ul}} + \sum_{i\in\mathcal{K}_s\setminus\{k\}} p_{i,s}^{\mathrm{ul}} |(\mathbf{g}_{i,s}^{\mathrm{ul}})^H \mathbf{c}_{k,s}^{\mathrm{ul}}|^2 |(\mathbf{d}_{i,s}^{\mathrm{ul}})^H \mathbf{w}_{i,s}^{\mathrm{ul}}|^2$.

Fig. 7 presents the ergodic sum rate performance of the proposed DL BSP and UL BSR for different fine factors. It can be observed that when the fines factors are set as one, the beam structured designs exhibit the worst performance, while configurations with $F_{\mathrm{x}}^{\mathrm{sat}} = F_{\mathrm{y}}^{\mathrm{sat}} = 2$ and $F_{\mathrm{x}}^{\mathrm{sat}} = F_{\mathrm{y}}^{\mathrm{sat}} = 4$ show a significant performance improvement. The ergodic sum rate difference between the two configurations of higher fine factors 2 and 4 is negligible. Although increasing the fine factors may cause stronger correlation among the columns of the beam matrix, such correlation does not compromise the effectiveness of the proposed decoupled beam structured designs.

Fig. 8 presents the ergodic sum rate performance of the proposed DL/UL DPR, DL BSP and UL BSR with $F_{\mathrm{x}}^{\mathrm{sat}} = F_{\mathrm{y}}^{\mathrm{sat}} = 2$. For comparison, we also provide the performance of single satellite precoding in [20] and [21], and the sum rate maximization approaches, i.e., joint precoder and receiver design (JPRD) [31] for DL and joint precoding and receiving vector design (JPRVD) [32] for UL. The results demonstrate that for both DL and UL, the proposed decoupled space domain designs with only local sCSI achieve negligible performance gaps compared to their optimal counterparts (JPRD/JPRVD), which require extensive inter-satellite sCSI sharing. Moreover, the beam structured designs achieve nearly 95% of the optimal sum rates. In contrast, the single satellite precoding ignoring inter-satellite interference exhibits the worst performance.

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Fig. 9 depicts the ergodic sum rate performance of the proposed DL BSP and UL BSR with different numbers of beams. The performance for different $B_{k,s}^{\mathrm{dia}}$ is achieved by the beam structured DL precoder and UL receiver with the corresponding beam index set $\mathcal{B}_{k,s}^{\mathrm{dia}}$. From the figure, we can observe that for both DL and UL, the sum rate performance can be improved with more selected beams.

Fig. 10 shows the sum rate performance of the proposed DL BSP and UL BSR with different update periods. The term "update period" refers to the case that the BSP/BSR is updated every $T_{\mathrm{upd}} \in \{100, 200, 400\}$ ms. Specifically, once the BSP/BSR is computed at time $t$, it is applied during the entire interval $[t, t+T_{\mathrm{upd}})$ and kept fixed until the next update. The term "ideal case" refers to the case that the BSP/BSR is updated per OFDM symbol. It can be observed that for both DL and UL, the sum rate performance degrade of the DL BSP and UL BSR is negligible with the update period of 100 ms.

Fig. 11 presents the ergodic sum rate performance of the proposed DL BSP and UL BSR for different values of $\Delta\theta$, which corresponds to the satellite array inclination angles relative to the x- and y- axes. We can observe that the proposed beam structured designs perform effectively for $\Delta\theta \leq 0.1^{\circ}$.

Fig. 12 presents the total complexities of our proposed space domain decoupled designs and beam structured designs as calculated in Sec. V, under different numbers of UTs and beams compared to their optimal counterparts (JPRD/JPRVD). From the figure, we can observe that the optimal designs have the highest complexity due to numerous iterations and sample average operations. The beam structured designs under three different beam index sets show low complexity compared to the space domain designs due to the utilization of the satellite channel sparsity.

## VII. Conclusion

We have investigated the decoupled designs of precoders and receivers for both DL and UL in massive MIMO multi-satellite systems. We first established the beam based satellite channel model with sampled steering vectors. Then, we proposed the DPR design for both DL and UL, where the precoders and receivers can be designed in a decoupled manner with only local sCSI, at each transmitter and receiver, respectively. Moreover, we transformed the design of DL precoder and UL receiver in the space domain into that of lower-dimensional vectors in the beam domain with local sCSI, and the resulting space domain precoder and receiver are beam structured. In addition, we proposed a low-complexity design and implementation for the decoupled beam structured DL precoder and UL receiver to reduce the computational complexity. Simulation results showed that the proposed space domain decoupled designs can significantly reduce complexity compared to their optimal counterparts (JPRD/JPRVD) with negligible performance loss. By utilizing the proposed beam structured designs, the complexity can be further reduced by an order of magnitude without sacrificing much performance. For large LEO constellations, the proposed decoupled designs and beam structured designs can significantly simplify the on-board processing while avoiding extensive inter-satellite information exchange. Future work can investigate the sCSI acquisition in the massive MIMO-OFDM multi-LEO satellite system with full consideration of time and frequency asynchronizations.

## Appendix A
## Proof of Theorem 1

From (28) and (30), we can observe that the coupled terms of the DL precoder and receiver designs are $\sum_{i\in\mathcal{K}}\sum_{j\in\mathcal{S}_i}(\mathbf{c}_{i,j}^{\mathrm{dl}})^H\mathbf{R}_{i,s}^{\mathrm{ut,dl}}\mathbf{c}_{i,j}^{\mathrm{dl}}\mathbf{g}_{i,s}^{\mathrm{dl}}(\mathbf{g}_{i,s}^{\mathrm{dl}})^H$ in $\bar{\mathbf{D}}_{k,s}^{\mathrm{dl}}$ and $\sum_{j=1}^{S}\sum_{i\in\mathcal{K}_j}p_{i,j}^{\mathrm{dl}}|(\mathbf{g}_{k,j}^{\mathrm{dl}})^H\mathbf{w}_{i,j}^{\mathrm{dl}}|^2\mathbf{R}_{k,j}^{\mathrm{ut,dl}}$ in $\bar{\mathbf{E}}_{k,s}^{\mathrm{dl}}$. For the case of $\rho_{k,s}^{\mathrm{dl}} \to \infty$, $\forall k,s$, define $a_{i,s}^{\mathrm{dl}} = \sum_{j=1}^{S_i}(\mathbf{c}_{i,j}^{\mathrm{dl}})^H\mathbf{R}_{i,s}^{\mathrm{ut,dl}}\mathbf{c}_{i,j}^{\mathrm{dl}}$, $\mathbf{A}_s^{\mathrm{dl}} = \mathrm{diag}(a_{1,s}^{\mathrm{dl}},\ldots,a_{K,s}^{\mathrm{dl}})$, $\mathbf{G}_s^{\mathrm{dl}} = [\mathbf{g}_{1,s}^{\mathrm{dl}},\ldots,\mathbf{g}_{K,s}^{\mathrm{dl}}]$, $\mathbf{W}_s^{\mathrm{dl}} = [\mathbf{w}_{1,s}^{\mathrm{dl}},\ldots,\mathbf{w}_{K,s}^{\mathrm{dl}}]$, and $\mathbf{B}_s^{\mathrm{dl}} = \mathrm{diag}(\eta_{1,s}^{\mathrm{dl}},\ldots,\eta_{K,s}^{\mathrm{dl}})$. The DL precoders in (28) can be rewritten as

$$\mathbf{W}_s^{\mathrm{dl}} = \mathbf{G}_s^{\mathrm{dl}}\big((\mathbf{G}_s^{\mathrm{dl}})^H\mathbf{G}_s^{\mathrm{dl}}\big)^{-1}\big(\mathbf{A}_s^{\mathrm{dl}}\big)^{-1}\big(\mathbf{B}_s^{\mathrm{dl}}\big)^{-1}. \tag{77}$$

The terms $\{a_{i,s}^{\mathrm{dl}}\}_{\forall i}$ depend on the knowledge of DL receiver $\{\mathbf{c}_{i,j}^{\mathrm{dl}}\}_{\forall i,j}$ and appear as diagonal elements in the matrix $\mathbf{A}_s^{\mathrm{dl}}$. Consequently, the power normalization operation can be directly applied to each column of $\mathbf{G}_s^{\mathrm{dl}}\big((\mathbf{G}_s^{\mathrm{dl}})^H\mathbf{G}_s^{\mathrm{dl}}\big)^{-1}$. Then, $\mathbf{W}_s^{\mathrm{dl}}$ in (77) can be rewritten as

$$\mathbf{W}_s^{\mathrm{dl}} = \mathbf{G}_s^{\mathrm{dl}}\big((\mathbf{G}_s^{\mathrm{dl}})^H\mathbf{G}_s^{\mathrm{dl}}\big)^{-1}\big(\mathbf{C}_s^{\mathrm{dl}}\big)^{-1}, \tag{78}$$

where $\mathbf{C}_s^{\mathrm{dl}} = \mathrm{diag}(c_{1,s}^{\mathrm{dl}},\ldots,c_{K,s}^{\mathrm{dl}})$ and $c_{k,s}^{\mathrm{dl}}$ is the corresponding power normalization factor for the $k$-th column. Notice that the precoder in (78) exhibits a zero-forcing precoder structure and satisfies that $|(\mathbf{g}_{k,s}^{\mathrm{dl}})^H\mathbf{w}_{i,s}^{\mathrm{dl}}| = 0$, $\forall k,i,s,k \neq i$. Then, $\mathbf{c}_{k,s}^{\mathrm{asinr,dl}}$ in (30) can be rewritten as

$$\mathbf{c}_{k,s}^{\mathrm{asinr,dl,de}} = \boldsymbol{\xi}_{\max}\left(\left(\sum_{j\in\mathcal{S}_k} g_{k,j}^{\mathrm{dl}}\mathbf{R}_{k,j}^{\mathrm{ut,dl}} + (\sigma_k^{\mathrm{dl}})^2\mathbf{I}\right)^{-1}\mathbf{R}_{k,s}^{\mathrm{ut,dl}}\right), \tag{79}$$

where $g_{k,j}^{\mathrm{dl}} = p_{k,j}^{\mathrm{dl}}(\mathbf{g}_{k,j}^{\mathrm{dl}})^H\mathbf{G}_j^{\mathrm{dl}}((\mathbf{G}_j^{\mathrm{dl}})^H\mathbf{G}_j^{\mathrm{dl}})^{-2}(\mathbf{G}_j^{\mathrm{dl}})^H\mathbf{g}_{k,j}^{\mathrm{dl}} \overset{(a)}{=} p_{k,j}^{\mathrm{dl}}$, and (a) follows from $((\mathbf{G}_j^{\mathrm{dl}})^H\mathbf{G}_j^{\mathrm{dl}})^{-1}(\mathbf{G}_j^{\mathrm{dl}})^H = (\mathbf{G}_j^{\mathrm{dl}})^+$, $(\mathbf{G}_j^{\mathrm{dl}})^+\mathbf{G}_j^{\mathrm{dl}} = \mathbf{I}_K$, and $(\mathbf{g}_{k,j}^{\mathrm{dl}})^H\mathbf{g}_{k,j}^{\mathrm{dl}} = 1, \forall k,j$. Then, $\mathbf{c}_{k,s}^{\mathrm{asinr,dl,de}}$ can be rewritten as

$$\mathbf{c}_{k,s}^{\mathrm{asinr,dl,de}} = \boldsymbol{\xi}_{\max}\left(\left(\sum_{j\in\mathcal{S}_k}\rho_{k,j}^{\mathrm{dl}}\mathbf{R}_{k,j}^{\mathrm{ut,dl}} + \mathbf{I}\right)^{-1}\mathbf{R}_{k,s}^{\mathrm{ut,dl}}\right). \tag{80}$$

Next, we derive the decoupled solution of the DL precoder. First, $\mathbf{c}_{k,s}^{\mathrm{asinr,dl,de}}$ in (80) is equivalent to

$$\mathbf{c}_{k,s}^{\mathrm{asinr,dl,de}} = \boldsymbol{\xi}_{\max}\left(\left(\sum_{j\in\mathcal{S}_k\setminus\{s\}}\rho_{k,j}^{\mathrm{dl}}\mathbf{R}_{k,j}^{\mathrm{ut,dl}} + \mathbf{I}\right)^{-1}\mathbf{R}_{k,s}^{\mathrm{ut,dl}}\right). \tag{81}$$

By utilizing $\mathbf{R}_{k,s}^{\mathrm{ut,dl}}\mathbf{R}_{k,j}^{\mathrm{ut,dl}} = \mathbf{0}$, we can obtain that $\left(\sum_{j\in\mathcal{S}_k\setminus\{s\}}\rho_{k,j}^{\mathrm{dl}}\mathbf{R}_{k,j}^{\mathrm{ut,dl}} + \mathbf{I}\right)\mathbf{R}_{k,s}^{\mathrm{ut,dl}} = \mathbf{R}_{k,s}^{\mathrm{ut,dl}}$, then the DL decoupled receiver design in (81) can be rewritten as

$$\mathbf{c}_{k,s}^{\mathrm{dl}} = \boldsymbol{\xi}_{\max}\big(\mathbf{R}_{k,s}^{\mathrm{ut,dl}}\big). \tag{82}$$

When $\mathbf{R}_{k,s}^{\mathrm{ut,dl}}\mathbf{R}_{k,j}^{\mathrm{ut,dl}} = \mathbf{0}$, $\forall k,s,j,s \neq j$, we have that $(\mathbf{c}_{k,s}^{\mathrm{dl}})^H\mathbf{R}_{k,j}^{\mathrm{ut,dl}}\mathbf{c}_{k,s}^{\mathrm{dl}} = 0$, for $j \neq s$, and $(\mathbf{c}_{k,s}^{\mathrm{dl}})^H\mathbf{R}_{k,s}^{\mathrm{ut,dl}}\mathbf{c}_{k,s}^{\mathrm{dl}} =$

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$\zeta_{\max}(\mathbf{R}_{k,s}^{\mathrm{ut,dl}})$, where $\zeta_{\max}(\mathbf{A})$ is the maximum eigenvalue of $\mathbf{A}$. Then $\bar{\mathbf{D}}_{k,s}^{\mathrm{dl}}$ in (28) can be rewritten as

$$\bar{\mathbf{D}}_{k,s}^{\mathrm{dl}} = \frac{(\sigma_k^{\mathrm{dl}})^2}{p_{k,s}^{\mathrm{dl}}}\mathbf{I} + \sum_{i\in\mathcal{K}} \xi_{i,s}^{\mathrm{dl}} \mathbf{g}_{i,s}^{\mathrm{dl}} (\mathbf{g}_{i,s}^{\mathrm{dl}})^H \triangleq \check{\mathbf{D}}_{k,s}^{\mathrm{dl}}. \tag{83}$$

Then, we can obtain the decoupled solution of the DL precoders as

$$\mathbf{w}_{k,s}^{\mathrm{aslnr,dl,de}} = \frac{1}{\eta_{k,s}^{\mathrm{aslnr,dl,de}}} \big(\check{\mathbf{D}}_{k,s}^{\mathrm{dl}}\big)^{-1} \mathbf{g}_{k,s}^{\mathrm{dl}}, \forall k, s, \tag{84}$$

where $\eta_{k,s}^{\mathrm{aslnr,dl,de}}$ is the power normalization factor.

For the case of $\rho_{k,s}^{\mathrm{dl}} \to 0$, the DL precoder in (28) and receiver in (30) become

$$\mathbf{w}_{k,s}^{\mathrm{dl}} = \mathbf{g}_{k,s}^{\mathrm{dl}} \tag{85}$$

and

$$\mathbf{c}_{k,s}^{\mathrm{dl}} = \boldsymbol{\xi}_{\max}\big(\mathbf{R}_{k,s}^{\mathrm{ut,dl}}\big), \tag{86}$$

which are special cases of (84) and (80), respectively.

For the case of $M \to \infty$, we have that $(\mathbf{g}_{i,s}^{\mathrm{dl}})^H \mathbf{g}_{k,s}^{\mathrm{dl}} = 0, \forall i, k, i \neq k$. We first rewrite the DL precoder in (28) as

$$\begin{aligned}\mathbf{w}_{k,s}^{\mathrm{dl}} &= \frac{1}{\eta_{k,s}^{\mathrm{dl}}} \left(\sum_{i\in\mathcal{K}} a_{i,s}^{\mathrm{dl}} \mathbf{g}_{i,s}^{\mathrm{dl}} (\mathbf{g}_{i,s}^{\mathrm{dl}})^H + \frac{1}{\rho_{k,s}^{\mathrm{dl}}}\mathbf{I}\right)^{-1} \mathbf{g}_{k,s}^{\mathrm{dl}} \\ &\overset{(a)}{=} \frac{1}{\eta_{k,s}^{\mathrm{dl}}} \frac{(\boldsymbol{\Phi}_{k,s}^{\mathrm{dl}})^{-1}\mathbf{g}_{k,s}^{\mathrm{dl}}}{1 + a_{k,s}^{\mathrm{dl}} (\mathbf{g}_{k,s}^{\mathrm{dl}})^H (\boldsymbol{\Phi}_{k,s}^{\mathrm{dl}})^{-1} \mathbf{g}_{k,s}^{\mathrm{dl}}} \overset{(b)}{=} \mathbf{g}_{k,s}^{\mathrm{dl}},\end{aligned} \tag{87}$$

where $\boldsymbol{\Phi}_{k,s}^{\mathrm{dl}} = \sum_{i\in\mathcal{K}\setminus\{k\}} a_{i,s}^{\mathrm{dl}} \mathbf{g}_{i,s}^{\mathrm{dl}} (\mathbf{g}_{i,s}^{\mathrm{dl}})^H + \frac{1}{\rho_{k,s}^{\mathrm{dl}}}\mathbf{I}$, (a) comes from the matrix inverse lemma, and (b) comes from $\boldsymbol{\Phi}_{k,s}^{\mathrm{dl}} \mathbf{g}_{k,s}^{\mathrm{dl}} = \frac{1}{\rho_{k,s}^{\mathrm{dl}}} \mathbf{g}_{k,s}^{\mathrm{dl}}$ by utilizing $(\mathbf{g}_{i,s}^{\mathrm{dl}})^H \mathbf{g}_{k,s}^{\mathrm{dl}} = 0, \forall i, k, i \neq k$ and $\eta_{k,s}^{\mathrm{dl}} = \frac{\rho_{k,s}^{\mathrm{dl}}}{1 + a_{k,s}^{\mathrm{dl}} (\mathbf{g}_{k,s}^{\mathrm{dl}})^H (\boldsymbol{\Phi}_{k,s}^{\mathrm{dl}})^{-1} \mathbf{g}_{k,s}^{\mathrm{dl}}}$. Note that the precoder in (87) is a special case of (84). With the precoder in (87), we can obtain the decoupled DL receiver as given in (80).

## Appendix B
## Proof of Theorem 3

Define $\mathcal{B}_s^{\mathrm{dia,dl}} = \mathcal{B}_{1,s}^{\mathrm{dia,dl}} \cup \mathcal{B}_{2,s}^{\mathrm{dia,dl}} \cup \cdots \cup \mathcal{B}_{K_s,s}^{\mathrm{dia,dl}}$, $|\mathcal{B}_s^{\mathrm{dia,dl}}| = B_s^{\mathrm{dia,dl}}$. Then, we can rewrite $\mathbf{g}_{k,s}^{\mathrm{dl}}$ as

$$\mathbf{g}_{k,s}^{\mathrm{dl}} = \hat{\mathbf{V}}_s^{\mathrm{dl}} \hat{\mathbf{g}}_{k,s}^{\mathrm{dl}}, \tag{88}$$

where $\hat{\mathbf{V}}_s^{\mathrm{dl}} = [\mathbf{V}^{\mathrm{dl}}]_{:,\mathcal{B}_s^{\mathrm{dia,dl}}} \in \mathbb{C}^{M\times B_s^{\mathrm{dia,dl}}}$, and $\hat{\mathbf{g}}_{k,s}^{\mathrm{dl}} = [\tilde{\mathbf{g}}_{k,s}^{\mathrm{dl}}]_{\mathcal{B}_s^{\mathrm{dia,dl}}}$. Substituting $\mathbf{g}_{k,s}^{\mathrm{dl}}$ in (88) into (27) yields

$$\mathrm{ASLNR}_{k,s}(\hat{\mathbf{w}}_{k,s}^{\mathrm{dl}}) = \frac{(\mathbf{c}_{k,s}^{\mathrm{dl}})^H \mathbf{R}_{k,s}^{\mathrm{ut,dl}} \mathbf{c}_{k,s}^{\mathrm{dl}} \left|(\hat{\mathbf{g}}_{k,s}^{\mathrm{dl}})^H \hat{\mathbf{w}}_{k,s}^{\mathrm{dl}}\right|^2}{(\hat{\mathbf{w}}_{k,s}^{\mathrm{dl}})^H \hat{\mathbf{D}}_{k,s}^{\mathrm{dl}} \hat{\mathbf{w}}_{k,s}^{\mathrm{dl}} + (\sigma_k^{\mathrm{dl}})^2/p_{k,s}^{\mathrm{dl}}}, \tag{89}$$

where $\hat{\mathbf{D}}_{k,s}^{\mathrm{dl}} = \sum_{j\neq s}^{S} \sum_{i\in\mathcal{K}_j} (\mathbf{c}_{i,j}^{\mathrm{dl}})^H \mathbf{R}_{i,s}^{\mathrm{ut,dl}} \mathbf{c}_{i,j}^{\mathrm{dl}} \hat{\mathbf{g}}_{i,s}^{\mathrm{dl}} (\hat{\mathbf{g}}_{i,s}^{\mathrm{dl}})^H + \sum_{i\in\mathcal{K}_s\setminus\{k\}} (\mathbf{c}_{i,s}^{\mathrm{dl}})^H \mathbf{R}_{i,s}^{\mathrm{ut,dl}} \mathbf{c}_{i,s}^{\mathrm{dl}} \hat{\mathbf{g}}_{i,s}^{\mathrm{dl}} (\hat{\mathbf{g}}_{i,s}^{\mathrm{dl}})^H$, and $\hat{\mathbf{w}}_{k,s}^{\mathrm{dl}} = (\hat{\mathbf{V}}_s^{\mathrm{dl}})^H \mathbf{w}_{k,s}^{\mathrm{dl}}$. Denote by $\hat{\mathbf{w}}_{k,s}^{\mathrm{dl},\star}$ the optimal solution to maximize the ASLNR in (89). Then, there exists an optimal $\mathbf{w}_{k,s}^{\mathrm{dl},\star}$, which satisfies $\hat{\mathbf{w}}_{k,s}^{\mathrm{dl},\star} = (\hat{\mathbf{V}}_s^{\mathrm{dl}})^H \mathbf{w}_{k,s}^{\mathrm{dl},\star}$ [39, Lemma 1]. Let $\mathbf{b}_{k,s}^{\mathrm{dl}} \in \mathbb{R}^{B_s^{\mathrm{dia,dl}}\times 1}$ be defined as

$$[\mathbf{b}_{k,s}^{\mathrm{dl}}]_b = \begin{cases} 1, & [\mathcal{B}_s^{\mathrm{dia,dl}}]_b \in \mathcal{B}_{k,s}^{\mathrm{dia,dl}}, \\ 0, & \text{otherwise}, \end{cases} \tag{90}$$

and define $\check{\mathbf{w}}_{k,s}^{\mathrm{dl},\star} = \mathbf{b}_{k,s}^{\mathrm{dl}} \odot \hat{\mathbf{w}}_{k,s}^{\mathrm{dl},\star}$. Because $\hat{\mathbf{w}}_{k,s}^{\mathrm{dl},\star}$ denotes the optimal solution for maximizing the ASLNR in (89), then

$$\mathrm{ASLNR}_{k,s}(\check{\mathbf{w}}_{k,s}^{\mathrm{dl},\star}) \leq \mathrm{ASLNR}_{k,s}(\hat{\mathbf{w}}_{k,s}^{\mathrm{dl},\star}). \tag{91}$$

Meanwhile, when calculating ASLNR by using (89) with $\hat{\mathbf{w}}_{k,s}^{\mathrm{dl},\star}$ replaced by $\check{\mathbf{w}}_{k,s}^{\mathrm{dl},\star}$, the denominator may either keep unchanged or decrease, while the numerator stays the same, since $\mathcal{B}_{k,s}^{\mathrm{dia,dl}} \cap \mathcal{B}_{i,s}^{\mathrm{dia,dl}} = \emptyset, \forall k, i, s$. Therefore, we can obtain that

$$\mathrm{ASLNR}_{k,s}(\hat{\mathbf{w}}_{k,s}^{\mathrm{dl},\star}) \leq \mathrm{ASLNR}_{k,s}(\check{\mathbf{w}}_{k,s}^{\mathrm{dl},\star}). \tag{92}$$

Combining (91) and (92) yields

$$\mathrm{ASLNR}_{k,s}(\hat{\mathbf{w}}_{k,s}^{\mathrm{dl},\star}) = \mathrm{ASLNR}_{k,s}(\check{\mathbf{w}}_{k,s}^{\mathrm{dl},\star}). \tag{93}$$

Therefore, we can focus on the design of non-zero elements in $\hat{\mathbf{w}}_{k,s}^{\mathrm{dl}}$. Let $\boldsymbol{\Omega}_{k,s}^{\mathrm{dl}} \in \mathbb{R}^{\bar{M}\times B_{k,s}^{\mathrm{dia,dl}}}$ be the beam selection matrix, which is given by

$$[\boldsymbol{\Omega}_{k,s}^{\mathrm{dl}}]_{m,n} = \begin{cases} 1, & m = [\mathcal{B}_{k,s}^{\mathrm{dia,dl}}]_n, \\ 0, & \text{otherwise}. \end{cases} \tag{94}$$

Define $\check{\mathbf{w}}_{k,s}^{\mathrm{dl}} = (\check{\mathbf{V}}_{k,s}^{\mathrm{dl}})^H \mathbf{w}_{k,s}^{\mathrm{dl}}$, where $\check{\mathbf{V}}_{k,s}^{\mathrm{dl}} = \mathbf{V}^{\mathrm{dl}} \boldsymbol{\Omega}_{k,s}^{\mathrm{dl}}$. For given $\check{\mathbf{w}}_{k,s}^{\mathrm{dl},\star}$, the minimum-norm solution $\mathbf{w}_{k,s}^{\mathrm{dl},\star}$ is given by $\mathbf{w}_{k,s}^{\mathrm{dl},\star} = \check{\mathbf{V}}_{k,s}^{\mathrm{dl}} ((\check{\mathbf{V}}_{k,s}^{\mathrm{dl}})^H \check{\mathbf{V}}_{k,s}^{\mathrm{dl}})^{-1} \check{\mathbf{w}}_{k,s}^{\mathrm{dl},\star}$ [40, Corollary 1]. If $||\mathbf{w}_{k,s}^{\mathrm{dl},\star}||^2$ is not minimum, $\mathbf{w}_{k,s}^{\mathrm{dl},\star}$ can be scaled up and increase the value of ASLNR, which contradicts the optimality. Because $\lim_{M\to\infty} (\check{\mathbf{V}}_{k,s}^{\mathrm{dl}})^H \check{\mathbf{V}}_{k,s}^{\mathrm{dl}} = \mathbf{I}$, then we have

$$\mathbf{w}_{k,s}^{\mathrm{dl},\star} = \check{\mathbf{V}}_{k,s}^{\mathrm{dl}} \check{\mathbf{w}}_{k,s}^{\mathrm{dl},\star}. \tag{95}$$

Since $(\mathbf{w}_{k,s}^{\mathrm{dl},\star})^H \mathbf{w}_{k,s}^{\mathrm{dl},\star} = 1$, we have $(\check{\mathbf{w}}_{k,s}^{\mathrm{dl},\star})^H \check{\mathbf{w}}_{k,s}^{\mathrm{dl},\star} = 1$. The optimal solution, $\check{\mathbf{w}}_{k,s}^{\mathrm{dl},\star}, \forall k \in \mathcal{K}_s, s \in \mathcal{S}$, is given by

$$\check{\mathbf{w}}_{k,s}^{\mathrm{dl},\star} = \arg\max_{\check{\mathbf{w}}_{k,s}^{\mathrm{dl}}} \frac{(\mathbf{c}_{k,s}^{\mathrm{dl}})^H \mathbf{R}_{k,s}^{\mathrm{ut,dl}} \mathbf{c}_{k,s}^{\mathrm{dl}} \left|(\check{\mathbf{g}}_{k,s}^{\mathrm{dl}})^H \check{\mathbf{w}}_{k,s}^{\mathrm{dl}}\right|^2}{(\check{\mathbf{w}}_{k,s}^{\mathrm{dl}})^H \check{\mathbf{D}}_{k,s}^{\mathrm{dl}} \check{\mathbf{w}}_{k,s}^{\mathrm{dl}}}, \tag{96}$$

where $\check{\mathbf{g}}_{k,s}^{\mathrm{dl}} = [\tilde{\mathbf{g}}_{k,s}^{\mathrm{dl}}]_{\mathcal{B}_{k,s}^{\mathrm{dia,dl}}}$, and

$$\begin{aligned}\check{\mathbf{D}}_{k,s}^{\mathrm{dl}} = &\sum_{j\neq s}^{S} \sum_{i\in\mathcal{K}_j} (\mathbf{c}_{i,j}^{\mathrm{dl}})^H \mathbf{R}_{i,s}^{\mathrm{ut,dl}} \mathbf{c}_{i,j}^{\mathrm{dl}} (\check{\mathbf{V}}_{k,s}^{\mathrm{dl}})^H \check{\mathbf{V}}_{i,s}^{\mathrm{dl}} \check{\mathbf{g}}_{i,s}^{\mathrm{dl}} (\check{\mathbf{g}}_{i,s}^{\mathrm{dl}})^H \\ &(\check{\mathbf{V}}_{i,s}^{\mathrm{dl}})^H \check{\mathbf{V}}_{k,s}^{\mathrm{dl}} + \sum_{i\in\mathcal{K}_s\setminus\{k\}} (\mathbf{c}_{i,s}^{\mathrm{dl}})^H \mathbf{R}_{i,s}^{\mathrm{ut,dl}} \mathbf{c}_{i,s}^{\mathrm{dl}} (\check{\mathbf{V}}_{k,s}^{\mathrm{dl}})^H \check{\mathbf{V}}_{i,s}^{\mathrm{dl}} \check{\mathbf{g}}_{i,s}^{\mathrm{dl}} \\ &(\check{\mathbf{g}}_{i,s}^{\mathrm{dl}})^H (\check{\mathbf{V}}_{i,s}^{\mathrm{dl}})^H \check{\mathbf{V}}_{k,s}^{\mathrm{dl}} + \frac{(\sigma_k^{\mathrm{dl}})^2}{p_{k,s}^{\mathrm{dl}}}\mathbf{I}.\end{aligned} \tag{97}$$

## Appendix C
## Proof of Theorem 4

According to the definitions of $\bar{\mathcal{B}}_{k,s}^{\mathrm{dia,dl}}$ and $\mathcal{B}_{k,s}^{\mathrm{dia,dl}}$, it can be verified that $\mathcal{B}_{k,s}^{\mathrm{dia,dl}}$ is a special case of $\bar{\mathcal{B}}_{k,s}^{\mathrm{dia,dl}}$, then we have

$$\mathrm{ASLNR}_{k,s}^{\mathrm{dl,b}}\big(\check{\mathbf{w}}_{k,s}^{\mathrm{aslnr,dl}}, \mathcal{B}_{k,s}^{\mathrm{dia,dl}}\big) \leq \mathrm{ASLNR}_{k,s}^{\mathrm{dl,b}}\Big(\check{\mathbf{w}}_{k,s}^{\mathrm{aslnr,dl}}, \bar{\mathcal{B}}_{k,s}^{\mathrm{dia,dl}}\Big). \tag{98}$$

Let $\check{\mathbf{w}}_{k,s}^{\star,\mathrm{dl}}$ be the optimal solution with beam index set $\bar{\mathcal{B}}_{k,s}^{\mathrm{dia,dl}}$, and let $\check{\mathbf{w}}_{k,s}^{\circ,\mathrm{dl}}$ be $\check{\mathbf{w}}_{k,s}^{\circ,\mathrm{dl}} = \mathbf{S}_{k,s}^{\circ,\mathrm{dl}} \check{\mathbf{w}}_{k,s}^{\star,\mathrm{dl}}$, where $\mathbf{S}_{k,s}^{\circ,\mathrm{dl}} \in \mathbb{R}^{\bar{B}_{k,s}^{\mathrm{dia,dl}}\times \bar{B}_{k,s}^{\mathrm{dia,dl}}}$ is defined as

$$[\mathbf{S}_{k,s}^{\circ,\mathrm{dl}}]_{q,q} = \begin{cases} 1, & [\bar{\mathcal{B}}_{k,s}^{\mathrm{dia,dl}}]_q \in \mathcal{B}_{k,s}^{\mathrm{dia,dl}}, \\ 0, & \text{otherwise}. \end{cases} \tag{99}$$

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Even though $\|\check{\mathbf{w}}_{k,s}^{\star,\mathrm{dl}}\|^2=1\geq\|\check{\mathbf{w}}_{k,s}^{\circ,\mathrm{dl}}\|^2$, $\check{\mathbf{w}}_{k,s}^{\circ,\mathrm{dl}}$ can be enlarged to satisfy $\|\check{\mathbf{w}}_{k,s}^{\circ,\mathrm{dl}}\|^2=1$ without affecting the ASLNR. When calculating ASLNR with (43), for the numerator, we have

$$\left|(\check{\mathbf{w}}_{k,s}^{\star,\mathrm{dl}})^H\mathbf{Q}_{k,k,s}^{\mathrm{dl}}[\tilde{\mathbf{g}}_{k,s}^{\mathrm{dl}}]_{\mathcal{B}_{k,s}^{\mathrm{dia,dl}}}\right|^2=\left|(\check{\mathbf{w}}_{k,s}^{\circ,\mathrm{dl}})^H\mathbf{Q}_{k,k,s}^{\mathrm{dl}}[\tilde{\mathbf{g}}_{k,s}^{\mathrm{dl}}]_{\mathcal{B}_{k,s}^{\mathrm{dia,dl}}}\right|^2, \tag{100}$$

where $\mathbf{Q}_{k,i,s}^{\mathrm{dl}}=[\mathbf{I}_{\bar{M}}]_{\bar{\mathcal{B}}_{k,s}^{\mathrm{dia,dl}},\mathcal{B}_{i,s}^{\mathrm{dia,dl}}}$. For the denominator of (43), it can be verified that when the elements $\{\bar{b}_{k,s,m}\}_{m=1}^{\bar{B}_{k,s}^{\mathrm{dia,dl}}-B_{k,s}^{\mathrm{dia,dl}}}$ in the newly selected beam index set $\bar{\mathcal{B}}_{k,s}^{\mathrm{dia,dl}}\setminus\mathcal{B}_{k,s}^{\mathrm{dia,dl}}$ satisfy

$$\bar{b}_{k,s,m}\notin\bigcup_{i\in\mathcal{K}_j,j\in\mathcal{S},\mathcal{B}_{i,s}^{\mathrm{dia,dl}}\cap\mathcal{B}_{k,s}^{\mathrm{dia,dl}}\neq\emptyset}\mathcal{B}_{i,s}^{\mathrm{dia,dl}},$$
$$\forall s\in\mathcal{S},k\in\mathcal{K}_s,m=1,2,\ldots,\bar{B}_{k,s}^{\mathrm{dia,dl}}-B_{k,s}^{\mathrm{dia,dl}}, \tag{101}$$

we have

$$(\check{\mathbf{w}}_{k,s}^{\star,\mathrm{dl}})^H\tilde{\mathbf{D}}_{k,s}^{\mathrm{dl}}\check{\mathbf{w}}_{k,s}^{\star,\mathrm{dl}}\geq(\check{\mathbf{w}}_{k,s}^{\circ,\mathrm{dl}})^H\tilde{\mathbf{D}}_{k,s}^{\mathrm{dl}}\check{\mathbf{w}}_{k,s}^{\circ,\mathrm{dl}}, \tag{102}$$

where

$$\tilde{\mathbf{D}}_{k,s}^{\mathrm{dl}}=\sum_{i\in\mathcal{K}_s}c_{i,s,s}^{\mathrm{dl}}\mathbf{Q}_{k,i,s}^{\mathrm{dl}}[\tilde{\mathbf{g}}_{i,s}^{\mathrm{dl}}]_{\mathcal{B}_{i,s}^{\mathrm{dia,dl}}}[\tilde{\mathbf{g}}_{i,s}^{\mathrm{dl}}]_{\mathcal{B}_{i,s}^{\mathrm{dia,dl}}}^H(\mathbf{Q}_{k,i,s}^{\mathrm{dl}})^T$$
$$+\sum_{j\neq s}^{S}\sum_{i\in\mathcal{K}_j}c_{i,j,s}^{\mathrm{dl}}\mathbf{Q}_{k,i,s}^{\mathrm{dl}}[\tilde{\mathbf{g}}_{i,s}^{\mathrm{dl}}]_{\mathcal{B}_{i,s}^{\mathrm{dia,dl}}}[\tilde{\mathbf{g}}_{i,s}^{\mathrm{dl}}]_{\mathcal{B}_{i,s}^{\mathrm{dia,dl}}}^H(\mathbf{Q}_{k,i,s}^{\mathrm{dl}})^T, \tag{103}$$

and $c_{i,j,s}^{\mathrm{dl}}=(\mathbf{c}_{i,j}^{\mathrm{dl}})^H\mathbf{R}_{i,s}^{\mathrm{ut,dl}}\mathbf{c}_{i,j}^{\mathrm{dl}}$. Then, we can obtain that

$$\mathrm{ASLNR}_{k,s}^{\mathrm{dl,b}}\left(\check{\mathbf{w}}_{k,s}^{\mathrm{aslnr,dl}},\mathcal{B}_{k,s}^{\mathrm{dia,dl}}\right)\geq\mathrm{ASLNR}_{k,s}^{\mathrm{dl,b}}\left(\check{\mathbf{w}}_{k,s}^{\mathrm{aslnr,dl}},\bar{\mathcal{B}}_{k,s}^{\mathrm{dia,dl}}\right). \tag{104}$$

Combining (98) and (104) yields

$$\mathrm{ASLNR}_{k,s}^{\mathrm{dl,b}}\left(\check{\mathbf{w}}_{k,s}^{\mathrm{aslnr,dl}},\mathcal{B}_{k,s}^{\mathrm{dia,dl}}\right)=\mathrm{ASLNR}_{k,s}^{\mathrm{dl,b}}\left(\check{\mathbf{w}}_{k,s}^{\mathrm{aslnr,dl}},\bar{\mathcal{B}}_{k,s}^{\mathrm{dia,dl}}\right). \tag{105}$$

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